
Faculty of Mathematical Sciences



University of Twente
The Netherlands

P.O. Box 217
7500 AE Enschede
The Netherlands

Phone: +31-53-4893400

Fax: +31-53-4893114

Email: memo@math.utwente.nl

www.math.utwente.nl/publications

MEMORANDUM No. 1610

Numerical modeling of magma-repository
interactions

O. BOKHOVE

DECEMBER 2001

ISSN 0169-2690

NUMERICAL MODELING OF MAGMA-REPOSITORY INTERACTIONS

Onno Bokhove

Faculty of Mathematical Sciences, Twente, The Netherlands

Abstract

This report explains the numerical programs behind a comprehensive modeling effort of magma-repository interactions. Magma-repository interactions occur when a magma dike with high-volatile content magma ascends through surrounding rock and encounters a tunnel or drift filled with either a magmatic gas or air at atmospheric pressure. The simplified mathematical model to describe these flow phenomena consists of compressible flow equations of one- or two-dimensional flow in a flow tube with a macroscopic equation of state relating the pressure to the density for a bubbly mixture of water in the liquid or vapor phase and the molten rock under isothermal conditions. These model equations are hyperbolic and consist of a conservative part and additional frictional and geometric terms. The latter geometric terms contain the influence of the varying cross section or width of the flow tube, and gravity. Two shock-capturing numerical methods, a Local Lax Friedrichs and an Essentially Non-Oscillatory scheme, are used to simulate these flows numerically. An extensive account is given of these two numerical methods and their validation for the magma flow application, including a comparison between various exact solutions and numerical solutions for idealized conditions. Finally, we give one- and two-dimensional reference simulations in more realistic flow geometries.

Keywords: magma-repository interactions, decompressing magma flows, hyperbolic equations, shock capturing Lax-Friedrichs and ENO numerical methods, flow tube models

AMS Subject Classification: 93A30, 35L65, 35L15, 74A50, 74S20

The research presented in this report has been funded in the year 1998 through a contract of the University of Bristol and in the year 2000-2001 through a contract of the Faculty of Mathematical Sciences of the University of Twente with the Center for Nuclear Waste Regulatory Analyses, Southwest Research Institute, San Antonio, Texas, U.S.A.

“Numerical modeling of magma-repository interactions”, by Onno Bokhove
version MagAir0.2 August, 2000
version MagRep0.0 April-December, 2001

Contents

1	Introduction	1
2	Flow-tube model for volatile-rich magma	5
2.1	Two fluid model: melt-volatiles and air	5
2.2	Scaling	8
2.3	One- and two-dimensional gas dynamics	8
3	Numerical schemes for conservation laws	12
3.1	Spatial discretization	12
3.2	Temporal discretization	13
3.3	Characteristic decomposition	13
3.4	Essentially Non-Oscillatory schemes	15
3.5	Convex Essentially Non-Oscillatory schemes	18
3.6	Forcing: gravity, geometry and dissipation	19
3.7	Boundary conditions	20
3.7.1	Open or extrapolating	20
3.7.2	Characteristic lower boundary condition	20
3.7.3	Solid boundaries	20
3.7.4	Interface between melt-volatile fluid and air	20
3.8	Structure of 1D programs	23
3.8.1	Input file	27
3.8.2	Core program	30
3.9	Structure of 2D programs	32
3.10	Portability	32
4	Numerical tests	34
4.1	Dynamics of magma 1D	34
4.1.1	Stationary shock	34
4.1.2	Moving and reflected shock	39
4.1.3	Shock tube with and without tunnel end	40
4.2	Dynamics in air 1D	47
4.2.1	Stationary shock	47
4.2.2	Moving and reflected shock	48
4.2.3	Shock tube	49

4.3	Dynamics of a melt-volatile fluid and air 1D	51
4.3.1	Shock tube	51
4.3.2	Resonating shocks	58
4.4	One block 2D magma dynamics tests	61
4.5	Two block 2D magma dynamics tests	65
5	Magma-repository interactions	75
5.1	Geometry and grid transformation in 1D	75
5.2	Reference 1D simulation	77
5.3	1D simulations for Science paper	78
5.4	Geometry and grid transformation in 2D	87
5.5	Future dike-drift geometries	89
6	Conclusion	90
A	Retrieval of package	93
B	Examples	94
B.1	Example 1: stationary shock in magma	94
B.2	Example 2: shock tube in air	95
B.3	Example 3: reference simulation run 101	95
B.4	Example 4: science simulation run 165	96
B.5	Example 5: one block 2D shocktube simulation run 202	96
B.6	Example 6: two block 2D magma simulation run 212	97

Chapter 1

Introduction

This report explains the numerical programs behind a comprehensive modeling effort of magma-repository interactions. Magma-repository interactions occur when a dike with high-volatile content magma ascends through surrounding rock and encounters a tunnel or drift filled with either a magmatic gas or air at atmospheric pressure. While magma-air interactions occur in natural systems such as lava tubes, the primary incentive has been to model volcanic flows in a magma dike and a closed repository drift in which nuclear waste is stored. One significant question is whether magma flows that explode into a repository drift are able to destroy canisters filled with nuclear waste and whether they are subsequently able to transport these canisters or small pieces thereof. Explosive fluid dynamical events which develop after a volcanic dike intrudes a series of repository drifts constitute one component in a hazard analysis for a nuclear repository site in a geological active area, such as the one proposed at Yucca Mountain [Connor *et al.*, 1997, 2000]. The present report describes in detail how these explosive volcanic flows have been modeled numerically.

The processes involved in such magma-repository interactions may be extremely complex. We therefore introduce the following five simplifying assumptions in order to develop some fundamental understanding of the basic magma physics. Our approach follows much of the established literature in which magma ascent is studied [Wilson *et al.*, 1980; Wilson and Head, 1981; Jaupart and Allègre, 1991; Woods, 1995].

i) Given the geological record, we assume that prior to breakthrough into the drift, the magma in a dike of $1 - 2\text{ m}$ width will be driven upward with a typical speed of order $1 - 2\text{ m/s}$ by a combination of chamber overpressure and magma buoyancy, typically of the order of $1 - 2\text{ MPa}$.

ii) We only consider the interaction of one section of the magma dike with one of the series of parallel repository tunnels which are planned in the Yucca Mountain site (see Fig. 1.1a). The proposed spacing between the tunnels is 80 m , and this defines the width of the relevant section of the magma dike. The proposed repository tunnel diameter is 5 m and the tunnels lie at a depth of about $300 - 400\text{ m}$. Furthermore, using the left-right symmetry in a dike that intersects a drift at a non-zero angle to its axis, we only consider one half of the dike-drift configuration.

iii) The fracture mechanics associated with the dike-repository system is complicated and poorly understood. We will not attempt to solve the combined rock and fluid mechanics problem but instead assume that the dike-repository geometry is prescribed or fixed from the onset. Our model is then initiated at the point when the dike, ascending through the crust, breaks through into the tunnel or drift. Crack propagation in an ascending dike is generally argued to be magma controlled. There appear to be two types of steady dike propagation described in the literature, in which the dike tip is either an under-pressured bulbous head [Lister, 1990; Lister and Kerr, 1991] or a thin crack lying ahead of the magma [Rubin, 1993, 1995]. More

recently, Mériaux *et al.* [1999] argued that crack propagation is dominated by the magma fluid dynamics, and that the damaged zone in the rock around the tip is narrow, in support of the original prediction of Lister [1990]. Following the results of Lister [1990], we consider a dike tip that narrows over about 40 m to a nozzle area at the dike-drift intersection. We present both simulations in which the nozzle area is fixed in time and ones in which the nozzle acts as an iris and opens from a very small value to its larger final value. The latter is a very simple parameterization to examine the effect of the opening or closing of the rock at the dike-drift intersection.

iv) The basaltic magma and exsolved volatiles form a multiphase mixture. Following Wilson and Head [1991], we parameterize this as a pseudo-one-phase fluid with a monotonic relationship between pressure p and density ρ . Hence the magma is effectively compressible from a macroscopic view point. Although there has been some analysis of the effects of the relative motion between the bubbles in a conduit, the bulk phenomena are comparable [Vergnolle and Jaupart, 1986]. For simplicity we therefore neglect these effects herein.

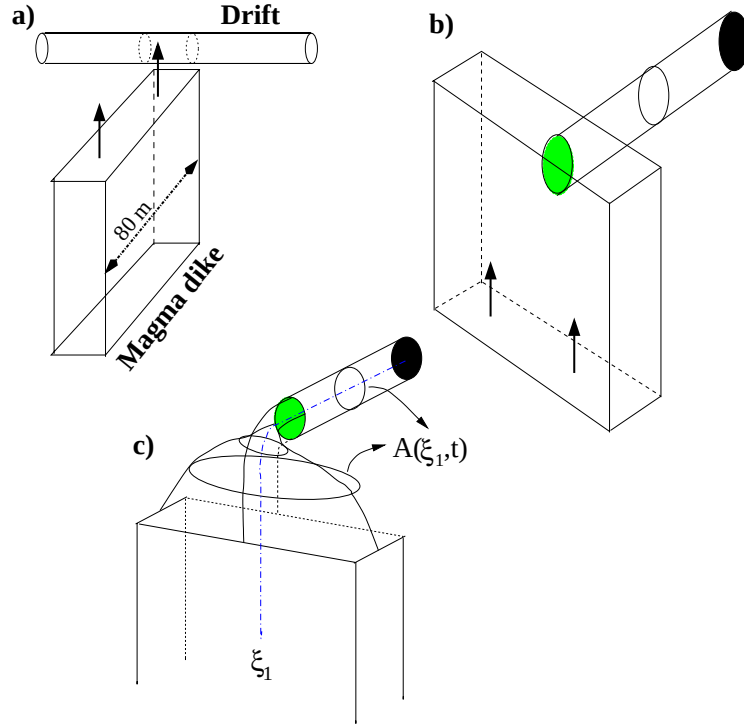


Figure 1.1: Sketches of the dike-drift system. (a) A magma dike is ascending towards a drift. (b) The idealized and experimental dike-drift configuration in which flow starts after a vertical diaphragm (green/grey dashed patterned) between dike and drift is removed instantaneously. In the experiment, a gum-rosin-acetone (GRA) mixture ascends in a vertical Hele-Shaw cell by imposing a vertical pressure gradient. GRA explodes into the under-pressured drift after removal of the diaphragm. (c) Sketch of a simplified configuration in a flow-tube model of the flow in dike and tunnel. The flow in this model depends only on a smooth curvilinear coordinate ξ_1 which follows the dike, and turns via an arc into the drift; variations of the cross-sectional area $A(\xi_1, t)$ of dike, connecting arc, and drift are captured in the model.

v) The complex three-dimensional fluid dynamics of magma and air is simplified into either a one-dimensional flow-tube model or a two-dimensional flow-area model in an idealized geometry. The one-

dimensional flow tube model describes the flow averaged over cross sections in the dike and the drift. It is sketched in Fig. 1.1c and we consider the flow from one sector of the dike into the tunnel on one side of the dike following the left-right symmetry of the problem (see also Fig. 5.1). The geometry sketched in Fig. 1.1b corresponds to analog experiments which are complementary to our overall investigations (Dr. A.M. LeJeune & Professor A.W. Woods, personal communication). While the high-volatile content magma with its explosive potential is the focus of this work, low-volatile compressible magma have been considered in Woods and Sparks [1998] and in corresponding analog experiments (Dr. A.M. LeJeune, personal communication and observation). Various two-dimensional flow-area configurations are defined in section 5.5 and shown in Fig. 5.8. In the flow-area models, the flow is either averaged laterally in the horizontal y -direction along the dike in both dike and drift, or it is averaged across the dike (x -direction) in the dike and across the drift in the drift (y -direction).

The programs described in this report consider transient compressible magma and air flows in separation, and in combination in a one dimensional flow tube model. In addition, we also consider magma dynamics without air in a two dimensional flow area model. Since shock waves and rarefaction waves are an essential feature in explosive volcanic dynamics, we have used established and modern Essentially Non-Oscillatory (ENO) shock capturing algorithms by Shu and Osher [1988, 1989] and Liu and Osher [1998]. The algorithms employed have been tested and used before by the author for the shallow-water equations [Bokhove and Peregrine, 1999]. The shallow-water equations are identical to the gas dynamics equations for an isothermal or isentropic gas in which pressure is proportional to density: depth in shallow-water dynamics plays a mathematical role that is similar to the role of density in gas dynamics. That is, the shallow-water and gas dynamics equations are both hyperbolic equations (if necessary in the absence of dissipation). The programs employed for compressible magma are thus adaptations of programs the author has used before in a shallow-water context; only the equation of state and ensuing interpretations have been changed. In short, numerical methods for hyperbolic equations have validity in various branches of (pseudo) compressible geophysical and geological fluid dynamics, including shallow-water wave modeling and compressible magma flows. Several preliminary tests have been performed with codes that model volcanic and air flows, in separation and together in horizontal pipes and in dike and drift systems. The dynamics of an interface is modeled when magma and air coexist and strongly influence one another. Varying cross-sections, gravity and parameterized frictional effects are ultimately all taken into account. The final code is validated by comparing idealized simulations in magma, air, and in magma and air separated by an interface, respectively, with exact (implicit) solutions of stationary shocks, moving and reflected shocks, and shock tube solutions. Moreover, two different algorithms and associated programs have been used in the validation phase.

The organization of this reports is as follows. The equations of motion for the compressible magma and air dynamics, and the geometry are presented in chapter 2. One-dimensional flow-tube equations and two-dimensional equations in a vertical plane are derived as simplifications of the three-dimensional equations of motion. These equations are rendered into dimensionless form to facilitate numerics.

Numerical algorithms are explained in detail in chapter 3. Two shock-capturing numerical algorithms, a Local Lax Friedrichs and Essentially Non-Oscillatory scheme, are introduced for hyperbolic and/or conservative systems. The conservative terms in the equations of motion are discretized carefully to model shock propagation and separately from the other, forcing, dissipative, gravitational and geometric terms. Boundary conditions and interface conditions between magma and air require careful attention because numerical interference and instability can occur between conservative and the other —gravitational or geometrical— terms. The description of the structure of the actual programs (section 3.8) is intended to be sufficiently detailed for learning how to use the programs. All programs have been written in the C-language, and have been used

with minor adaptation of the input and output procedures on cc- and gcc-compilers on a Sun Ultrasparc 2200 with two CPU's running at 200 MHz , a HP9000 with one CPU, and on PC's with Pentium-II/III processors and one CPU running at 250 – 350 MHz or 650 MHz , respectively. Matlab graphics programs are available to plot the produced data; the latter are stored separately to allow use of other graphics packages. The matlab programs are far from optimal: creating a Graphical User Interface (GUI) would be desirable.

In chapter 4, a suite of numerical tests has been considered to validate the final codes in one and two dimensions and these tests also form an introduction to learning how to use the programs. Numerical tests consist of comparing numerical exact (inviscid) solutions in a horizontal tube. Further comparison with stationary solutions of volcanic flows in hydraulically controlled horizontal nozzles, in vertical conduits [Woods, 1995] or curved conduits forms an as yet unexploited possibility.

In chapter 5, the irregular numerical grids in one and two dimensions are defined. In addition, several magma-repository interactions in one and two dimensions are explored, including a reference simulation from Bokhove and Woods [2000, 2001] and the transient simulations in Woods *et al.* [2001]. We also sketch some two-dimensional dike-drift geometries we wish to consider in future research.

Finally, several Appendices contain instructions on retrieval of the numerical package, and worked examples for running the codes and plotting a selected few output files.

It is unnecessary to read this report in chronological order when the goal is to use or modify the developed numerical programs. To get acquainted with the code, it is best to start with chapter 2; continue with sections 3.1, 3.2, 3.8, 3.9, and 3.10; and then follow the instructions in Appendices A and B while browsing through the Makefiles and the remainder of this report.

Chapter 2

Flow-tube model for volatile-rich magma

The fluid equations of the compressible dynamics of magma and air are introduced in turn. These equations form the basis for subsequent analytical and numerical investigations of explosive dynamics of magma-repository interactions.

2.1 Two fluid model: melt-volatiles and air

The multiphase basaltic fluid of melt and volatiles is modeled as an isothermal compressible fluid with a parameterized equation of state. The density ρ of this fluid equals the reciprocal of the volume occupied by a unit mass of the mixture of exsolved volatiles (gas), dissolved volatiles (liquid) and melt. The mass fraction of exsolved volatiles is $n(p)$. Dissolved volatiles and melt are lumped together as an incompressible mixture of mass fraction $1 - n(p)$ and density σ , giving

$$\rho(p) = \left(\frac{n(p) R_v T}{p} + \frac{1 - n(p)}{\sigma} \right)^{-1}, \quad (1)$$

where $R_v \approx 462 \text{ J kg}^{-1} \text{ K}^{-1}$ is the gas constant for H_2O in the basaltic mixture, the mixture temperature $T = 1000 - 1200 \text{ K}$, and lumped melt-liquid density $\sigma \approx 2500 \text{ kg m}^{-3}$. Volatiles, bubbles and melt are assumed to be in chemical equilibrium modeled using Henry's law [Sparks, 1978]

$$n(p) = n_0 - n_{s_H} \equiv n_0 - s_H p^\beta. \quad (2)$$

Here the total volatile content $n_0 = 1.0 - 2.5$ weight percent ($wt\%$), the solubility constant $s_H \approx 3 \times 10^{-6} \text{ Pa}^{-1/2}$ and $\beta \approx 1/2$ for basaltic magmas. The model is a leading order approximation to the complicated physics of the multi-phase melt-volatile mixture. Supersaturation effects are neglected [Sparks, 1978; Woods, 1995], and we assume there is negligible slip between the phases.

Eventually all the volatiles become dissolved when the pressure reaches a critical pressure p_c where $n(p = p_c) = 0$ [equation (2)]. For example, for gas mass fractions $n_0 = 0.01, 0.025 \text{ wt}\%$ the critical pressure $p_c = (n_0/s_H)^{1/\beta} \approx 1.11, 6.9 \text{ MPa}$. For $p > p_c$, the purely liquid flow is “nearly” incompressible with a typical sound speed of about $c_i = 1400 - 2000 \text{ m/s}$. In the present study we only consider pressures $p < p_c$, for which there is a highly compressible exsolved volatile phase, although one could in principle couple the nearly incompressible purely liquid region with $\rho = \sigma + (p - p_c)/c_i^2$ for $p > p_c$ and the compressible region

of the pseudo one-phase fluid modeled by (1) and (2) for $p < p_c$. This has been done with constant $\rho = \sigma$ for $p > p_c$ in models with stationary flow [Wilson *et al.*, 1980; Woods, 1995].

For typical volcanic parameters, the values $\rho(p)$ and sound speed $a(p) = \sqrt{\partial p / \partial \rho}$ have been drawn in Fig. 2.1 for a) a wet basaltic magma and b) a gum-resin-acetone mixture used in analog laboratory experiments. In the analog laboratory experiment the melt-volatile mixture is replaced by a gum-rosin-acetone (GRA) multiphase fluid. The gum-rosin has properties similar to rhyolite and the acetone is the volatile. Gum rosin is a brittle amorphous solid at room temperature that changes to a liquid with a small amount of dissolved acetone [Phillips *et al.*, 1995]. The properties of GRA are incompletely understood (A.W. Woods and R.S.J. Sparks, personal communication) but to a certain extent reproduce the phase behavior and volatile dependence of hydrated magmas. While the volatile fractions in the gaseous and liquid phase in basaltic magma change as function of pressure, assuming that an instantaneous equilibrium is established between the two phases, GRA appears to be a boiling mixture in which a large fraction of dissolved acetone changes almost instantaneously into exsolved form when pressure decreases quickly, and vice versa (presumably with hysteresis effects). After this initial transition further exsolution appears to be negligible on the time scales of rapidly decompressing flows we are likely to encounter in a laboratory experiment. Pending more knowledge on the equation of state for the GRA multiphase fluid, GRA is modeled with fixed mass fraction $n(p) = n_0$ in which s_H and β are the free parameters induced from measurements. Based on a figure in Phillips *et al.* [1995], displaying the dissolved mass fraction of volatiles versus the pressure, *i.e.*, in our model $n_s = s_h p^\beta$ versus p , we have used a least square fit to determine $\log(s_H)$ and β in $\log(n_s) = \log(s_H) + \beta \log(p)$. The sound speed c_i in the pine-resin we have used in Fig. 2.1 is just a mere guess. Elaborate laboratory experiments are required to determine the precise $\rho(p)$ -characteristics of the gum-resin-acetone mixture. The relationships graphed in Fig. 2.1b indicate either that we need more data in order to determine the effective and macroscopic equation of state of GRA or that the density of GRA may change quickly as a function of pressure.

The three-dimensional compressible momentum and continuity equations for magma with the equation of state (1) are

$$\begin{aligned} \frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla_3) \mathbf{u} + \frac{1}{\rho} \nabla_3 p &= -g \hat{\mathbf{z}} - \mu \mathbf{F}, \\ \frac{\partial \rho}{\partial t} + \nabla_3 \cdot (\rho \mathbf{v}) &= 0, \end{aligned} \quad (3)$$

with three-dimensional velocity $\mathbf{u}$ and gradient ∇_3 , gravity g , vertical unit vector $\hat{\mathbf{z}}$, frictional or forcing parameter μ and friction or forcing $\mathbf{F}$.

The compressible equations of motion for air (e.g. Batchelor 1967) are

$$\begin{aligned} \frac{\partial \mathbf{u}_a}{\partial t} + (\mathbf{u}_a \cdot \nabla_3) \mathbf{u}_a + \frac{1}{\rho_a} \nabla_3 p_a &= -g \hat{\mathbf{z}} - \mathbf{F}_a, \\ \frac{\partial \rho_a}{\partial t} + \nabla_3 \cdot (\rho_a \mathbf{u}_a) &= 0, \\ \frac{\partial s}{\partial t} + (\mathbf{u}_a \cdot \nabla_3) s &= 0 \end{aligned} \quad (4)$$

with subscripts “a” distinguishing variables in air from those in the basaltic fluid, friction of forcing $\mathbf{F}_a$, entropy s and gas constant $R = 287.04 \text{ J kg}^{-1} \text{ K}^{-1}$. To close the system, we need the first law of thermodynamics

$$dU = T_a ds - p d(1/\rho) \quad (5)$$

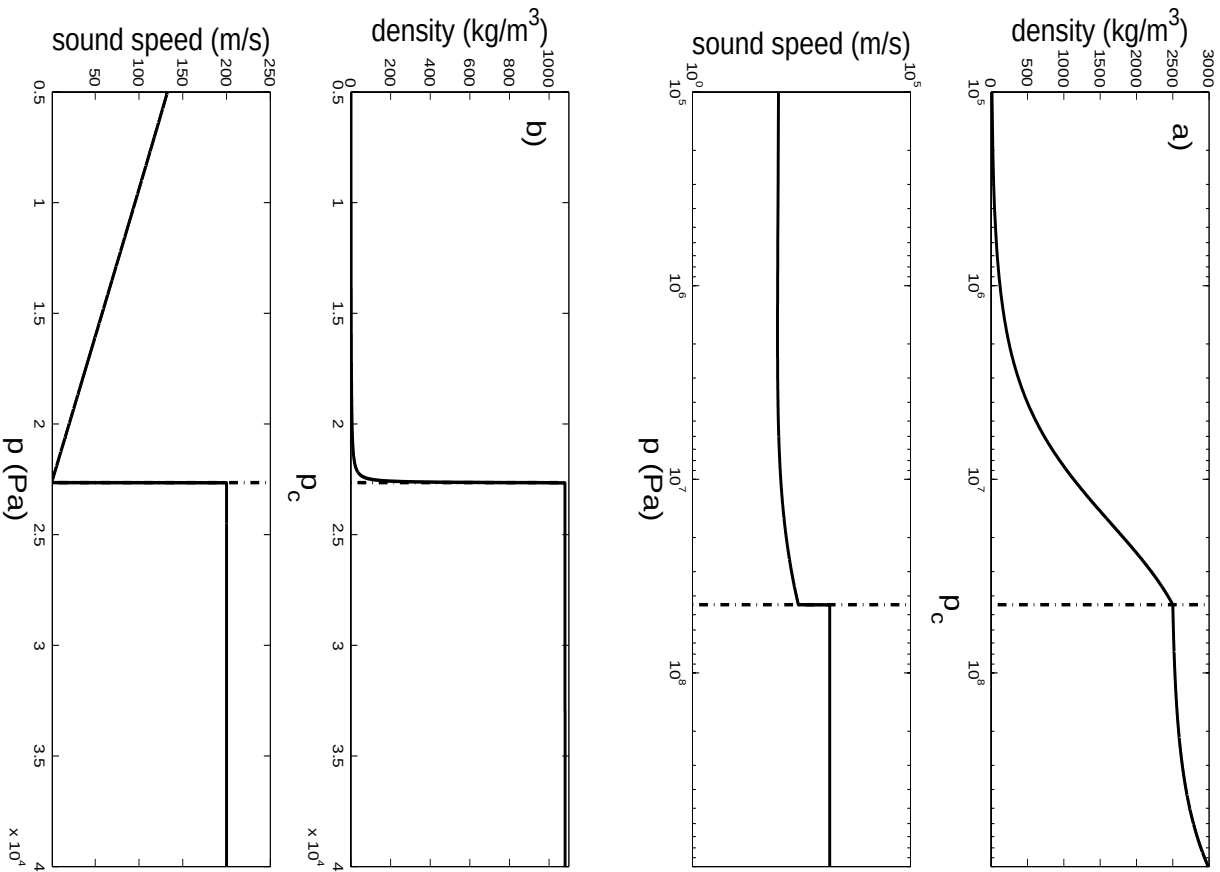


Figure 2.1: Graphs of idealized equations of state for: (a) a basaltic high-volatile content magma with $T = 1000\text{ K}$, $\sigma = 2500\text{ kg m}^{-3}$, $n_0 = 2\text{ wt}\%$, $s_H = 3 \times 10^{-6}\text{ Pa}^{-1/2}$, $R_v = 462\text{ J kg}^{-1}\text{ K}^{-1}$, $\beta = 0.5$, $c_i = 1400\text{ m/s}$ (horizontal log scale) and (b) gum-resin-acetone mixture with $T = 293\text{ K}$, $\sigma = 1080\text{ kg m}^{-3}$, $n_0 = 20\text{ wt}\%$, $s_H = 6.1165 \times 10^{-6}\text{ Pa}^{-1/2}$, $R_v = 482\text{ J kg}^{-1}\text{ K}^{-1}$, $\beta = 1.0366$, $c_i = 100\text{ m/s}$. (i) Density and (ii) sound speed are drawn as function of pressure. Graphs made with Matlab program `densityp.m`, *i.e.* use `densityp(nhc)` with $nch = 0, 1$ for basaltic magma and GRA, respectively. See also `densityscaled.m`.

with air temperature T_a and an equation of state, for which we choose the ideal gas law

$$p_a = \rho_a R T_a. \quad (6)$$

2.2 Scaling

Numerical integration is performed using dimensionless equations. Numerical values can then be chosen to lie around unity to improve numerical stability. By considering the equations of motion and state for magma, the following scalings for various parameters are suggested:

$$\begin{aligned} p &= P_0 p', & \rho &= \rho_0 \rho', & t &= (L/U) t', & \mathbf{x} &= L \mathbf{x}', \\ \mathbf{u} &= U \mathbf{u}, & n(p) &= n_0 n'(p'), & \sigma &= \rho_0 \sigma' \end{aligned}$$

with $P_0 = n_0 R_v T \rho_0$ and $U^2 = P_0/\rho_0$. Upon dropping the primes, the dimensionless (scaled) melt-volatile equations read

$$\begin{aligned} \frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla_3 \mathbf{u} + \frac{1}{\rho} \nabla_3 p &= -\frac{1}{F_l} \hat{\mathbf{z}} - \nu_f \mathbf{F}, \\ \frac{\partial \rho}{\partial t} + \nabla_3 \cdot (\rho \mathbf{v}) &= 0 \end{aligned} \quad (7)$$

with equation of state

$$\rho(p) = \left(\frac{n(p)}{p} + \frac{1 - n_0 n(p)}{\sigma} \right)^{-1} \quad (8)$$

and Henry's law

$$n(p) = 1 - \epsilon p^\beta, \quad (9)$$

Froude number $F_l = P_0/(\rho_0 g L)$, friction number $\nu_f = \mu_0/(L \sqrt{P_0 \rho_0})$, and $\epsilon = s P_0^\beta/n_0$. The density function (8) is implemented in the core code, see section 3.8, as function `Density(pp)`. For a given density the pressure is determined by a root finding routine which uses the function `Dens(pp, d0)` in the core program. A cursory examination of Fig. 2.1a) around pressure values of $10^6 Pa$ and $3 * 10^4 Pa$, respectively, suggests that values of $\rho_0 \approx 100 kg m^{-3}$ and $\rho_0 \approx 10 kg m^{-3}$, respectively, are reasonable for the volcanic fluid and GRA. Length scale L is chosen to be the radius d of the tunnel.

2.3 One- and two-dimensional gas dynamics

Consider a system of curvilinear coordinates ξ_i ($i = 1, 2, 3$) with scale factors h_i [Batchelor, 1967]. An infinitesimal cube in such a coordinate system then has sides $h_i d\xi_i$ and volume $d\tau$ given by $(d\tau)^2 = \sum_{i=1}^3 (h_i d\xi_i)^2$.

In the dike-drift configuration a representative $\xi_1(x, y, z)$ -coordinate may be identified with corresponding cross-sectional area $A(\xi_1, t)$, normal to the ξ_1 -isolines, as is illustrated in Fig. 1.1c by the dashed-dotted lines. The time-dependence of the cross-sectional area A corresponds to the response of the rock walls to pressures in the magma dike or drift, or to the movement of objects in the tunnel as a consequence of pressure gradients. A one-dimensional system, obtained by averaging over cross-sections and by neglecting flows in cross-sectional planes, is derived from (7) by considering mass and momentum density conservation in a

control volume $h_1 A(\xi_1, t) d\xi_1$. Momentum density is defined as ρu_1 and $h_1 = h_1(\xi_1)$ is the scale factor associated with the curvilinear coordinate. Pressure in the ξ_1 -direction acts both on the slices at ξ_1 and $\xi_1 + \Delta\xi_1$, giving a contribution $\sim -\partial(pA)/\partial\xi_1$, and on the flow tube walls between these slices, giving a contribution $\sim p \partial A/\partial\xi_1$. The gravitational force is a volume force. For the basaltic fluid one then finds the following equations of motion (e.g. Whitham 1974):

$$\begin{aligned} \frac{\partial(\rho A u)}{\partial t} + \frac{1}{h_1} \frac{\partial}{\partial \xi_1} \left(\rho A u^2 + p A \right) &= -A F_1 + \frac{p}{h_1} \frac{\partial A}{\partial \xi_1} - \frac{\rho A}{F_l h_1} \frac{\partial z(\xi_1)}{\partial \xi_1}, \\ \frac{\partial(\rho A)}{\partial t} + \frac{1}{h_1} \frac{\partial}{\partial \xi_1} \left(\rho A u \right) &= 0, \end{aligned} \quad (10)$$

where $u = u_1$ is the ξ_1 -component of the velocity, and $-F_1$ the parameterized forcing and dissipation. In a vertical dike or conduit, we have $\xi_1 = z$ and $h_1 = 1$, and in a horizontal tunnel $\xi_1 = x$ and $h_1 = 1$. Corner effects at the tunnels entrance are simplified by considering geometries with smooth dike-drift transitions (cf. Fig. 1.1b). In essence we follow the flow along the (dashed-dotted) centerline $\xi_1 = 0$ in Fig. 1.1c). The evolution of the energy density of magma is governed by

$$\begin{aligned} \frac{\partial}{\partial t} \left(\frac{1}{2} \rho A u^2 + \rho A [U_b(\rho) + \frac{1}{F_l} (z - z_0)] \right) &+ \frac{1}{h_1} \frac{\partial}{\partial \xi_1} \left[\left(\frac{1}{2} \rho A u^2 + \rho A [U_b(\rho) + \frac{1}{F_l} (z - z_0)] + p A \right) u \right] \\ &= -\rho^2 \frac{\partial U_b}{\partial \rho} \frac{\partial A}{\partial t} - A u F_1, \end{aligned} \quad (11)$$

where

$$U_b = \int_{p_0}^p d\tilde{p} \frac{1}{\rho(\tilde{p})} \quad (12)$$

is the internal energy of the basaltic flow relative to some reference pressure p_0 , and z_0 is a reference level. Total, kinetic, and potential plus internal energy are used to monitor some of the simulations. The measurement routine `Measure()` calculates and stores total, kinetic, and potential plus internal energy.

It is not well understood how to model the frictional forces of the bubbly magmatic liquid owing to its complex rheology. The effective viscosity has been shown to increase with volatile exsolution and also with the pressure of bubbles [Jaupart and Allègre, 1991]. These authors proposed an empirical parameterization of frictional dissipation in a high-viscosity magmatic foam, averaged over the cross-sectional area, and proportional to the velocity:

$$F_1 = \frac{3 \mu_0 e^{(5-100 n_0)} (1 - \phi)^{-5/2}}{2 L_e^2} u, \quad (13)$$

where L_e is a typical length scale, for example the width of the dike or the radius of drift or conduit, and $\mu_0 = 10 - 100 \text{ Pa s}$. Relation (13) has validity when the void fraction of the mixture, ϕ , given by

$$\phi = \frac{1}{1 + (1 - n) p / (n R_v T \sigma)} \quad (14)$$

remains below the fragmentation threshold $\phi < \alpha$, where $\alpha = 0.7 - 0.9$ [Woods, 1995]. As ϕ evolves through this regime, the gas becomes the continuous phase and frictional forces diminish. When $\phi > \alpha$, a simple parameterization for turbulent flow in a pipe was proposed by Wilson *et al.* [1980]:

$$F_1 = 0.0025 \frac{\rho |u| u}{L_e}. \quad (15)$$

We can, of course, view these frictional parameterizations as a very crude, leading order (turbulent) closure for the unknown Reynolds stress terms. Both parameterizations refer through the length scale L_e to the geometry of the cross section. In the inviscid limit when $F_1 = 0$ the shape of the cross sections is arbitrary while their area is specified, but when the frictional terms described above are added the area shape is specified through a characteristic cross-sectional width L_e . In our dike-drift system L_e is five times smaller in the dike than in the drift; friction in the dike is thus five to 25 times larger than in the drift. Both parameterizations are implemented in the more involved core program in the function `Forcing()`.

The scaled compressible equations of motion for air in a flow tube are:

$$\begin{aligned} \frac{\partial(\rho_a A u_a)}{\partial t} + \frac{1}{h_1} \frac{\partial}{\partial \xi_1} \left(\rho_a A u_a^2 + p_a A \right) &= -A F_a + \frac{p_a}{h_1} \frac{\partial A}{\partial \xi_1} - \frac{\rho_a A}{F_l h_1} \frac{\partial z(\xi_1)}{\partial \xi_1}, \\ \frac{\partial(\rho_a A)}{\partial t} + \frac{1}{h_1} \frac{\partial}{\partial \xi_1} \left(\rho_a A u_a \right) &= 0, \\ \frac{\partial \Omega}{\partial t} + \frac{1}{h_1} \frac{\partial}{\partial \xi_1} \left((\Omega + p_a) u_a \right) &= -p \frac{\partial A}{\partial t} - u_a A F_a \end{aligned} \quad (16)$$

in which

$$\Omega = \rho_a A \left(\frac{1}{2} u_a^2 + e + \frac{1}{F_l} (z - z_0) \right) \quad (17)$$

is the energy density of air with internal energy e . Air is modeled as an ideal gas with $p_a = \kappa(s) \rho_a^\gamma = (\gamma - 1) \rho_a e$, where $\kappa = \kappa(s)$ is a function of entropy s and $\gamma = c_p/c_v = 1.4$ is the ratio of specific heats at constant pressure and volume, respectively. Viscous forces in air will be ignored, i.e. $F_a = 0$.

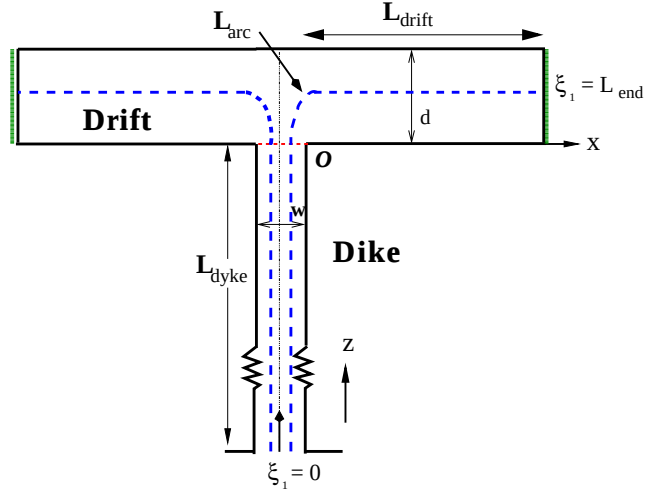


Figure 2.2: A vertical cross section of the magma dike and drift system defines the various length scales involved in typical flow-tube model and two-dimensional simulations. Using a symmetric configuration, we only consider the right half of the flow into the drift.

A similar approach may be followed to derive equations for representative flow in a vertical cross section with width $\Delta y = b(x, z, t)$, see Fig. 2.2. The corresponding two-dimensional equations of motion for basaltic magma are

$$\frac{\partial(\rho b u)}{\partial t} + \frac{\partial(\rho b u^2 + p b)}{\partial x} + \frac{\partial(\rho b u w)}{\partial z} = p \frac{\partial b}{\partial x} - F_1,$$

$$\begin{aligned}
\frac{\partial(\rho b)}{\partial t} + \frac{\partial(\rho u b)}{\partial x} + \frac{\partial(\rho w b)}{\partial z} &= 0, \\
\frac{\partial(\rho b w)}{\partial t} + \frac{\partial(\rho b u w)}{\partial x} + \frac{\partial(\rho b w^2 + p b)}{\partial z} &= p \frac{\partial b}{\partial z} - \frac{1}{F_l} \rho - F_3,
\end{aligned} \tag{18}$$

and similar equations emerge for air.

Chapter 3

Numerical schemes for conservation laws

3.1 Spatial discretization

Essentially non-oscillatory (ENO) finite-difference numerical schemes have been used to discretize (10) and (16) on a uniform grid. The last decades have seen major improvements in shock-capturing numerical schemes. We primarily used a Local Lax Friedrichs (LLF2; Liu and Osher, 1998), which is second order in space and time, and have compared this scheme with a third-order Essentially Non-Oscillatory (ENO) scheme by Shu and Osher [1988, 1989] in all numerical tests. The speed of the LLF2 scheme favored its use in the simulations of magma-repository interactions in a dike-drift geometry. An extension of the LLF2 scheme to a third-order Convex ENO scheme, as developed by Liu and Osher [1998], is straightforward but has not yet been completed.

All the above-mentioned shock-capturing schemes integrate the equations in conservative form. Consider a system of conservation laws

$$\mathbf{q}_t + \sum_{k=1}^d \mathbf{f}_k(\mathbf{q})_{x_k} = \mathbf{q}_t + \mathcal{L}(\mathbf{q}) = \mathbf{F}^\dagger(q, x, t) \quad (1)$$

with initial condition $\mathbf{q}(\mathbf{x}, 0) = \mathbf{q}_0(\mathbf{x})$, $\mathbf{q} = (q_1, \dots, q_n)^T$, $\mathbf{x} = (x_1, \dots, x_d)^T$, and additional forcing, dissipative and/or geometric terms are denoted by $\mathbf{F}^\dagger$. On the computational grid $\mathbf{x}_j = j \Delta \mathbf{x}$, $t_n = n \Delta t$, and $\mathbf{q}_j^n$ is the computed approximation of the exact solution $\mathbf{q}(\mathbf{x}_j, t_n)$. We continue our description of the algorithms for simplicity in one spatial dimension, where we have $\mathbf{q}_t + \mathbf{f}(\mathbf{q})_x = \mathbf{F}^\dagger$. Non-uniform grids may be transformed to uniform ones under a smooth transformation $\phi = \phi(x)$. This is of interest when we wish to model the flow far in the dike on a coarse grid and the flow around the entrance of the tunnel and the flow in the tunnel on a fine grid in order to enhance the resolution of shock waves around the tunnel. All numerical methods presented here remain valid in this stretched curvilinear framework [Shu, 1997], where we have $\mathbf{q}_t + \phi_x f(\mathbf{q})_\phi = \mathbf{F}^\dagger$.

The conservative schemes considered have a discretization of the form

$$\frac{dq}{dt} = q_j^n - \lambda (\hat{f}_{j+1/2} - \hat{f}_{j-1/2}) + F_j^\dagger \quad (2)$$

with $\lambda = \Delta t / \Delta x$, for an appropriate higher-order flux $\hat{\mathbf{f}}$ such that $L(q) = \mathcal{L}(q) + O((\Delta x)^m)$. (When no confusion arises vector notation is dropped.)

The LLF2, ENO, or Convex ENO schemes use an adaptive stencil to calculate the higher-order flux $\hat{f}_{j+1/2}$. An adaptive stencil uses information on r grid points to the left and s grid points to the right of the central grid point j . The stencil is determined independently at each grid point j , which fixes r and s , based on information at surrounding grid points such that discontinuities are avoided. Information of at least one upwind grid point is taken into account. ENO and Convex ENO schemes use higher-order interpolating formulae, while a wave limiter is used in LLF2 to choose the smoothest stencil.

Discretization in two dimensions on a regular grid is a line-by-line extension of the one-dimensional ENO and convex ENO algorithms.

3.2 Temporal discretization

Either a second-order or third-order total variation diminishing (TVD) Runge-Kutta time discretization [Shu and Osher, 1988, 1989] has been used:

$$\begin{aligned} q^{(1)} &= q^n + \Delta t [L(q^n) + F^{\dagger n}] \\ q^{n+1} &= \frac{1}{2} \left(q^n + q^{(1)} + \Delta t [L(q^{(1)}) + F^{\dagger(1)}] \right) \end{aligned} \quad (3)$$

and

$$\begin{aligned} q^{(1)} &= q^n + \Delta t [L(q^n) + F^{\dagger n}] \\ q^{(2)} &= \frac{1}{4} \left(3q^n + q^{(1)} + \Delta t [L(q^{(1)}) + F^{\dagger(1)}] \right) \\ q^{n+1} &= \frac{1}{3} \left(q^n + 2q^{(2)} + 2\Delta t [L(q^{(2)}) + F^{\dagger(2)}] \right). \end{aligned} \quad (4)$$

When the terms denoted by $F^{\dagger}$ contain no spatial derivative of the variables q , their discretization in time is straightforward. When (part of) the flux $\hat{f}$ balances $F^{\dagger}$ in stationary flows, which occurs for example in quiescent hydrostatic balanced flows, such a straightforward discretization may violate that balance at higher order and an adapted strategy may be required (see the discussion in section 3.6).

3.3 Characteristic decomposition

Systems of conservation laws can in general be written in the form

$$\mathbf{q}_t + \mathbf{A} \mathbf{q}_x + \mathbf{B} \mathbf{q}_z = 0. \quad (5)$$

When such a system is hyperbolic, it can be decomposed into characteristic form by using the left $\mathbf{R}_{A,B}^{-1}$ and right $\mathbf{R}_{A,B}$ eigenvector matrices of $\mathbf{A}$ and $\mathbf{B}$ in each dimension respectively. In one dimension, multiplication by $\mathbf{R}_A^{-1}$ brings (5) into characteristic form:

$$\mathbf{R}_A^{-1} \mathbf{q}_t + \mathbf{\Lambda} \mathbf{R}_A^{-1} \mathbf{q}_x = 0 \quad (6)$$

with diagonal eigenvalue matrix $\mathbf{\Lambda}$ and identity matrix $\mathbf{I} = \mathbf{R}_A^{-1} \mathbf{R}_A$.

Matrix $\mathbf{\Lambda}$ is locally taken as constant and fluxes are evaluated at cell edges in ENO interpolations. The fluxes $\hat{\mathbf{f}}_{j+1/2}$ in the original formulation are found by projection. back from eigenvector space using the right

eigenvector matrix $\mathbf{R}_{A,B}$. Roe-averaged quantities [Shu and Osher, 1989] are used to calculate variables at the cell edge $j + 1/2$ from the variables at finite-difference points j and $j + 1$:

$$\begin{aligned} u_{j+1/2} &= \frac{u_j \sqrt{\rho_j} + u_{j+1} \sqrt{\rho_{j+1}}}{\sqrt{\rho_j} + \sqrt{\rho_{j+1}}} \\ \rho_{j+1/2} &= \frac{1}{2} (\rho_j + \rho_{j+1}) \\ H_{j+1/2} &= \frac{H_j \sqrt{\rho_j} + H_{j+1} \sqrt{\rho_{j+1}}}{\sqrt{\rho_j} + \sqrt{\rho_{j+1}}} \end{aligned} \quad (7)$$

with $H = (\tilde{\Omega} + p)/\rho$ and $\tilde{\Omega} = (1/2) \rho u^2 + \rho e$ used only for air dynamics. Projection of the fluxes and variables onto the space of eigenvectors and back is computationally expensive. Alternatively, we could define the required edge values of u, ρ, H by using a HLLC solver [Toro, 1997].

With base variables $q = \{m_1 = \rho u, \rho m_2 = \rho v\}$ for magma, taking $A = h_1 = 1$ momentarily, matrices $\mathbf{A}, \mathbf{B}, \mathbf{R}_{A,B}^{-1}$ and $\mathbf{R}_{A,B}$ have the form:

$$\mathbf{A} = \begin{pmatrix} 2u & a^2 - u^2 & 0 \\ 1 & 0 & 0 \\ v & -uv & u \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} v & -uv & u \\ 0 & 0 & 1 \\ 0 & a^2 - u^2 & 2v \end{pmatrix}, \quad (8)$$

$$\mathbf{R}_A = \begin{pmatrix} u - a & u + a & 0 \\ 1 & 1 & 0 \\ v & v & 1 \end{pmatrix}, \quad \mathbf{R}_A^{-1} = \begin{pmatrix} -\frac{1}{2a} & \frac{u+a}{2a} & 0 \\ \frac{1}{2a} & \frac{a-u}{2a} & 0 \\ 0 & -v & 1 \end{pmatrix}, \quad (9)$$

$$\mathbf{R}_B = \begin{pmatrix} u & u & 1 \\ 1 & 1 & 0 \\ v - a & v + a & 0 \end{pmatrix}, \quad \mathbf{R}_B^{-1} = \begin{pmatrix} 0 & \frac{v+a}{2a} & \frac{-1}{2a} \\ 0 & \frac{a-v}{2a} & \frac{1}{2a} \\ 1 & -u & 0 \end{pmatrix} \quad (10)$$

with sound speed $a(\rho) = a(p) = \sqrt{\partial p / \partial \rho}$. The eigenvalues $\Lambda_{A,B}$ are $u - a, u + a, u$ and $v - a, v + a, v$, respectively. The characteristic equations in one spatial dimension for the x -direction are:

$$\begin{aligned} \frac{\partial(u \pm c)}{\partial t} + (u \pm a) \frac{\partial(u \pm c)}{\partial x} &= 0, \\ \frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} &= 0 \end{aligned} \quad (11)$$

with $c(\rho) = c(p) = \int^p d\tilde{p} [1/\rho(\tilde{p})] \sqrt{\partial \rho / \partial \tilde{p}}$.

With base variables $q = \{m_a = \rho_a u_a, \rho_a, \tilde{\Omega}\}$ for air, taking $A = h_1 = 1$, matrices $\mathbf{A}, \mathbf{R}^{-1}$ and $\mathbf{R}$ have the form (subscripts “a” have been dropped momentarily):

$$\mathbf{A} = \begin{pmatrix} (2 - \gamma)u & \frac{1}{2}(\gamma - 3)u^2 & (\gamma - 1) \\ 1 & -u & 0 \\ [\frac{\gamma \tilde{\Omega}}{\rho} - \frac{3}{2}(\gamma - 1)u^2] & [(\gamma - 1)u^3 - \frac{\gamma \tilde{\Omega} u}{\rho}] & (\gamma - 1)u \end{pmatrix}, \quad (12)$$

$$\mathbf{R} = \begin{pmatrix} u - c_a & u + c_a & u \\ 1 & 1 & 1 \\ (H - u c_a) & (H + u c_a) & \frac{1}{2}u^2 \end{pmatrix}, \quad \mathbf{R}^{-1} = \begin{pmatrix} -\frac{1}{2}(\frac{1}{c_a} + u b_1) & (\frac{1}{2}b_2 + \frac{u}{c}) & \frac{1}{2}b_1 \\ \frac{1}{2}(\frac{1}{c_a} - u b_1) & (\frac{1}{2}b_2 - \frac{u}{c}) & \frac{1}{2}b_1 \\ u b_1 & (1 - b_2) & -b_1 \end{pmatrix} \quad (13)$$

with sound speed $c_a = \sqrt{\gamma p / \rho}$, $b_1 = (\gamma - 1)/c_a^2$, $b_2 = \frac{1}{2}b_1 u^2$ and $H = (\tilde{\Omega} + p)/\rho$. The eigenvalues Λ are $\{u \pm c_a, u\}$ and the characteristic equations are:

$$\begin{aligned} \frac{\partial p}{\partial t} + (u \pm c_a) \frac{\partial p}{\partial x} \pm \rho c_a \left(\frac{\partial u}{\partial t} + (u \pm c_a) \frac{\partial u}{\partial x} \right) &= 0, \\ \frac{\partial s}{\partial t} + u \frac{\partial s}{\partial x} &= 0. \end{aligned} \quad (14)$$

In the actual code, we use $q = \{m_1 = \rho A u; \rho A; m_2 = \rho A v; \hat{\Omega} = A \tilde{\Omega}\}$ where v is the y -velocity. For one-dimensional flows v remains zero and for isothermal magma the variable $\hat{\Omega}$ is superfluous and set to zero. For uniform pipes one takes $A = 1$. Lateral momentum m_2 is necessary in a two-dimensional version of the code in which A is replaced b . The involved matrices have been updated accordingly.

3.4 Essentially Non-Oscillatory schemes

The ENO algorithm for finite difference grids was developed by Shu and Osher [1988, 1989]. (The presented description of the ENO algorithm is a compilation of material in these references, Shu [1997] and Rogerson [1999].) The ENO calculation of higher-order flux $\hat{f}_{i+1/2}$ in one dimension is based on the use of adaptive “stencils”, in which variables on r grid points left and s grid points to the right of grid point i are used to find the flux at cell edge $i + 1/2$. A higher-order interpolating formula is constructed at each time step by using information from surrounding grid points such that discontinuities are avoided. At least one upwind grid point occupies the stencil.

ENO schemes use the concept of divided differences. Divided differences $V[\cdot]$ of a function $V(x)$ are defined recursively. Starting with

$$V[x_{i-1/2}] = V(x_{i-1/2}), \quad (15)$$

the k^{th} -degree divided differences for $k \geq 1$ are defined by

$$V[x_{i-1/2}, \dots, x_{i+k-1/2}] = \frac{V[x_{i+1/2}, \dots, x_{i+k-1/2}] - V[x_{i-1/2}, \dots, x_{i+k-3/2}]}{x_{i+k-1/2} - x_{i-1/2}}. \quad (16)$$

An important property of divided differences is that

$$V[x_{i-1/2}, \dots, x_{i+k-1/2}] = \frac{V^{(k)}(\xi)}{k!}, \quad (17)$$

where $V^{(k)}$ is the k^{th} -derivative of $V(x)$, for some ξ inside the stencil $[x_{i-1/2}, x_{i+k-1/2}]$ provided function $V(x)$ is smooth in this stencil. When $V(x)$ is discontinuous in the above stencil, then

$$V[x_{i-1/2}, \dots, x_{i+k-1/2}] = O\left(\frac{1}{\Delta x}\right); \quad (18)$$

divided differences can thus be used to detect discontinuities and to choose the smoothest stencil around or at a certain grid point.

When a function $h(x)$ satisfies

$$f(q(x)) = \frac{1}{\Delta x} \int_{x-\Delta x/2}^{x+\Delta x/2} d\tilde{x} h(\tilde{x}) \quad (19)$$

then we have

$$\frac{\partial f(q(x))}{\partial x} = \left(h(x + \Delta x/2) - h(x - \Delta x/2) \right) / \Delta x. \quad (20)$$

Flux $\partial f / \partial x$ is approximated by a numerical flux

$$\left. \frac{\partial f}{\partial x} \right|_i = \frac{\hat{f}_{i+1/2} - \hat{f}_{i-1/2}}{x_{i+1/2} - x_{i-1/2}} \quad (21)$$

and so $\hat{f}_{i+1/2} \approx h(x_{i+1/2})$. Consider the primitive function $H(x)$ of $h(x)$

$$H(x) = \int_{-\infty}^x d\tilde{x} h(\tilde{x}) \Rightarrow \partial H / \partial x = h. \quad (22)$$

Hence, at $x_{i+1/2}$ we find

$$H(x_{i+1/2}) = \int_{-\infty}^{x_{i+1/2}} d\tilde{x} h(\tilde{x}) = \sum_{k=-\infty}^i \int_{x_{k-1/2}}^{x_{k+1/2}} d\tilde{x} h(\tilde{x}) = \Delta x_k \sum_{k=-\infty}^i f(q_k) \quad (23)$$

with $\Delta x_k = x_{k+1/2} - x_{k-1/2}$. The lower limit of the integral is irrelevant and can be replaced by $x_{j_0+1/2}$ for some integer j_0 . Since $H(x_{i+1/2}) - H(x_{i-1/2}) = (x_{i+1/2} - x_{i-1/2}) f(q_i)$, we find that the first divided difference is

$$H_{i-1/2}^1 = H[x_{i-1/2}, x_{i+1/2}] \equiv f[q_i] \quad (24)$$

and in general

$$H_{i-1/2}^k = H[x_{i-1/2}, \dots, x_{i+k-1/2}] = \frac{1}{k} f[q_i, \dots, q_{i+k-1}] \quad \text{and} \quad f_i^k = f[q_i, \dots, q_{i+k}]. \quad (25)$$

Using (16), the first couple of divided differences of f are

$$\begin{aligned} H_{i-1/2}^2 = H[x_{i-1/2}, \dots, x_{i+3/2}] &= \frac{H[x_{i+1/2}, x_{i+3/2}] - H[x_{i-1/2}, x_{i+1/2}]}{x_{i+3/2} - x_{i-1/2}} \\ &= \frac{f[q_{i+1}] - f[q_i]}{x_{i+3/2} - x_{i-1/2}} \end{aligned} \quad (26)$$

$$= \frac{1}{2} f[q_i, q_{i+1}] \quad (\text{on an equally spaced grid}) \quad (27)$$

$$\begin{aligned} H_{i-1/2}^3 = H[x_{i-1/2}, \dots, x_{i+5/2}] &= \frac{H[x_{i+1/2}, \dots, x_{i+5/2}] - H[x_{i-1/2}, \dots, x_{i+3/2}]}{x_{i+5/2} - x_{i-1/2}} \\ &= \frac{1}{3} f[q_i, q_{i+2}] \quad (\text{on an equally spaced grid}). \end{aligned} \quad (28)$$

The utilized portion of

$$H_{i-1/2}^k = H[x_{i-1/2}, \dots, x_{i-1/2+k}] = \frac{H_{i+1/2}^{k-1} - H_{i-1/2}^{k-1}}{x_{i+k-1/2} - x_{i-1/2}}$$

consists of the stencil of grid points used in calculating a divided difference. The sketch in Fig. 3.1 helps to clarify the stencil involved in each divided difference.

So for every i we need to calculate

$$H_{i-v-1/2}^k, H_{i-v+1-1/2}^k, \dots, H_{i+1/2}^k, \quad \text{for } k = 1, \dots, Q_1 + 1 \text{ and } v = k - 1, \quad (29)$$

since for positive Roe speed/eigenvalues we need

$$H_{i-v-1/2}^k, H_{i-v+1-1/2}^k, \dots, H_{i-1/2}^k, \quad (30)$$

and for negative Roe speed/eigenvalues we also need $H_{i+1/2}^k$, that is, at least one upwind point is always involved.

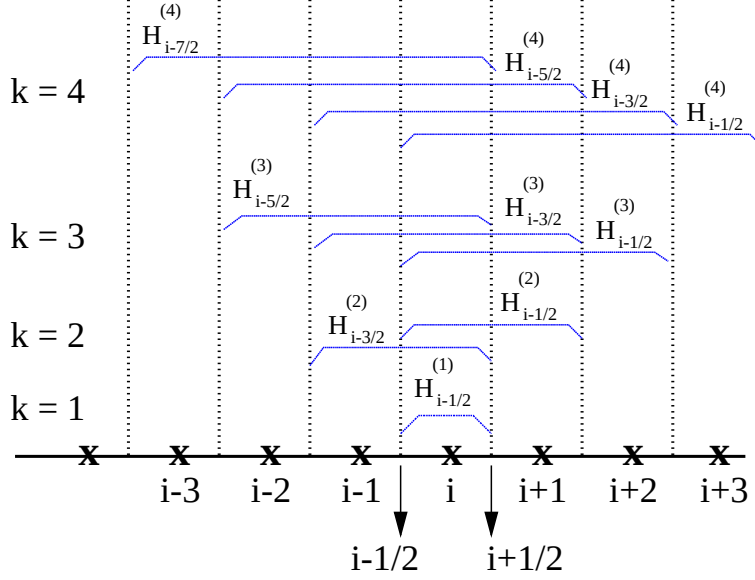


Figure 3.1: The utilized portion of divided differences is defined by their reach, which is sketched up to third order in a space diagram of the grid. Crosses indicate grid points at integer values of i_o and cell edges of grid point i are denoted by dotted lines at half values $i_o \pm 1/2$.

The next step is to obtain $H(x)$ at order Q_1 by a reconstruction via a primitive function technique, for which a Q_1 -point stencil is required. Consequently, to obtain $h(x)$ at order Q_1 , a $(Q_1 + 1)$ -point stencil is needed for its primitive, $H(x)$.

The first point in the stencil is chosen according to the local sign of $f'(q)$ at $x_{i+1/2}$. We may use the Roe speed

$$\bar{a}_{i+1/2} = \frac{f(q_{i+1}) - f(q_i)}{q_{i+1} - q_i} \quad (31)$$

or eigenvalues determined from these Roe speeds for each component of the characteristic equations for a system of partial differential equations, e.g. (11) and (14). In algorithmic form, this so-called ENO-Roe scheme [Shu and Osher, 1989] has the form:

*** Choose first point upwind. ***

if $a_{i+1/2} \geq 0$ then $k_{min}^{(1)} = i$;
else $k_{min}^{(1)} = i + 1$;

$$H^1(x) = H_{k_{min}^{(1)} - 1/2}^{(1)}(x - x_{k_{min}^{(1)} - 1/2})$$

for $k = 2, Q_1 + 1$

*** Smoothest choice gives smallest divided difference. ***

if $|H_{k_{min}^{(k-1)} - 1/2}^{(k-1)}| \geq |H_{k_{min}^{(k-1)} - 3/2}^{(k-1)}|$
then $k_{min}^{(k)} = k_{min}^{(k-1)} - 1$;
else $k_{min}^{(k)} = k_{min}^{(k-1)}$;

$$H^{(k)}(x) = H^{(k-1)}(x) + H_{k_{min}^{(k)}-1/2}^k \prod_{k=k_{min}^{(k-1)}}^{k_{min}^{(k-1)}+k-1} (x - x_{k-1/2})$$

It follows that

$$H(x) \approx H^{(Q_1+1)}(x) = H_{k_{min}^{(1)}-1/2}^1 (x - x_{k_{min}^{(1)}-1/2}) + \sum_{k=2}^{Q_1+1} H_{k_{min}^{(k)}-1/2}^k \prod_{\alpha=0}^{k-1} (x - x_{k_{min}^{(k-1)}+\alpha-1/2}), \quad (32)$$

$$\begin{aligned} h(x) \approx h^{Q_1}(x) &= \frac{\partial H^{(Q_1+1)}}{\partial x} \\ &= H_{k_{min}^{(1)}-1/2}^1 + \sum_{k=2}^{Q_1+1} H_{k_{min}^{(k)}-1/2}^k \sum_{\alpha=0}^{k-1} \prod_{\beta=0(\beta \neq \alpha)}^{k-1} (x - x_{k_{min}^{(k-1)}+\beta-1/2}). \end{aligned} \quad (33)$$

Shu and Osher [1989] also define such an algorithm for Local Lax Friedrichs (ENO-LLF) and a mixture of ENO-Roe and ENO-LLF. Our ENO codes use the mixed scheme. ENO-LLF is slower but more dissipative and is able to handle entropy-violating expansion shocks. ENO-LLF is a successor of the LLF2 scheme presented next. The mixed scheme takes the best of both worlds. In the ENO programs, the ENO-Roe and ENO-LLF schemes are found in the core program in subroutines `EnoRoe()` and `EnoLLF()`. Both routines have been simplified for use on a regular grid with the aid of another routine `Setcoeff()` in which the factors under the multiplication symbol in (33) are calculated.

3.5 Convex Essentially Non-Oscillatory schemes

So far, only a standard second-order local Lax-Friedrichs flux (LLF2) has been implemented. Strictly speaking, this is not the convex ENO implementation by Liu and Osher [1998] but only one of its building blocks. LLF2 consists of the following [Liu and Osher, 1998]:

$$\begin{aligned} \hat{f}_{j+1/2}^{LLF2} &= \frac{1}{2} \left(f(q_{j+1}) + f(q_j) - \alpha_{j+1/2} (q_{j+1} - q_j) \right) + \\ &\quad \frac{1}{4} m \left[\Delta_+ f(q_j) + \alpha_{j+1/2} \Delta_+ q_j, \Delta_- f(q_j) + \alpha_{j+1/2} \Delta_- q_j \right] - \\ &\quad \frac{1}{4} m \left[\Delta_+ f(q_{j+1}) - \alpha_{j+1/2} \Delta_+ q_{j+1}, \Delta_- f(q_{j+1}) - \alpha_{j+1/2} \Delta_- q_{j+1} \right] \end{aligned} \quad (34)$$

with

$$\Delta_{\pm} p_j = \pm (p_{j\pm 1} - p_j) \quad (35)$$

and a canonical minmod wave limiter $m(\cdot, \cdot)$

$$m(x, y) = \begin{cases} (\text{sign } x) \min(|x|, |y|) & \text{if } xy > 0, \\ 0 & \text{otherwise.} \end{cases} \quad (36)$$

There are other choices for the wave limiter, but the minmod one is very robust. For a scalar equation, dissipation $\alpha_{j+1/2}$ is the maximum of $|f'(q)|$ in cell j and $j+1$. For a system the largest eigenvalue (as an upper bound of the largest local eigenvalue) fixes $\alpha_{j+1/2}$ for all components at the cell edge $j+1/2$ between adjacent cells j and $j+1$. Higher-order extensions of convex ENO are found in Liu and Osher (1998). A small modification of the ENO-LLF algorithm in Shu and Osher (1989), as is used in the ENO codes, will yield a faster but more dissipative convex algorithm.

The simplicity of LLF2 and its higher-order convex ENO extensions and the associated computational speed, explains its appeal. None of the time-consuming characteristic decompositions of Shu and Osher's ENO-schemes and other schemes in the literature [*e.g.* Toro, 1990; Leveque, 1997] are required. The numerical simulations with the LLF2 scheme will testify that numerical diffusion and dissipation causes shocks to be smoothed more than in the ENO methods.

3.6 Forcing: gravity, geometry and dissipation

The conservative flux terms in the equations of motion have been preserved in the discretization. The remaining terms are either geometric, gravitational or dissipative and are parameterized as functions of the integrated variables $\{m = \rho u, \rho\}$ and $\{m_a = \rho_a u_a, \rho_a, \tilde{\Omega}\}$ without further spatial discretization. Once the flux terms are discretized, we are left with a system of $Npts$ ordinary differential equations (ODE's), which include forcing and discretized conservative terms, and one of the proposed TVD Runge-Kutta time discretizations described in section 3.2 suffices to integrate these ODE's forward in time.

For a quiescent fluid, the momentum balance

$$\frac{1}{h_1} \frac{\partial(pA)}{\partial \xi_1} - \frac{p}{h_1} \frac{\partial A}{\partial \xi_1} + \frac{\rho A}{F_l} \frac{\partial z}{\partial \xi_1} = 0 \quad (37)$$

is not automatically guaranteed to be preserved under the discretization and the fluid may start artificial flow while it should remain at rest. For solid lower boundaries, hydrostatic balance seems to be preserved in the simulations, but open boundaries require more care. At the lower boundary, the characteristic equations for the magma melt are used. The eigenvalues $u \pm a$ and left and right eigen matrices are calculated from variables at the last interior grid point. Depending on the sign of the eigenvalue, matter either flows out of the domain, in which case exterior ghost points obtain the values of the last interior grid point, or flows into the domain, in which case the ghost points obtain the initial values of the last interior grid point. (One could place this last interior grid point an infinitesimal distance into the domain.) The above method to preserve hydrostatic balance is reasonably satisfactory, but does create a spurious flow which is fortunately small relative to the rarefaction and shock waves of interest. In a horizontal pipe with varying cross-sectional area A artificial flow of a similar kind can arise. At present, the resolution is chosen high enough to minimize artificial flows to an acceptable level. To the best of my knowledge, no general method has been proposed to avoid these artificial flows, although several options exist. One option is to adjust the last two terms in (37) with higher-order corrections that eliminate the discrepancy, another method of Leveque [1998] is specially designed for quasi-steady flows.

In the LLF2 numerical scheme, a division of the pressure into hydrostatic, p_h , and perturbation pressure, p' , with $p = p_h + p'$ allows one to absorb the gravitational term by defining

$$\frac{1}{h_1} \frac{\partial p_h}{\partial \xi_1} = - \frac{\rho}{h_1 F_l} \frac{\partial z}{\partial \xi_1}. \quad (38)$$

Rearranging (38), we find an ordinary differential equation for p_h as function of z at a fixed time

$$\frac{dp_h}{dz} = -\rho(p_h + p'(z)), \quad (39)$$

which can be solved numerically. This division and absorption of the gravitational term takes care of spurious flows but I have not been able to deal properly with the coupling between melt-volatiles and air at the interface.

The numerical treatment of the parameterized viscous forces appears to be good. It would be desirable to test the numerical model and improve our understanding of discretized boundary conditions by considering the hydraulically-controlled stationary flows in horizontal, vertical and curved conduits. Stationary volcanic flows in vertical conduits obey an ODE and have been solved in Woods [1995].

Geometric, gravitational and frictional terms are implemented in the core programs in the function `Forcing()`.

3.7 Boundary conditions

Boundary conditions are implemented in a function called `BounCond()`. For the moment boundary conditions are set (`#define`) in the core numerical program which bears the name of the code. A search for `#define` will quickly locate these definitions near the top of the core program.

3.7.1 Open or extrapolating

Extrapolating boundaries are open boundaries in which interior values are extrapolated in first or fourth order fashion. They are named `EXTRAP`, `EASTEXTRAP` and `WESTEXTRAP`.

3.7.2 Characteristic lower boundary condition

A decomposition of the magma equations into characteristic form reveals which wave components are leaving and entering the domain. The eigenvalues $u \pm a$ in magma at the lower boundary determine whether the corresponding characteristic exits the domain or not. For outgoing waves, exterior ghost points are defined by extrapolating values of the last interior grid point for each characteristic wave separately. For incoming waves, exterior grid points are defined by specifying a wave, for example based on the initial value of the interior grid point. In the presence of forcing, gravity and geometric terms, characteristics no longer stay constant on characteristic curves and an adjustment is required. I have not made these adjustments and the current inviscid option works reasonably well. The characteristic boundary routine is implemented under the name `WESTCHAR`.

3.7.3 Solid boundaries

Solid boundaries are implemented by extrapolating or reflecting the values of density and energy density around the boundary, while taking the opposite value for the momentum density. The last grid point in the fluid has one cell edge coinciding with the boundary. In the function `BounCond()` solid boundaries are defined as `SOLID` (domain closed), `EASTSOLID` (right side of domain closed) and `WESTSOLID` (left side of domain closed).

3.7.4 Interface between melt-volatile fluid and air

Ghost fluid method

Numerical schemes for gas dynamics on Eulerian grids that propagate interfaces between two or more fluids are difficult to construct. Recently, Fedkiw *et al.* [1999] have used a level set approach and a ghost fluid method to model interface dynamics in one and two dimensions. Fedkiw *et al.* [1999] used a level set equation

$$\frac{\partial \Phi}{\partial t} + u \frac{\partial \Phi}{\partial x} + v \frac{\partial \Phi}{\partial y} = 0 \quad (40)$$

to demarcate the interface as the zero level of level set function Φ . Initially, Φ is the signed distance function, which is advected around by (40) using the methods described in Fedkiw *et al.* [1999] and then reinitialized using

$$\frac{\partial \Phi}{\partial t} + S(\Phi) \left(\sqrt{\Phi_x^2 + \Phi_y^2} \right) = 0 \quad (41)$$

to keep Φ approximately equal to the distance function near the interface. $S(\Phi_0)$ is a smoothed sign function of Φ_0 . A wave limiter approach Fedkiw *et al.* [1999] prohibits Φ from developing variability at subgrid scales. The level set approach appears to be more complicated than necessary and does not exactly follow the interface movement. Instead of tracking the level set function, I decided to track the interface contour by particles. In two dimensions, particle advection will have to be filtered to limit subgrid scale motion. Contour advection simulations show that fine-scale Lagrangian structures can be recovered by advection with a coarse-grained Eulerian velocity, so particle advection will be reasonably accurate.

One dimension

In one dimension the interface between melt-volatiles on one and air on the other side is given by a fluid particle at $\xi = \Xi_I(t)$. The Lagrangian dynamics of the particle is governed by

$$\begin{aligned} \frac{d\Xi_I}{dt} &= \frac{u_i}{h_1}, \\ \frac{dU_{int}}{dt} &= -\frac{1}{\rho h_1} \frac{\partial p}{\partial \xi} \Big|_{\xi=\Xi_I(t)} = -\frac{1}{\rho_a h_1} \frac{\partial p_a}{\partial \xi} \Big|_{\xi=\Xi_I(t)}. \end{aligned} \quad (42)$$

Both the interface velocity $U_{int}(t) = u(\xi = \Xi_I(t), t)$ and pressure are continuous across the interface, while density is generally not.

The dynamics of the compressible fluid on each side the interface is modeled with any of the schemes proposed sofar. ENO fluxes, as we have seen, are calculated by using the smoothest higher-order polynomial in a stencil $[x_{i-r}, x_{i+s}]$ around grid point x_i with $r, s > 0$. Velocity $U_{int}(t)$ is constructed from the smoothest polynomial of the velocity $u(\xi, t)$ around $\xi = \xi_i(t)$ in ENO-Roe fashion. The detailed algorithm is a trivial modification of Shu and Osher's [1989] general algorithm.

*** Choose first grid point closest to interface $\xi_{i_0} < \Xi_I < \xi_{i_0+1}$. ***

*** Construct divided differences $H_{i-1/2}^{(k)}$ for velocity $u(\xi)$. ***

if $|\Xi_I - \xi_{i_0}| \leq |\Xi_I - \xi_{i_0+1}|$ then $k_{min}^{(1)} = i_0$

else $k_{min}^{(1)} = i_0 + 1$

$H^{(1)}(x) = H_{k_{min}^{(1)}-1/2}^{(1)}(x - x_{k_{min}^{(1)}-1/2})$

for $k = 2, Q_1 + 1$

if $|H_{k_{min}^{(k-1)}-1/2}^{(k-1)}| \geq |H_{k_{min}^{(k-1)}-3/2}^{(k-1)}|$

then $k_{min}^{(k)} = k_{min}^{(k-1)} - 1$;

else $k_{min}^{(k)} = k_{min}^{(k-1)}$;

$H^{(k)}(x) = H^{(k-1)}(x) + H_{k_{min}^{(k)}-1/2}^{(k)} \prod_{k=k_{min}^{(k-1)}}^{k_{min}^{(k-1)}+k-1} (x - x_{k-1/2})$

It follows that

$$H(x) \approx H^{(Q_1+1)}(x) = H_{k_{min}^{(1)}-1/2}^{(1)}(x - x_{k_{min}^{(1)}-1/2}) + \sum_{k=2}^{Q_1+1} H_{k_{min}^{(k)}-1/2}^{(k)} \prod_{\alpha=0}^{k-1} (x - x_{k_{min}^{(k-1)}+\alpha-1/2}), \quad (43)$$

$$\begin{aligned}
u(x) \approx u^{Q_1}(x) &= \frac{\partial H^{Q_1+1}}{\partial x} \\
&= H_{k_{min}-1/2}^1 + \sum_{k=2}^{Q_1+1} H_{k_{min}-1/2}^k \sum_{\alpha=0}^{k-1} \Pi_{\beta=0(\beta \neq \alpha)}^{k-1} (x - x_{k_{min}+\beta-1/2}^{(k-1)}). \quad (44)
\end{aligned}$$

Only a limited number of divided differences are required around the interface location, so the storage requirements are small.

The interface dynamics is thus determined by particle advection, but to calculate the fluxes at the cell edges in the fluid on either side of the interface the ghost fluid method of Fedkiw *et al.* [1999] is used. We will denote the magmatic fluid left of the interface by FL1 and the air to the right by FL2. Since pressure and velocity are generally continuous across the interface, ghost values of the ghost fluid are obtained by equating values of pressure and velocity of the ghost fluid FL1 right of the interface to those of real fluid FL2, and vice versa left of the interface. While the melt ghost fluid FL1 is thus fully determined, the air ghost fluid FL2 is not. The entropy s of the ghost fluid FL2 left of the interface is determined by extrapolating the entropy of FL2 at grid point $i_I + 2$, the second grid point to the right of the interface, to grid points i_I, i_{I-1}, i_{I-2} (see Fig. 3.2). Since $p_a = \kappa(s) \rho_a^\gamma$, $\kappa_{I+2} = \kappa(s_{I+2})$ is readily obtained from pressure and density values and vice versa. Ghost values for energy and air density can thus be found via κ_{I+2} . Fedkiw *et al.* [1999] call this procedure an isobaric fix, which does prevent local “overheating” and corresponding numerical instabilities. The isobaric fix clearly lacks mathematical rigor—it does not conserve momentum or mass—but works reasonably well. The obvious advantage is simplicity and with local grid refinement the method becomes accurate and remains efficient.

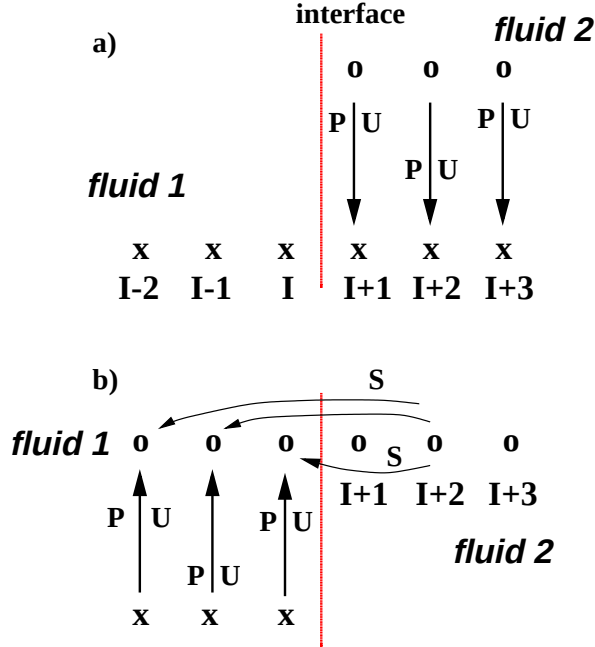


Figure 3.2: (a) By requiring continuity across the interface, pressure P and velocity U of real fluid 2 determine P and U of ghost fluid 1 right of the interface, and vice versa left of the interface. (b) Ghost entropy of fluid 2 (air) is determined by extrapolation of the entropy S of the second grid point $i_I + 2$ in air to grid points left of the interface. This is an isobaric fix. [After Fedkiw *et al.*, 1999.]

I have tried to match the LLF2 and ENO schemes with a time-dependent finite-volume treatment near

the interface, which conserves mass and momentum, but so far I have not been able to overcome numerical instabilities. Since the interface is a fluid particle, discretization in a Lagrangian framework appears to be most efficient. An attempt to discretize the equations in a Lagrangian framework was promising, but the large density difference across the interface lead to undesirable large Eulerian grid spacings in air after the regular Lagrangian grid was mapped back into Eulerian space. Lagrangian grid adjustment posed a new set of presently unresolved problems.

The interface dynamics is set by `#define INTERFACE` and is implemented in the core program in a couple of places. The velocity of the interface is determined by ENO-interpolation in the main loops as `ub[nrk] = Uinterpol(nrk)` and the interface position with notation `xb[nk]` advances in time in the function `Runge-Kutta()`.

3.8 Structure of 1D programs

A series of test programs and final target programs have been developed. Table 3.1 gives an overview of the developed C-programs and their main achievements, and Tables 3.2 and 3.3 list the accompanying Matlab plotting programs and their options for comparison with a few exact solutions. The matlab programs are largely self-explanatory. One-dimensional test programs generally have two to four versions: there are two Eulerian codes, with LLF2 (*e.g.* `encxxx1`) and/or ENO (*e.g.* `enoxxx1`) versions, and one or two Lagrangian codes, with LLF2 (*e.g.* `enclxxx1`) and/or ENO (*e.g.* `enolxxx1`) versions.

Each code consists of several files which are compiled and linked together into one single executable. That executable is named after the core program which contains the numerical algorithm. The header file for both LLF2 and ENO codes is `iso1.h`. For example, the test program with executable `enciso1` for a magmatic fluid in one horizontal dimension consists of five programs:

- `enciso1.c` is the central core program with function `TimeStep()` which provides the name of the executable,
- `enciso1cpu.c` contains the `Main()` C-function which directs to a function `Enoinput()` dealing with reading the input file `enciso1.in`, a function `lncon()` creating general output files `elceXXX`, `elctXXX`, `elcqXXX`, `elciXXX`, `elcdXXX` with run number XXX, and the core function `TimeStep()`,
- `enciso1in.c` creates the output file given a run number XXX, which is specified in input file `enciso1.in`,
- `enciso1inpu.c` reads parameters from the input file `enciso1.in`, and
- `enoiso1ext.c` contains external functions, *e.g.*, functions from Numerical Recipes in C which allocate variable arrays, convey error messages and find roots of provided functions.

The central line in the Makefile for code `enciso1` reads

```
ENCISO1 = enciso1.o enciso1cpu.o enciso1in.o enciso1inpu.o enoiso1ext.o
```

and indicates that programs `enciso1.c` `enciso1cpu.c` `enciso1in.c` `enciso1inpu.c` `enoiso1ext.c` are used in compiling and linking. Except for the core program, the remaining programs are used in several codes based on the same algorithm, that is LLF2 or ENO. For example, for `enoiso1` we find:

```
ENOISO1 = enoiso1.o enoiso1cpu.o enoiso1in.o enoiso1inpu.o enoiso1ext.o.
```

ENO versions of the LLF2 programs are largely similar:

- `enoiso1.c` is the central core program which provides the name of the executable,

ENC	ENO	Characteristics	Exact solutions or comments
enciso1.c	enoiso1.c	melt-volatiles, Eulerian, x -grid	stationary/moving/reflected shock, shock tube
encair1.c	enoair1.c	air, Eulerian, x -grid	stationary/moving/reflected shock, shock tube
encliso1.c	-	melt-volatiles, Lagrangian, a -grid	stationary/moving/reflected shock, shock tube
enclair1.c	-	air, Lagrangian, a -grid	stationary/moving/reflected shocks, shock tube
encmval.c	enomval.c	melt-volatiles & air, Eulerian, x -grid	shock tube, resonating/reflected shock
encmax1.c	-	melt-volatiles & air, Eulerian, ξ_1 -grid varying cross section & arc	-
encmvsl.c	-	melt-volatiles & air, Eulerian stretched ϕ -grid	reduction to ODE for hydraulic flow (not implemented)
encmvt1.c	-	melt-volatiles & air, Eulerian stretched ϕ -grid	time dependent $A(\xi, t)$
enctisl.c	-	magma only & air, Eulerian stretched ϕ -grid	-

Table 3.1: The various available programs and their characteristics. ENO is Essentially Non-Oscillatory [Shu and Osher, 1988, 1989; Shu, 1997]. ENC is second-order Local Lax-Friedrichs [Liu and Osher, 1998]. The a -grid refers to the fluid label a as coordinate in the Lagrangian framework.

Plot program	Fluid	Comments
mva1.m	magma & air	stack plots of u, ρ, p in dike-drift system
mva1sudet.m	magma & air	same with fancy vertical axes
mva1sience.m	magma & air	same with air region indicated in grey
isot1bfilm.m	magma & air	animations of u, p
isoair1bfilm.m	air, magma & air	animations of u, p
diagn.m	magma & air	plots of largest shocks, particles & canisters
mva1stend.m	magma & air	initial and final profiles of several runs
diadp.m	magma & air	plots of largest shocks, particles & canisters of several runs

Table 3.2: Summary of matlab programs for making animations, stack plots for melt-volatiles in curved geometry, and diagnosing the positions in space and time of the largest shocks in air, tracer particles and canisters.

- enoisolcpu.c contains the Main() c-function which directs to a function Enoinput() dealing with reading the input file enoisol.in, a function lnicon() creating general output files elseXXX, elstXXX, elsqXXX with run number XXX, and the core function TimeStep(),
- enoisol.in.c creates the output file given a run number XXX, which is specified in input file enoisol.in,
- enoisolinp.c reads parameter from the input file enoisol.in, and
- enoisolext.c see above.

Observe that the ENO input file and output files have different names. After this introduction, the Makefile should be self-explanatory.

The functions referred to in Table 3.3 are used in the Matlab programs which calculate various exact reference solutions:

$$F_{sair}(M) = \sqrt{\gamma} \kappa^{1/(2\gamma)} p_1^{(\gamma-1)/(2\gamma)} \left(1 + \frac{2\gamma(M^2-1)}{\gamma+1} \right)^{(\gamma-1)/(2\gamma)} + \frac{a_1(M^2-1)}{(\gamma+1)M} - \frac{a_4}{\gamma-1}, \quad (45)$$

$$F_{ndcp}(p) = \frac{1}{\rho^2} \left(\frac{\partial p}{\partial \rho} \right)^2 = \sqrt{\frac{1-\epsilon\sqrt{p}}{p^2} + \frac{\epsilon}{2p\sqrt{p}} - \frac{\epsilon n_0}{2\sqrt{p}}}, \quad (46)$$

$$F_{pmva}(M) = u_2 + \int_{p_1}^{p_2} d\tilde{p} F_{ndcp}(\tilde{p}) - \int_{p_1}^{p_4} d\tilde{p} F_{ndcp}(\tilde{p}), \quad (47)$$

$$p_2 = p_1 \left(1 + 2\gamma \frac{(M^2-1)}{(\gamma+1)} \right), \quad (48)$$

Plot program	Fluid	Exact solution program	Function	Formula
isoair1.m	air	airschok.m shocktube	airshocktu.m	$F_{sair}(M)$
	air	reflsair.m moving & reflected shock	-	-
	magma & air	shtumeltair.m shocktube	fndcp.m preshtmeltair.m	$F_{ndcp}(p)$ $F_{pmva}(M)$
isot1.m	magma	shtuiso.m shocktube	preshtiso.m	$F_{piso}(p)$
	magma	rflshiso.m moving & reflected shock	rflisov3.m	$F_{mriso}(p)$
	magma magma		densiso.m soundiso.m	$\rho(p)$ $a(\rho) = a(p)$

Table 3.3: Programs isot1.m and isoair.m are matlab plotting programs for magma only, air only, and magma and air together, respectively. Various test simulations have been used in which numerical results are compared with exact solutions. Most of these exact solutions are implicit and require some numerical evaluation; the myriad of Matlab-programs that perform this task are presented in this table.

$$u_2 = 2 a_1 \frac{(M^2 - 1)}{(\gamma + 1) M}, \quad (49)$$

$$F_{piso}(p) = \sqrt{\frac{(p - p_1)(\rho(p) - \rho(p_1))}{\rho(p)\rho(p_1)}} + \int_{p_1}^p d\tilde{p} F_{ndcp}(\tilde{p}) - \int_{p_1}^{p_4} d\tilde{p} F_{ndcp}(\tilde{p}), \quad (50)$$

$$F_{mriso}(p) = p - p_2 - \left(1 + \frac{\rho(p_2)}{\rho(p) - \rho(p_2)}\right) \rho(p_2) u_2^2, \quad (51)$$

$$u_2 = \left(1 - \frac{\rho(p_1)}{\rho(p_2)}\right) U_i, \quad (52)$$

$$U_i = \sqrt{\frac{\rho(p_2)(p_2 - p_1)}{\rho(p_1)(\rho(p_2) - \rho(p_1))}}. \quad (53)$$

There are two Matlab-programs that create animations from given input data: `isot1bfilm.m` deals with data from magmatic fluids generated in the `enc/eno/encl/enolisol` programs; and `isoairbfilm` deals with data from simulations with air only and magma and air together generated by programs `enc/encl/eno/enolair1` and `enc/enomva1`, `encmax1`, `encmvs1`, `enctis1.c` and `encmvt1`, respectively.

3.8.1 Input file

The structure of the input file `enciso1.in` or `enoiso1.in` is partly fixed and partly flexible. After the run number `runno` is specified, see the input files below, a coefficient `Nrcoef` specifies how many coefficients will be defined. The first eight 0–7 are necessary to run a basic simulation, but the remaining coefficients, if any, can be used to specify application specific coefficients. Program `encmvs1.c` has 29 of these coefficients, while program `encmvt1.c` has 37 coefficients. It is straightforward to add more, but coefficients can only be taken away once they are cleared in the program.

For example, the input file for the program `enciso1.c` with executable `enciso1` for the stationary shock in magma presented in section 4.1 is:

Variable	Value	Comment
Nor:	3	only for ENO: spatial order of scheme
Nrk:	3	only for ENO: time order of scheme
Nt:	500	number of time steps
Nmeas:	50	cycle for measurements
Nppts:	100	number of grid points
Nsys:	3	number of variables
dt:	0.05	time step
cfl:	0.1	<i>cfl</i> condition
Lx:	0.5	scaled length of domain
c0:	1.0	(not in use)
icno:	0	initial condition choice number
runno:	000	three-digit run number to label output files
Nrcoef:	7	number of coefficients to read
n0:	0.02	volatile content of magma
Rv:	462	gas constant of magma
T:	1000	isothermal temperature magma
sigma:	2500	melt density
s:	3.e06	Henry's constant
Ra:	287.04	gas constant of air
dampn1:	0.e0	damping (not in use)
rho0:	100.0	scaling factor for density

* stationary shock simulation enciso1.c 000 *

Together with the time step $\Delta t = cfl \Delta x$, the number of time steps determines the final simulation time in case a variable time step is used. Codes `encmvs1.c` & `encmvt1.c` uses a variable time step based on the maximum speed present.

The input file for the program `encmvt1.c` with executable `encmvt1` for canister movement in a dike-drift system with magma and air [§5.3 in Bokhove, 1999] differs from the previous example:

Variable	Value	Comment
⋮	⋮	
Lx:	20	scaled length domain
N0:	100	
⋮	⋮	
Nrcoef:	36	number of coefficients to read
⋮	⋮	
7-rho0:	100.0	scaling factor for density ($kg\ m^{-3}$)
8-RADIUSARC:	1.0e0	scaled radius of arc/void
9-ATUNNEL:	0.7854e0	scaled area of drift
10-ADIKE:	3.2e0	scaled area of dike
11-Nudamp:	0.e0	damping factor (void)
12-1/FI:	0.0054e0	inverse Froude number
13-pmagma:	2.e07	initial dike tip pressure (Pa)
14-Ldike:	215.e0	scaled length dike
15-ptun:	1.e05	pressure in drift (Pa)
16-gamma:	1.4	γ
17-mu0/L:	2.e0	μ_o/L
18-lambda:	0.0025	turbulent friction factor
19-alpha:	0.7e0	critical vesicularity (%)
20-diametertunnel:	1.0e0	scaled diameter d of drift
21-Amin:	0.2e0	not in use
22-widthdike:	0.2e0	scaled width w dike
23-Ltunnel:	40.e0	scaled L_{drift}
24-alphae:	0.25e0	geometric factor for coarse grid in dike (5.1)
25-dlr:	40.e0	scaled factor ΔL_r for coarse grid
26-Lr-dlr:	40.e0	scaled factor L_r for coarse grid
27-pshock:	3.e05	not in use
28-noparticles:	33	number of released tracer particles
29-timepastart:	9.6154e0	scaled time particles start advection
30-ramptime0:	19.2308e0	scaled ramp time for time-dependent nozzle
31-Lcan/L:	1.4e0	scaled, L_{can}/L
32-rcan/L:	0.2e0	scaled, r_{can}/L
33-rhocan/rho0:	110e0	scaled, ρ_{can}/ρ_0
34-staticfr:	1.0	static friction λ_s
35-dynamicfr:	0.5	dynamics friction λ_d
36-CD:	1.0	drag coefficient C_D

* stationary shock simulation enciso1.c 085 *

Note that $c0$ has been changed to $N0$. $N0$ is the amount of time steps between special measurements, just like $Nmeas$; for variable timestep it defines the time between special measurements.

The input files `enciso1.in` and `enoiso1.in` contain all the runs for testing the various programs in this report as well as most of the runs in Bokhove [1999]. Only the top lines of these input files will be read. With a simple cut and paste action, we can choose the input file we like or modify an existing one. Appendices A and B contain instructions on installing and using the programs.

3.8.2 Core program

The core program is the most important program. It contains the coded version of the numerical scheme, while all the other programs take care of initial IO or contain support functions. The boundary conditions are `#` defined in the top lines of the program. One has to set these boundary conditions in the program and then compile.

Several of the most important functions of the core program are:

- Function `TimeStep()` is the main routine with the time and Runge-Kutta loop.
- Function `Initial_Condition()` defines the initial conditions. The domain length and grid points are calculated, and flexible array and matrix lengths are defined. Different choices of initial condition are specified by the input parameter `icno`. One can either use the initial conditions available, just browse through and read the comments in the program, or create one.
- Function `Measure(nnt, time, Ni)` measures profiles of the key variables after `Nmeas` time steps or a time equivalent—in case variable time steps are used.
- Boundary conditions are dealt with in function `BounCond(nrk)`.
- Functions `Fluxiso(nrk)` and `Fluxeul(nrk)` calculate the conservative terms in the equations. In the older programs for magma simulations this is sometimes done within the time and Runge-Kutta loop.
- Function `Runge-Kutta(nk)` takes care of time stepping and of canister movements in program `encmvt1.c`.
- All terms other than fluxes are considered in function `Forcing(nrk, Nz)`.
- Functions `Uinterpol(nrk)`, `Updateface(0)`, and `Diagnos(nnt, time)` only appear in magma-air programs. Function `Uinterpol(nrk)` calculates the velocity at the interface using ENO interpolations; `Updateface(0)` updates the interface, magma, and air indices after each time step; and function `Diagnos(nnt, time)` takes care of specialized measurements at (shorter) cycles after `N0` time steps (or a time interval equivalent).
- Particle advection in `encmvt1.c` is done in function `Particles_Forward(nrk)`.

A flow chart of the algorithm is given in Fig. 3.3.

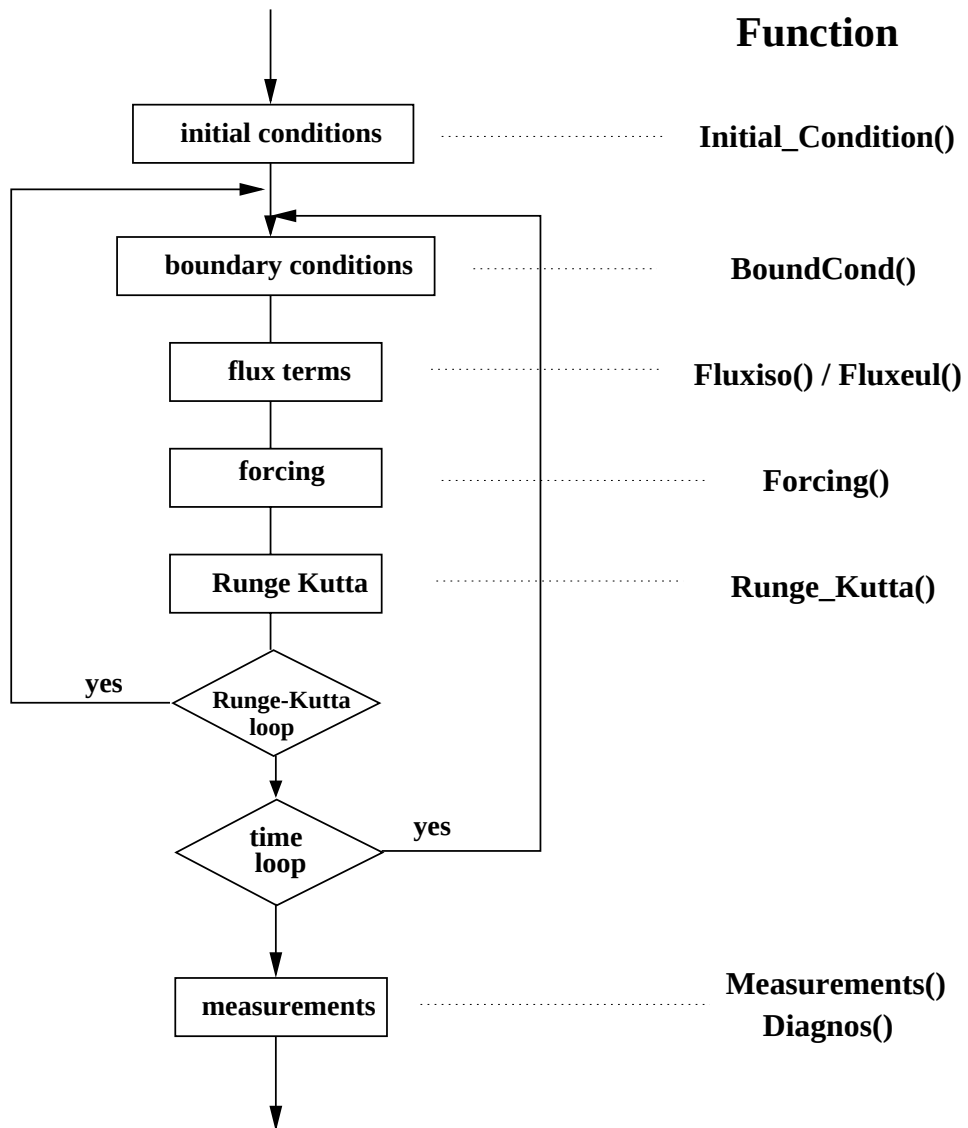


Figure 3.3: Flow chart and the associated functions in core function `TimeStep()`.

3.9 Structure of 2D programs

A series of test programs and final target programs have been developed. Table 3.4 summarizes the developed C-programs and the accompanying Matlab plot programs are found in Table 3.5. Each code consists of several programs which are linked together and compiled using a Makefile, as can be seen from the following line from the Makefile: `ENCMAC2 = encmac2.o enciso2cpu.o enciso2in.o enciso2inpu.o enoiso2ext.o`. So the program `encmac2.c` consists of the header file `iso2.h` and the following five C-files:

- `enciso2cpu.c` is the central core program with three functions which refer to other files: `Enoinput()`, `lnicon()` and `TimeStep()`,
- `enciso2inpu.c` contains the function `Enoinput()` which reads in the input file `enciso2.in`,
- `enciso2in.c` contains function `lnicon()` which opens the output files `e2cqXXX`, `e2ceXXX`, `e2ctXXX`, `e2ciXXX`, `e2cdXXX`, `e2cuXXX` with the run number `XXX` supplied in the input file,
- `encmac2.c` contains the numerical algorithm in function `TimeStep()` —the name of the executable refers to this most important central function, and finally
- `enoiso2ext.c` contains various external functions (from Numerical Recipes for C) used in the previous files.

There is also a header file, included in each program, `iso2.h`, which includes several global variables. The flow chart in Fig. 3.3 is also applicable to the two-dimensional LLF2 and ENO programs, because the two-dimensional versions are line-by-line extensions of their one-dimensional counterparts.

The output files have the following purpose:

- `e2cuXXX` contains information about the amount of data stored in the bulk data file `e2cqXXX` and in the later simulations it also contains the time before and after the time loop in the main function `Timestep()`,
- `e2cqXXX` stores the data in bulk,
- `e2ceXXX` stores the potential, kinetic and total energy at specified times,
- `e2ctXXX` may store the cross-sectional area but is not in use at the moment,
- `e2ciXXX` may store additional (interface) data but is not in use at the moment.
- `e2cdXXX` may store additional data, but is not in use at the moment.

3.10 Portability

The developed C-programs compile with minor modifications with the `cc` or `gcc` compilers on an HP9000, Linux PC's and Suns. The definition of character variables and manipulation with strings has so far been platform dependent. These string operations occur in the files `enciso1in.c`, `enoiso1in.c`, `enciso2in.c` or `enoiso2in.c` which belong each to different programs. The purpose of these character definitions is to define output files, such as `e1cqXXX`, with a three digit extension `XXX` which is specified in the input file. For each computer platform the character definitions are slightly different, and the before mentioned files and the header files `iso1.h`, `iso2.h` need to be modified appropriately and recompilation is required. Working versions for various systems are found in these files as comments.

ENC	ENO	Characteristics	Exact solutions
encmac2.c	-	magma, Eulerian, x-grid & stretched grid	stationary/moving/reflected shock, shock tube

Table 3.4: The various available 2D programs and their characteristics. ENO is Essentially Non-Oscillatory [Shu and Osher, 1988, 1989; Shu, 1997]. ENC is second-order Local Lax-Friedrichs [Liu and Osher, 1998].

Plot program	Fluid	C program	Comments
magtws.m	magma	encmac2.c	1 & 2 rectangular blocks

Table 3.5: Overview of Matlab plotting programs for two-dimensional simulations.

Chapter 4

Numerical tests

To develop and test the numerical codes, the basaltic fluid flows (7) and air flows (16) have been considered in separation in one spatial dimension before the interface dynamics was added. In all these three cases, for the two separate fluids and the two fluids together, specific simulations in one-dimension are compared with exact (implicit) solutions of stationary shocks, moving and reflected shocks, and shock tube solutions. A summary of the one- and two-dimensional numerical tests in this section is found in Tables 4.1 and 4.2. The two-dimensional magma code was tested by rerunning some of the one-dimensional tests or modifications thereof.

4.1 Dynamics of magma 1D

4.1.1 Stationary shock

The stationary shock solution of (10) (with scale factor $h_1 = 1, \xi_1 = x, g = \nu_f = 0$) follows directly from momentum density and mass conservation across the discontinuity:

$$[\rho v] = 0, \quad [\rho v^2 + p] = 0, \quad (1)$$

where $v \equiv u - U$ is the speed relative to the moving shock with speed U , and where $[m]$ denotes the jump in quantity m across the discontinuity. If the pressures left and right of a stationary shock are given by p_2 and p_1 with left state 2 and right state 1, then the velocities are $u_2 = \rho_1 u_1 / \rho_2$ and

$$u_1 = \sqrt{\frac{\rho_2}{\rho_1} \frac{(p_2 - p_1)}{(\rho_2 - \rho_1)}},$$

respectively. We start with values $p_2 = 5 * 10^6 Pa$ and $p_1 = 10^5 Pa$ and a discontinuity at $x = 1.25 m$.

The stack plots in Fig. 4.2 and hereafter should be interpreted as follows. The initial profile of density or velocity is plotted versus space. Subsequent profiles of density and velocity, arising after subsequent time-intervals, are plotted in the same graph but the value on the vertical axis is shifted upward each time by a specified amount; in other words it is “stacked”. Stack plots thus become density or velocity plots versus space and time. Sometimes the vertical axis is denoted as time axis but I prefer to display the amplitude of the variable.

The LLF2 simulation is seen, in the stack plots of Fig. 4.2a, to shed an acoustic wave before it adjusts to a stationary numerical shock solution, while ENO appears to be perfect (see Fig. 4.2b) because the steepness

Program	Description & icno	Run #	CFL	Δx	N_{pts}	T_{end}	T_{sim}
encisol.c	stationary shock 0	e1c*000	0.1	0.05	100	0.25	<i>s</i>
encisol.c	shock tube 1	e1c*001	0.1	0.03	400	3.3	<i>s</i>
encisol.c	shock tube & wall 2	e1c*002	0.1	0.03	400	6	4 : 44
encisol.c	moving & reflected shock 3	e1c*003	0.1	0.5/400	400	0.5	<i>s</i>
enoisol.c	stationary shock 0	e1s*000	0.1	0.05	100	0.25	<i>s</i>
enoisol.c	shock tube 1	e1s*001	0.1	0.03	400	3.3	2 : 37
enoisol.c	shock tube & wall 2	e1s*002	0.1	0.03	400	6	4 : 44
enoisol.c	moving & reflected shock 3	e1s*003	0.1	0.5/400	400	0.5	9 : 17
encairl.c	stationary shock 0	e1c*010	0.1	0.05	100	0.2	<i>s</i>
encairl.c	shock tube 1	e1c*011	0.1	0.015	800	1.2	<i>s</i>
encairl.c	moving & reflected shock 2	e1c*012	0.1	0.5/400	400	0.1875	<i>s</i>
enoairl.c	stationary shock 0	e1s*010	0.1	0.05	100	0.2	<i>s</i>
enoairl.c	shock tube 1	e1s*011	0.1	0.015	800	1.2	4 : 15
enoairl.c	moving & reflected shock 2	e1s*012	0.1	0.5/400	400	0.1875	3 : 57
encmval.c	shock tube 1	e1c*020	0.05	0.03	400	1.68	-
encmval.c	shock tube 1	e1c*021	0.05	0.012	1000	1.68	5 : 58
enomval.c	shock tube 1	e1s*020	0.05	0.03	400	1.68	7 : 23
enomval.c	shock tube 1	e1s*021	0.05	0.012	1000	1.68	50 : 28
encmval.c	resonating shocks 3	e1c*030	0.05	0.012	1000	15	41 : 44
encmvs1.c	magma-repository 10	e1c*056	0.05	-	3000	21.91	113 : 7.8
encmvs1.c	magma-repository 10	e1c*101	0.05	-	3000	52.57	2.734 <i>s</i>
encmvs1.c	magma-repository 10	e1c*060	0.05	-	3000	219.03	842 : 71.64
encmvt1.c	magma-repository 13 (canister)	e1c*091	0.05	-	6000	39.43	208 : 25.6
encmvt1.c	magma-repository 11 dike-drift $A(\xi_1, t)$ 10 <i>s</i>	e1c*097	0.05	-	3000	219.03	-
enctis1.c	magma-repository 14 magma only	e1cx161 utmtu2	0.05	-	3000	99.93	-
encmvs1.c	magma-repository 15	e1cx165	0.05	-	3000	175.22	-
encmvs1.c	magma-repository 15	e1cx166	0.05	-	6000	175.22	-
encmvs1.c	magma-repository 15	e1cx167	0.05	-	12000	175.22	-

Table 4.1: Summary of numerical tests. Most of the basaltic runs, air runs, and melt-volatile-air runs have been scaled with $P_0 = 924000 \text{ Pa}$, $L = 5 \text{ m}$, $\rho_0 = 100 \text{ kg m}^{-3}$, $T_0 = L/U_0 = 0.052 \text{ s}$. T_{end} refers to dimensionless time units. Variable icno refers to the type of initial condition. Computational times concern computation on a Sun Ultrasparc 2200 with two CPU's running at 200 MHz for all but the `encmvs1` and `encmvt1` programs, which were run on a PC with Pentium-III processor. Simulation 165167 were performed on a 650 MHz Pc under Linux. Files `e1c*XXX` refer to LLF2 output and files `e1s*XXX` refer to ENO output.

Program	Block	Description & icno	Run #	CFL	Time steps	$N_{pts} \times$ N_{pts}	T_{end} & T_{sim}	Stretch
encmac2.c	1	stationary shock 0	e2c*200	0.1	500	100×20	0.25	no
encmac2.c	1	moving & reflected shock 1	e2c*201	0.4	1000	400×10	0.5	no
encmac2.c	1	shock tube 2	e2c*202	0.1	2000	400×10	6.0	no
encmac2.c	2	shock tube 4	e2c*203	0.1	200	200×20 50×200	0.5	no
encmac2.c	2	shock tube 5	e2c*204	0.1	1500	200×20 50×200	6.0	no
encmac2.c	2	moving shock 3	e2c*205	0.1	1500	100×20 50×50	0.5	no
encmac2.c	2	shock tube 6	e2c*212	0.1	320	40×40 -	3.2	yes
encmac2.c	2	shock tube 6	e2c*213	0.1	10^4	40×40 -	10.0	yes

Table 4.2: Summary of numerical tests in two dimensions. Simulations occurred on a PC with Pentium-III 650 *MHz* processor. Variable icno refers to the type of initial condition and has to be specified in the input file. Files e1c*XXX refer to LLF2 output and files e1s*XXX refer to ENO output.

of the wave is determined by one grid spacing only and because no adjustment occurs. For shocks moving into still fluid ahead of the shock both ENO and CENO schemes show an acoustic wave adjustment. The acoustic adjustments leads to a decrease and subsequently an increase in energy for the LLF2 simulation, while energy is constant for ENO (Fig. 4.1).

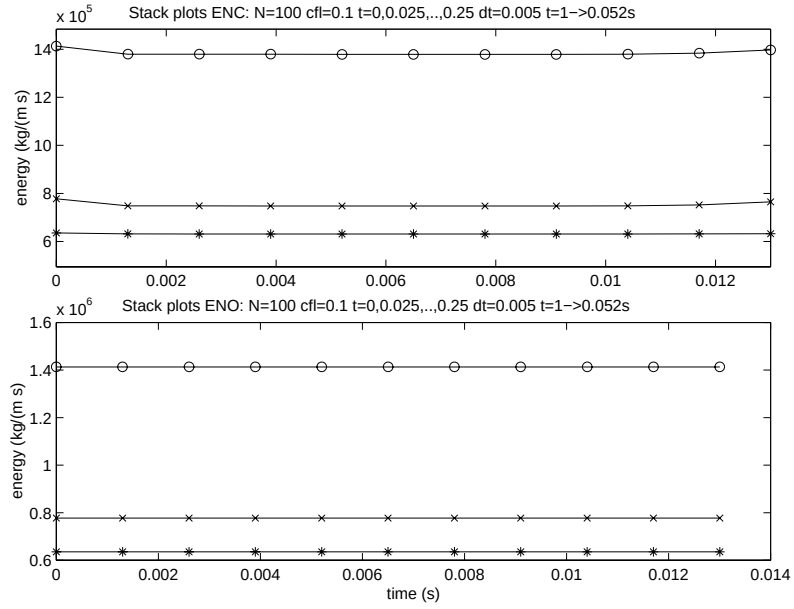


Figure 4.1: Kinetic (*), potential (x), and total (o) energy in a simulation of a stationary shock in melt-volatile fluid: (a) LLF2 code, and (b) ENO code.

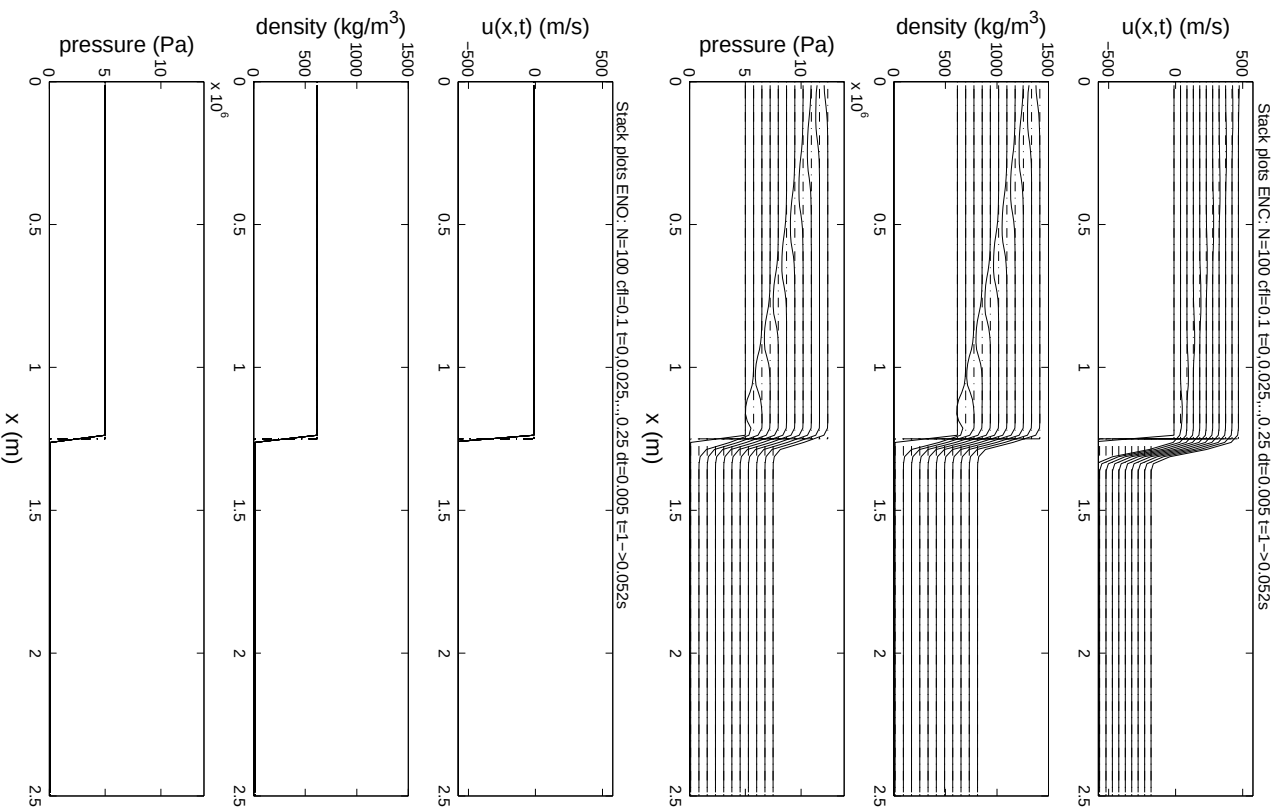


Figure 4.2: Velocity, density and pressure profiles against space are shown for a stationary shock of a melt-volatile fluid: (a) LLF2 code, stack plots with offset $0.8 P_0 = 7.4 \cdot 10^5 Pa$ in p , $0.8 \rho_0 = 80 kg m^{-3}$ in ρ and $0.5 U = 48.06 m/s$ in u , length scale $L = 5 m$ and time spacing is $0.025 s$, and (b) ENO code. Dashed-dotted lines indicate the exact and solid lines the numerical solution.

4.1.2 Moving and reflected shock

For a shock moving into quiescent fluid on the right in state 1, shock relations (1) give

$$v_1 = U_i = \sqrt{\frac{\rho_2}{\rho_1} \frac{p_2 - p_1}{\rho_2 - \rho_1}}, \quad u_2 = (1 - \rho_1/\rho_2) U_i \quad (2)$$

for $p_2 > p_1$. After reflection, the region 3 between the reflected shock and the walls is at rest. Evaluation of the shock relations (1) with left state 2 and right state 3 gives the implicit relation

$$p_3 - p_2 - \frac{(p_2 - p_1)(\rho_2 - \rho_1)}{\rho_1} \left(1 + \frac{\rho_2}{(\rho(p_3) - \rho_2)} \right), \quad (3)$$

in terms of the initially known pressures p_1 and p_2 which determine —note that $\rho_{1,2} \equiv \rho(p_{1,2})$. Once p_3 is known, density ρ_3 follows as $\rho(p_3)$ and the reflected shock speed is negative, i.e.

$$U_r = -\frac{\rho_2 u_2}{(\rho_3 - \rho_2)}. \quad (4)$$

The strength of the shock reflection is given by the ratio of pressure jumps of reflected over incident shock $S_{rm} = (p_3 - p_2)/(p_2 - p_1)$ which is a function of p_1 and p_2 . S_{rm} is plotted versus p_1 and p_2 in Fig. 4.3 and depends on p_1 and p_2 in a nonlinear fashion. Moreover, the magnitude of the reflected shock is always higher, sometimes significantly so, than that of the incoming shock.

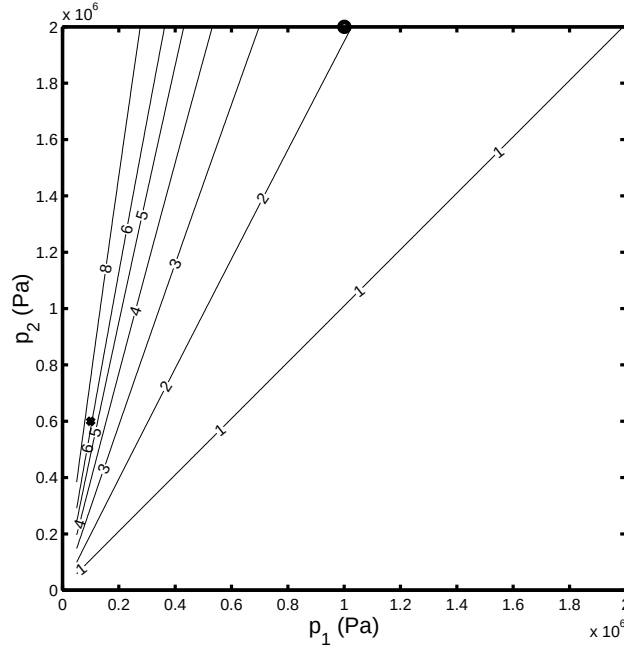


Figure 4.3: Contour plot of S_{rm} versus p_1, p_2 for $p_2 > p_1$. This figure is created with C-program `p123.c`, output file `p123.out` and Matlab program `p123con.m`.

The simulations shown in Fig. 4.4 and Fig. 4.5 compare well with the exact solutions. The ENO simulation captures the moving and reflected shock better than the LLF2 simulation.

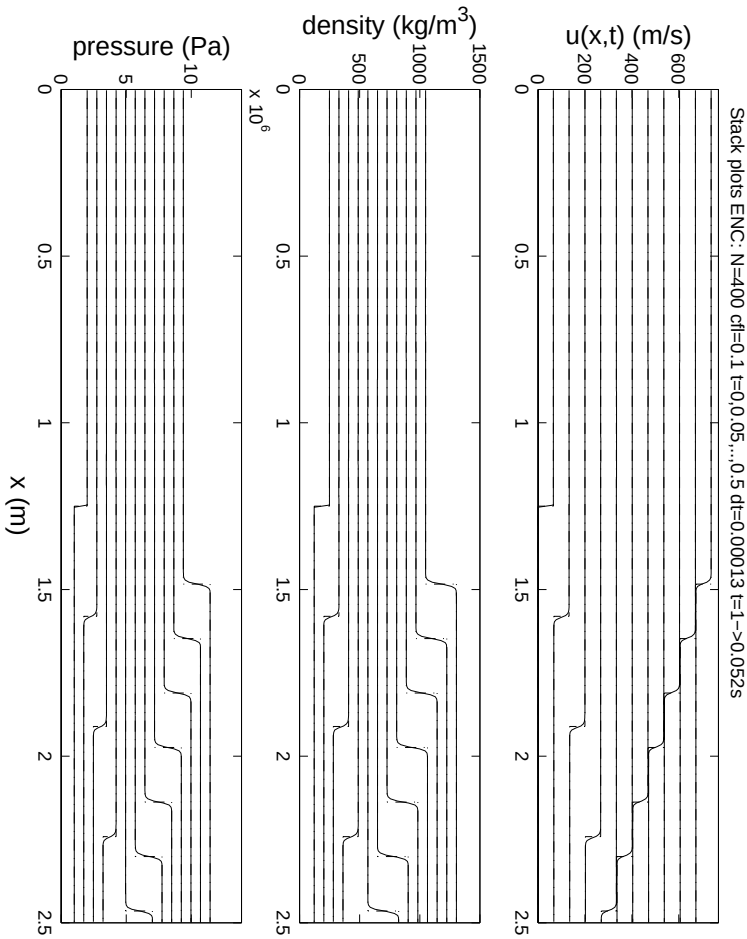


Figure 4.4: Velocity, density and pressure profiles are shown for a moving and reflected shock of a melt-volatile fluid: (a) LLF code, and (b) ENO code. Dashed-dotted lines indicate the exact and solid lines the numerical solution. (Stack plots with offset $0.8 P_0 = 7.4 \cdot 10^5 Pa$ in p , $0.8 \rho_0 = 80 kg m^{-3}$ in ρ and $0.7 U = 67.29 m/s$ in u . Length scale $L = 5 m$. Time spacing is $0.5 s$.)

4.1.3 Shock tube with and without tunnel end

The shock tube solution of (10) (with scale factor $h_1 = 1, \xi_1 = x, g = \nu_f = 0$ and) with pressure $5 * 10^6 Pa$ on the left and $10^5 Pa$ on the right is obtained by matching a shock to a backward propagating rarefaction wave (similar calculations are found in Whitham [1974]).

The initial condition for a shock tube consists of a quiescent state with pressure p_4 (on the left) and p_1 on the right ($p_4 > p_1$). The space-time diagram Fig. 4.6 explains the evolution: the left state 4, for $x < a_4 t$, is connected via a rarefaction, for $-a_4 t < x < (u_3 - a_3) t$ to a constant region 3, for $(u_3 - a_3) t < x < U t$, behind the shock. After the shock $x > U t$ and the speed of sound is $a(p)$, so $a_3 = a(p_3)$. From the shock relations (1) we find

$$u_3 = \sqrt{\frac{(p_3 - p_1)(\rho_3 - \rho_1)}{\rho_1 \rho_3}}, \quad (5)$$

while the rarefaction wave is a simple wave. The C_+ -characteristic is constant on $dx/dt = u + a$: $u + c(p) = c(p_4)$ and

$$c(p) = \int^p d\tilde{p} \frac{1}{\rho(\tilde{p})} \left(\frac{\partial \rho(\tilde{p})}{\partial \tilde{p}} \right)^{1/2}.$$

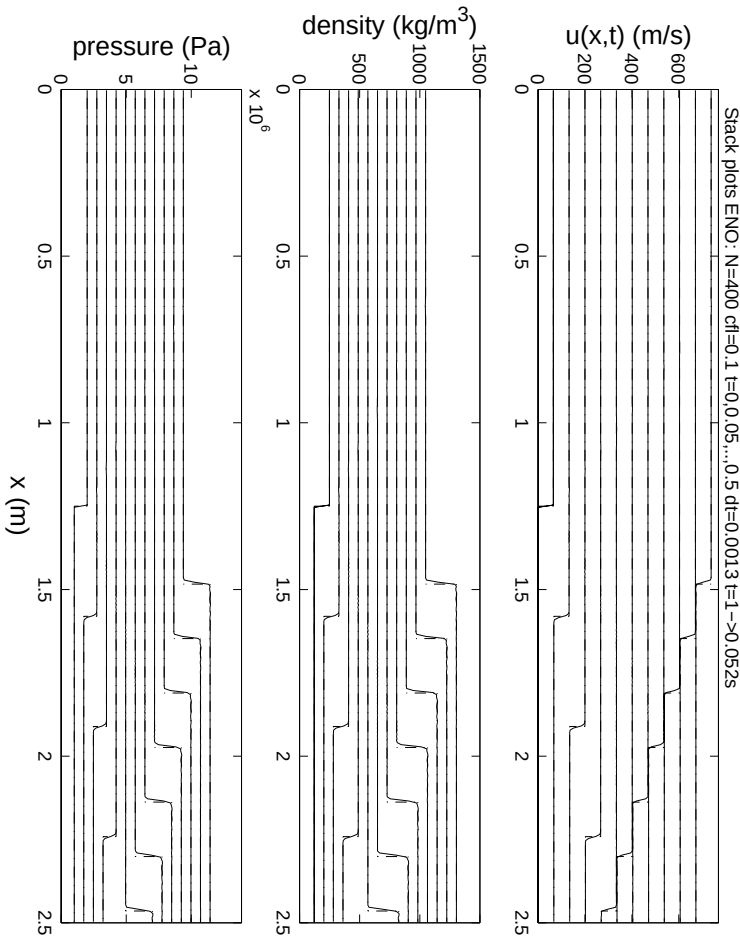


Figure 4.4: Continued

Matching this constant characteristic to the constant region 3 gives an implicit relation:

$$u_3 + c(p_3) - c(p_4) = \sqrt{\frac{(p_3 - p_1)(p_3 - p_1)}{\rho_1 p_3}} + c(p_3) - c(p_4) = 0 \quad (6)$$

from which p_3 can be determined, and hence u_3 follows from (5). For the simple expansion wave we have $dx/dt = x/t = u - a(p)$ and with $u + c = c(p_4)$ we find $x = (c(p_4) - c(p) - a(p))t$ and $u = c(p_4) - c(p)$ for $p_3 < p < p_4$.

In the following simulations $p_1 = 10^5 Pa$ and $p_4 = 5 * 10^6 Pa$. The evolution of velocity, density and pressure are given in Fig. 4.7 for the exact solution (dashed-dotted lines) and the numerical solution (solid lines). The rarefaction wave is traveling to the left, moving away from the discontinuous initial condition. This smooth rarefaction wave connects to a forward propagating shock wave, i.e. to the right, which is apparent in the numerical solution as a steep pressure change of about $5 * 10^5 Pa$. The steepness of the shock wave is limited by the grid size and the numerical scheme. In particular, the ENO simulation resolves the shock better than the LFF2 simulation which has smoothed the shock stronger. Energy is dissipated in the shock and more energy is lost after the shock leaves the domain around $t = 0.14 s$ (see Fig. 4.8).

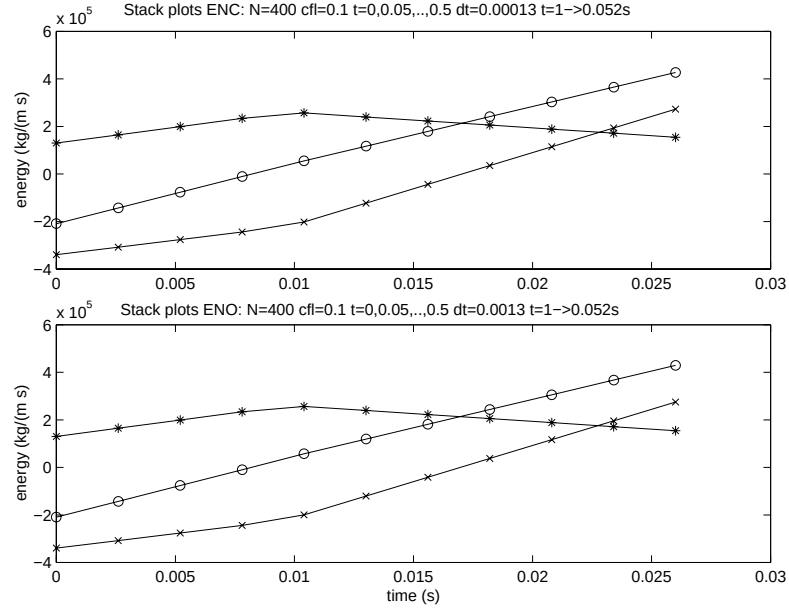


Figure 4.5: Kinetic (*), potential (x), and total (o) energy in simulation of moving and reflected shock in a melt-volatile fluid: (a) LLF2 code and (b) ENO code.

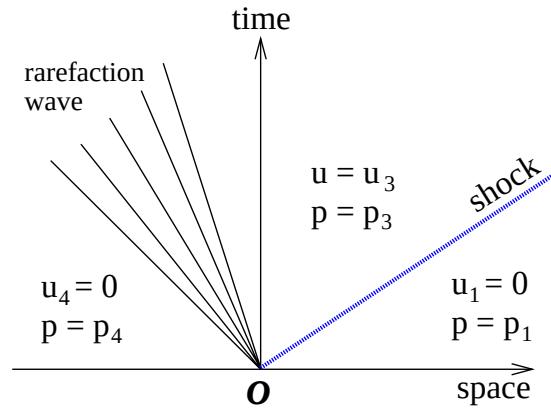


Figure 4.6: Space-time diagram of shock tube flow evolution for magma.

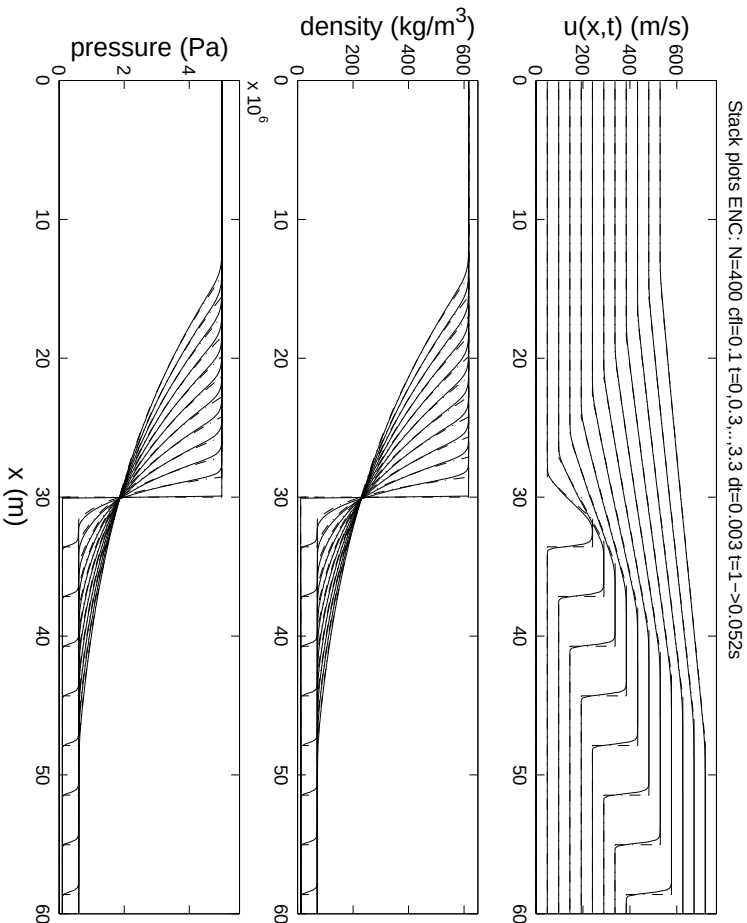


Figure 4.7: Shock tube tests for melt-volatile flow with open boundary conditions. Velocity, density, and pressure stack plots are shown for (a) a LLF2 code and (b) an ENO code. Dashed-dotted lines indicate exact and solid lines numerical solutions. Stack plots have offset $0.5U = 48.0625 \text{ m/s}$ in u . $L = 5 \text{ m}$. Time spacing 0.3.

When a wall is added at $x = 60 \text{ m}$ the incoming shock in the shock tube reflects and leads to a reflected shock which is much larger (Fig. 4.9) than one might expect from the moving and reflected shock calculations in Fig. 4.4. The difference between the two shock reflection problems is that the rarefaction wave feeds into the reflected shock and, more importantly, that the pressures before (p_2) and after (p_1) the incoming shock are an order of magnitude lower. According to (3) or the lower left cross in Fig. 4.3 the reflected shock has a pressure $p_3 = 3.74 * 10^6 \text{ Pa}$ for $p_1 \approx 10^5 \text{ Pa}$ and $p_2 \approx 6 * 10^5 \text{ Pa}$, giving $S_{rm} \approx 6$. This is in agreement with the simulations and should be contrasted with the calculations in Fig. 4.4 where for $p_1 = 10^6 \text{ Pa}$ and $p_2 = 2 * 10^6 \text{ Pa}$ the reflected shock has $p_3 = 4.056 * 10^6 \text{ Pa}$, giving a shock reflection amplitude $S_{rm} \approx 2$ that is in agreement with Fig. 4.3—marked by the upper solid circle. The energy dissipation in the reflected shock is much stronger than the dissipation in the incoming one, as is evident from the change in slope at $t \approx 0.14 \text{ s}$ in Fig. 4.10. The ENO code gives sharper shock resolution but is in the reflection process not devoid of small-scale oscillations. The large shock amplification in reflection is expected to be generic and largely independent of the detailed equation of state.

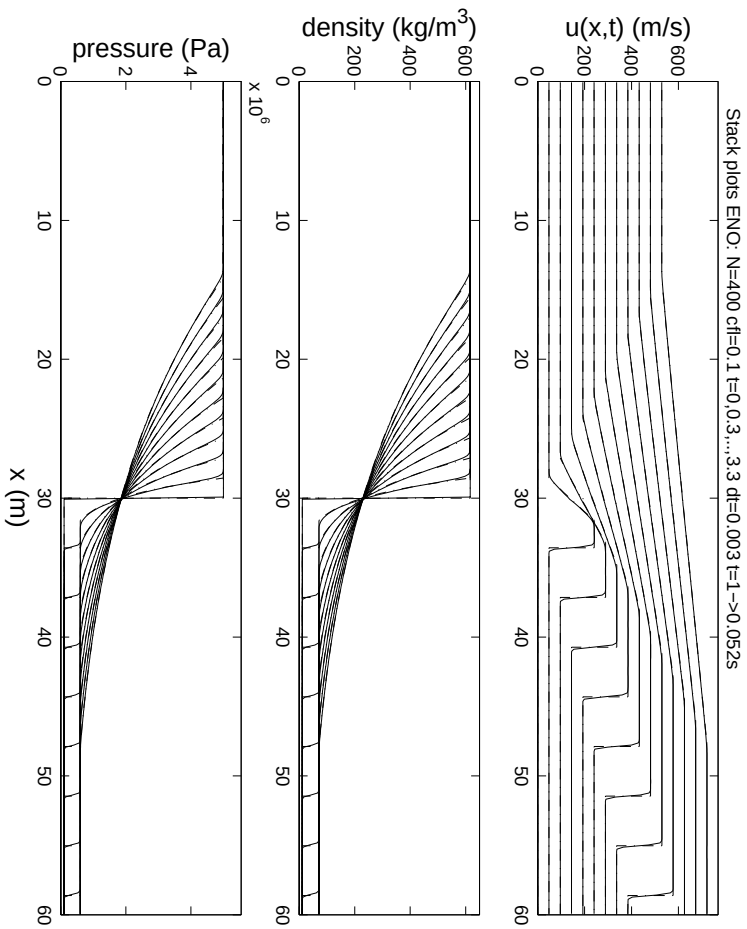


Figure 4.7: Continued.

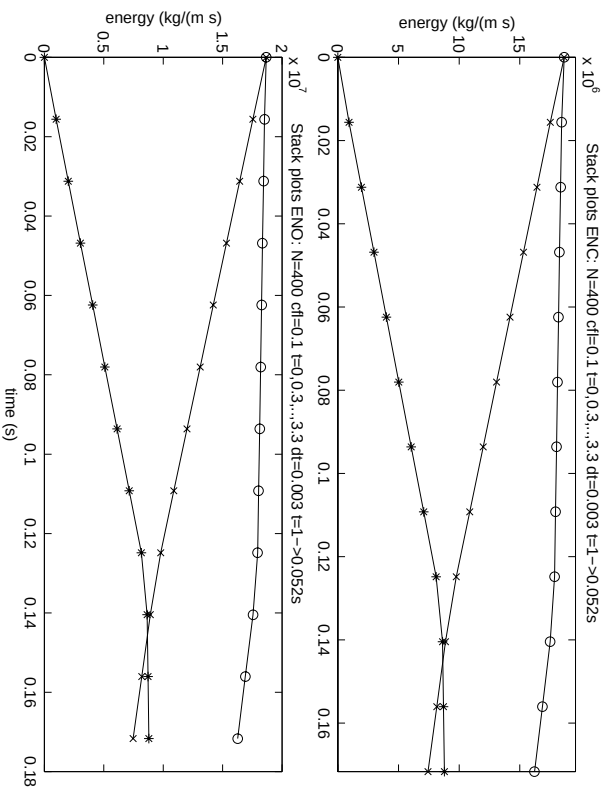


Figure 4.8: Shock tube tests for melt-volatile flow with open boundary conditions. Kinetic (*), potential (x) and total energy (o) are shown for a) a LLF2 code and (b) an ENO code.

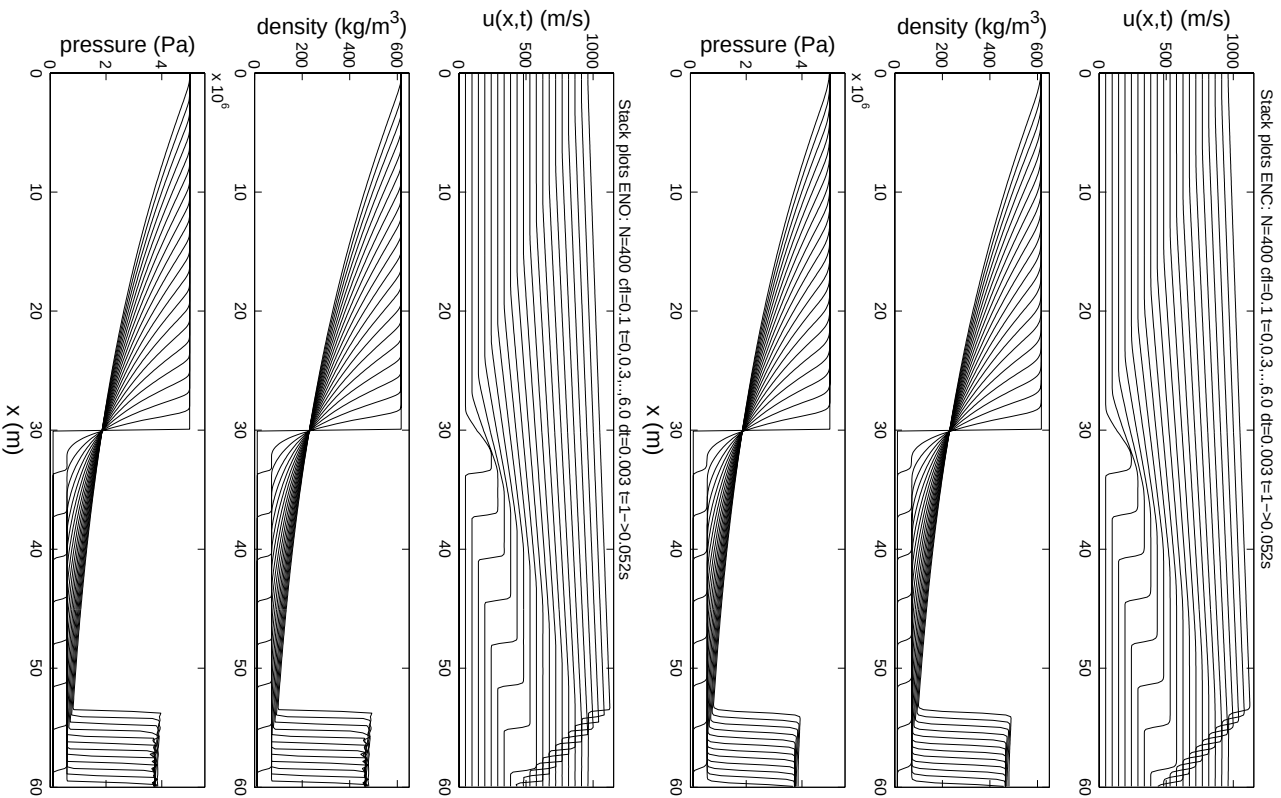


Figure 4.9: Shock tube tests for melt-volatile flow with a right wall. Velocity, density, and pressure stack plots are shown for (a) a LLF2 code and (b) an ENO code. Stack plots have offset $0.5U = 48.0625 \text{ m/s}$ in u . $L = 5 \text{ m}$. Time spacing 0.3.

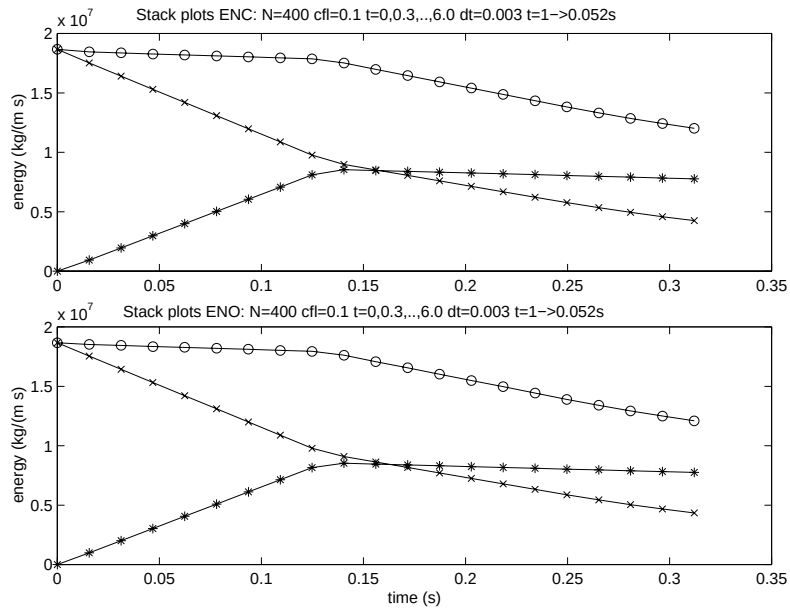


Figure 4.10: Shock tube tests for melt-volatile flow with a right wall. Kinetic (*), potential (x) and total energy (o) are shown for a) a LLF2 code and (b) an ENO code.

4.2 Dynamics in air 1D

The shock relations for air are [Whitham, 1974]

$$[\rho_a v_a] = 0, \quad [p_a + \rho_a v_a^2] = 0, \quad [\rho_a v_a (h + \frac{1}{2} v^2)] = 0 \quad (7)$$

with $h = e + p_a/\rho_a = (\gamma/(\gamma-1))(p_a/\rho_a)$ and $v_a \equiv u_a - U$. We will drop the subscript “a” as long as no confusion arises.

4.2.1 Stationary shock

Given constant pressure, density and velocity in left state 2 and stationarity of the shock, the right state 1 is found from (7) as:

$$u_1 = \frac{2\gamma p_2 + (\gamma-1)\rho_2 u_2^2}{(\gamma+1)\rho_2 u_2}, \quad p_1 = p_2 + \rho_2 u_2^2 - \rho_2 u_2 u_1, \quad \rho_1 = \frac{\rho_2 u_2}{u_1}. \quad (8)$$

For an initial condition with $p_2 = 1.8 \cdot 10^5 \text{ Pa}$, $\rho_2 = p_2/(\rho_0 T_{ia})$, $T_{ia} = 293 \text{ K}$, $u_2 = -0.8 c_{air}$ and sound speed $c_{air} = \sqrt{\gamma p/\rho}$, the LLF2 and ENO simulations correspond well with the exact solution, as is evident from Fig. 4.11. The ENO simulation is again perfect, while the shock in the LLF2 simulation is less well resolved and the energy grows slightly (not visible in Fig. 4.12).

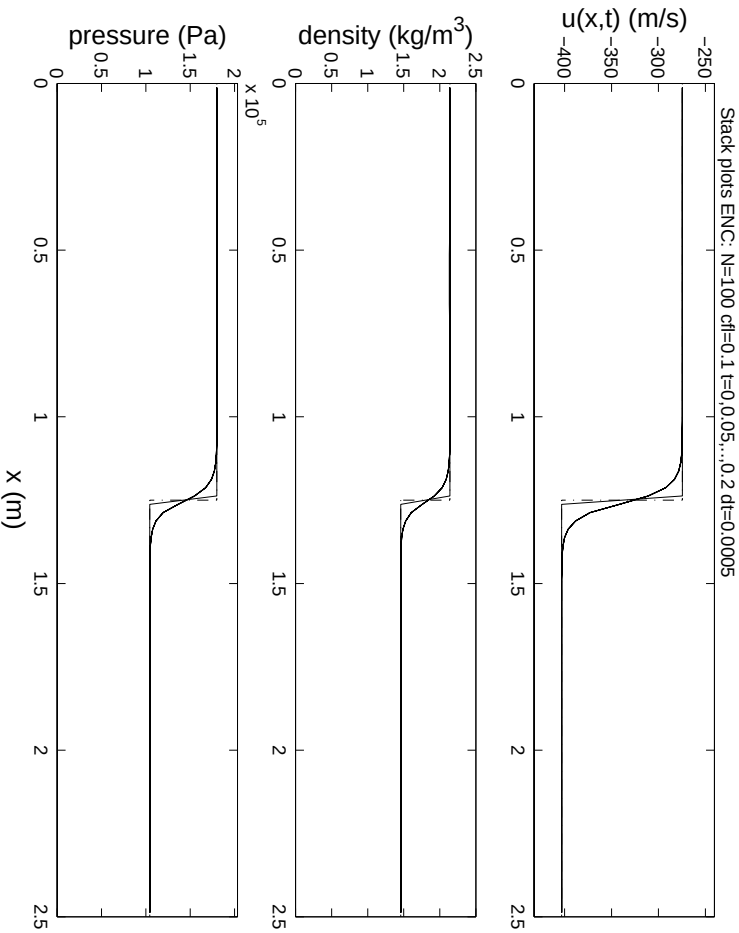


Figure 4.11: For a stationary shock in air, velocity, density, and pressure profiles are shown for (a) a LLF2 code and (b) an ENO code. Dashed-dotted lines represent exact and solid lines numerical solutions.

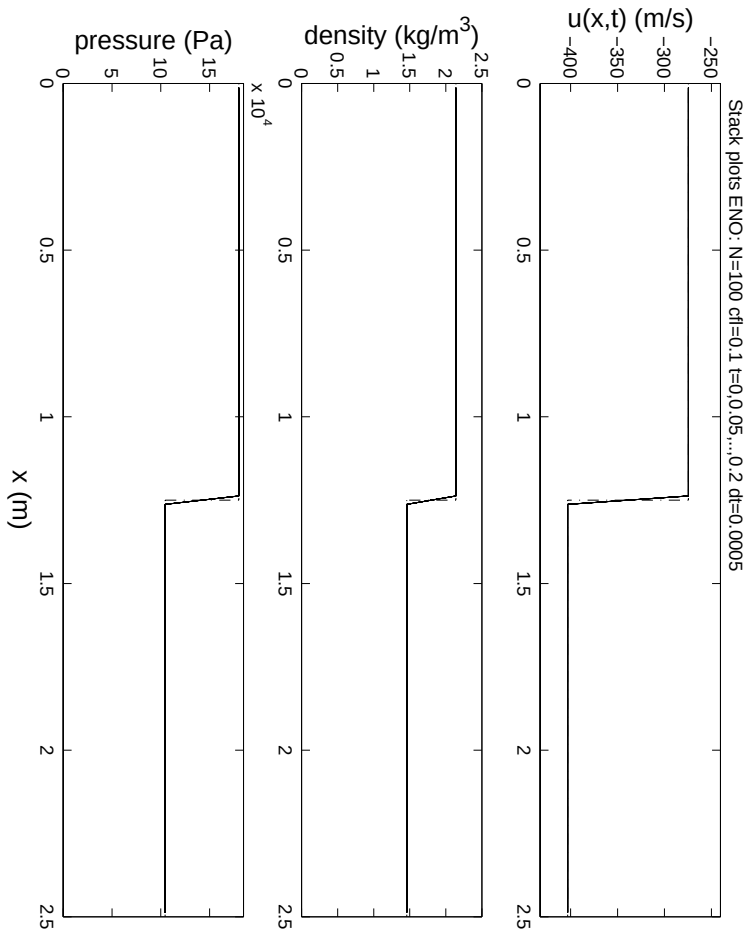


Figure 4.11: Continued.

4.2.2 Moving and reflected shock

Consider the space-time plot Fig. 4.13 for a moving and reflected shock in air. Given p_1, ρ_1 , and p_2 , several relations follow from the shock condition for the incoming shock [Whitham, 1974]:

$$\begin{aligned}
 z_i &\equiv \frac{(p_2 - p_1)}{p_1} = \frac{2\gamma(M_i^2 - 1)}{(\gamma + 1)}, \\
 M_i &\equiv \frac{U_i - u_1}{a_1} = \sqrt{1 + \frac{(\gamma + 1)z_i}{2\gamma}}, \\
 \frac{u_2 - u_1}{a_1} &= \frac{2(M_i^2 - 1)}{(\gamma + 1)M_i} = \frac{z_i}{\gamma \sqrt{(1 + (\gamma + 1)z_i/(2\gamma))}}, \\
 \frac{\rho_2}{\rho_1} &= \frac{(\gamma + 1)M_i^2}{2 + (\gamma - 1)M_i^2} = \frac{1 + (\gamma + 1)z_i/(2\gamma)}{1 + (\gamma - 1)z_i/(2\gamma)}, \\
 \frac{a_2}{a_1} &= \sqrt{\frac{(1 + z_i)(1 + (\gamma - 1)z_i/(2\gamma))}{1 + (\gamma - 1)z_i/(2\gamma)}}
 \end{aligned} \tag{9}$$

with $a_{1,2}$ the speed of sound in state 1, 2, respectively.

By a change of sign the relations above can be altered for the reflected shock. Since $u_1 = u_3 = 0$ the expressions for $u_2/a_1, -u_2/a_2$ and a_2/a_1 determine z_r and the values for M_r, ρ_3, p_3 follow:

$$z_r \equiv \frac{(p_3 - p_2)}{p_2} = \frac{z_i}{1 + (\gamma - 1)z_i/(2\gamma)},$$

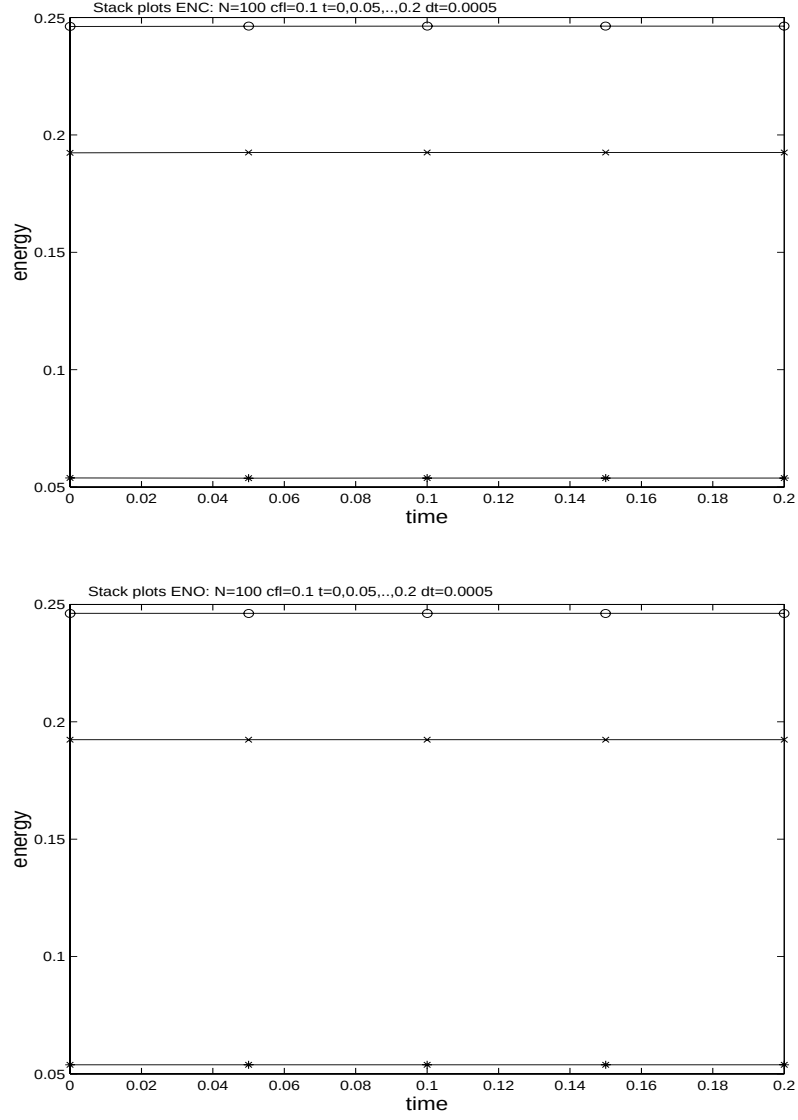


Figure 4.12: Kinetic (*), potential (x) and total energy (o) are shown for a) a LLF2 code and (b) an ENO code for a stationary shock in air.

$$\begin{aligned}
 M_r &\equiv \frac{U_r - u_2}{a_2} = -\sqrt{1 + \frac{(\gamma + 1) z_r}{2\gamma}}, \\
 \frac{u_3 - u_2}{a_2} &= \frac{2(M_r^2 - 1)}{(\gamma + 1)M_r} = \frac{-z_r}{\gamma \sqrt{(1 + (\gamma + 1) z_r / (2\gamma))}}, \\
 \frac{\rho_3}{\rho_2} &= \frac{(\gamma + 1) M_r^2}{2 + (\gamma - 1) M_r^2}.
 \end{aligned} \tag{10}$$

The comparison between analytical and numerical solutions is good (Fig. 4.14 and Fig. 4.15), although the ENO simulation resolves the shock better —as usual.

4.2.3 Shock tube

The shock tube problem in air adds a new feature to the solution. The space time sketch Fig. 4.16 for a shock tube in air requires an extra region 2 between an interface and the shock.

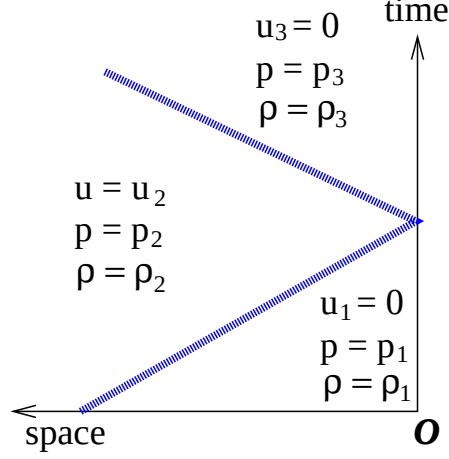


Figure 4.13: Space time sketch for shock reflection against a wall, at the right, in air.

The characteristics in air are [*e.g.* Whitham, 1974]

$$\frac{dp}{dt} \pm \rho a \frac{du}{dt} = 0 \quad (11)$$

on $C_{\pm}$: $dx/dt = u \pm a$. When we consider a shock tube problem the constant states at either side of the interface are isentropic, even although there is a jump in the entropy value across the interface. For a γ -law gas $p = \kappa(s) \rho^{\gamma}$ with a constant entropy s left of the interface and $a^2 = \partial p / \partial \rho = \gamma p / \rho$, the (isentropic) characteristics are

$$\frac{2a}{\gamma - 1} \pm u = \text{constant} \quad (12)$$

on $C_{\pm}$. Given u_1, p_1, ρ_1 , relations (9) yield u_2, p_2, ρ_2 in terms of M_i . Across the interface pressures and velocities are continuous. A simple wave connects the given state 4 to region 3 because entropy is conserved. We thus find that

$$\kappa = p_4 / (\rho_4)^{\gamma} = p_3 / (\rho_3)^{\gamma}. \quad (13)$$

The positive characteristic $2a / (\gamma - 1) + u = 2a_4 / (\gamma - 1)$ is constant in the simple-wave region and matching it to region 3 gives

$$2a_3 / (\gamma - 1) + u_3 = 2a_4 / (\gamma - 1). \quad (14)$$

Combining (13), (14) with $u_3 = u_2(M_i), p_3 = p_2(M_i)$ from (9), an implicit relation determining $M = M_i$ arises

$$\frac{\sqrt{\gamma} \kappa^{1/(2\gamma)} p_1^{(\gamma-1)/(2\gamma)}}{(\gamma-1)} \left(1 + \frac{2\gamma(M^2-1)}{(\gamma+1)} \right)^{(\gamma-1)/(2\gamma)} p_1^{(\gamma-1)/(2\gamma)} + \frac{a_1(M^2-1)}{(\gamma+1)M} - \frac{a_4}{(\gamma-1)} = 0. \quad (15)$$

Once M is known, the remaining variables can be determined. In the simple wave region $-a_4 t < x < (u_3 - a_3) t$, one has

$$u - a = \frac{x}{t}, \quad u + \frac{2a}{(\gamma-1)} = \frac{2a_4}{(\gamma-1)}. \quad (16)$$

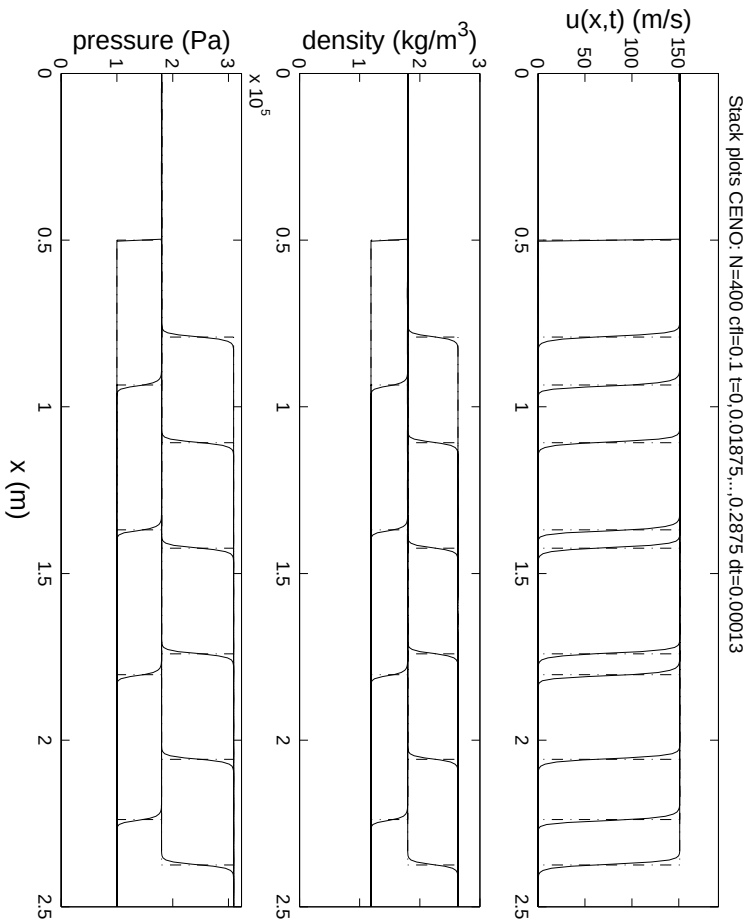


Figure 4.14: For a moving and reflected shock in air, velocity, density, and pressure profiles are shown for (a) a LLF2 code and (b) an ENO code.

After rearrangement of (16) and by using (13), we find

$$\begin{aligned}
 a &= \frac{2a_4}{(\gamma+1)} - \frac{(\gamma-1)}{(\gamma+1)} \frac{x}{t}, \\
 \rho &= \left(\frac{a^2}{\gamma\kappa} \right)^{1/(\gamma-1)}, \\
 p &= \kappa \left(\frac{1}{\gamma\kappa} \frac{2a_4}{(\gamma+1)} - \frac{(\gamma-1)}{(\gamma+1)} \frac{x}{t} \right)^{\gamma/(\gamma-1)}, \\
 u &= \frac{2}{(\gamma+1)} \left(\frac{x}{t} + a_4 \right). \tag{17}
 \end{aligned}$$

Exact and numerical solutions compare well (see Fig. 4.17 and Fig. 4.18); the ENO simulations clearly give sharper interface and shock resolution; the position of the interface is indicated by open circles in Fig. 4.17.

4.3 Dynamics of a melt-volatile fluid and air 1D

4.3.1 Shock tube

The same space time sketch Fig. 4.16 as in air applies with the modification that the fluid left of the interface, in regions 3 and 4, consists of a basaltic magma fluid, and the fluid to the right of the interface, in regions 1 and 2, is air. While the shock relations (9) in air again determine state 2, given state 1, the left states 3 and

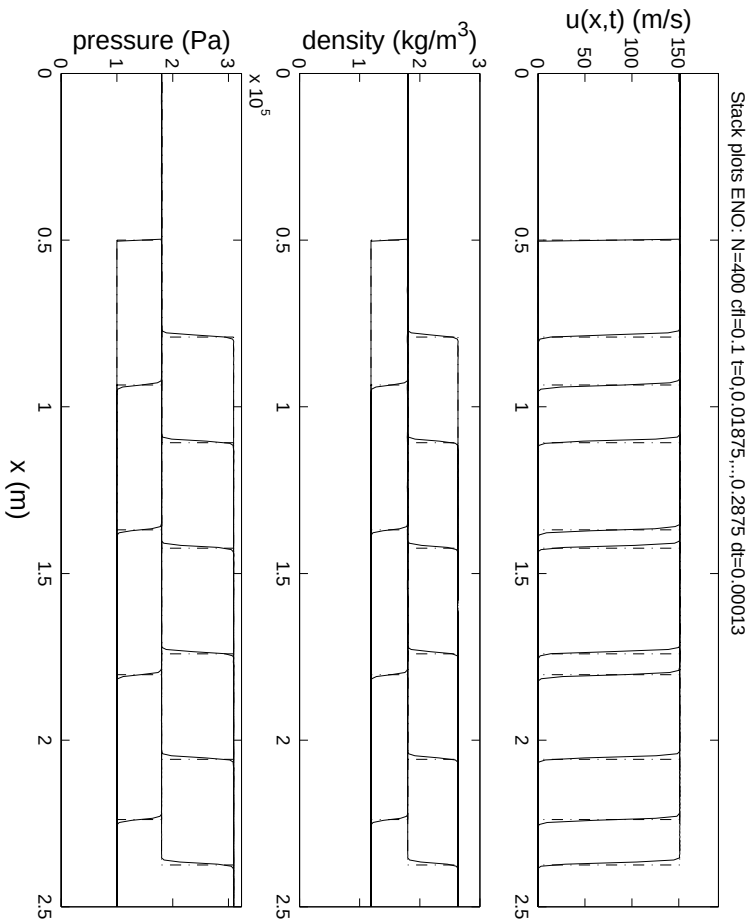


Figure 4.14: Continued.

4 follow directly from the shock-tube solution for melt only, and continuity of pressure and velocity across the interface. The implicit relations

$$w_2 + c(p_2) - c_4 = 0, \quad (18)$$

$$w_2 = \frac{2a_1(M^2 - 1)}{(\gamma + 1)M}, \quad (19)$$

$$p_2 = \left(1 + \frac{2\gamma(M^2 - 1)}{(\gamma + 1)}\right)p_1 \quad (20)$$

determine $M = U/a_1$, and the remaining variables follow. Note that a is defined differently in the melt ($a_{3,4}$) and air ($a_{1,2}$) regions. In the simple wave region $-a_4 t < x < (u_3 - a_3)t$ velocity and pressure can be given in terms of space and time, but is more convenient to specify

$$x = (c_4 - c(p) - a(p))t, \quad u = c_4 - c(p) \quad (21)$$

as function of pressure in the range $p_3 < p < p_4$ and time.

The comparison between exact and numerical shock tube solutions is problematic for initial conditions with the large density and pressure differences that are relevant for the study of magma-repository interactions. The comparison in Fig. 4.19 reveals that the interface is modeled reasonably well, but that the numerical shock moves ahead of the exact shock, which happens in the first few time steps and over the first few grid cells. A reduction of only the time step yields no improvement. A ghost fluid method (see section 3.7.4) is used to match the two fluids across the interface. The simple extrapolation of entropy values

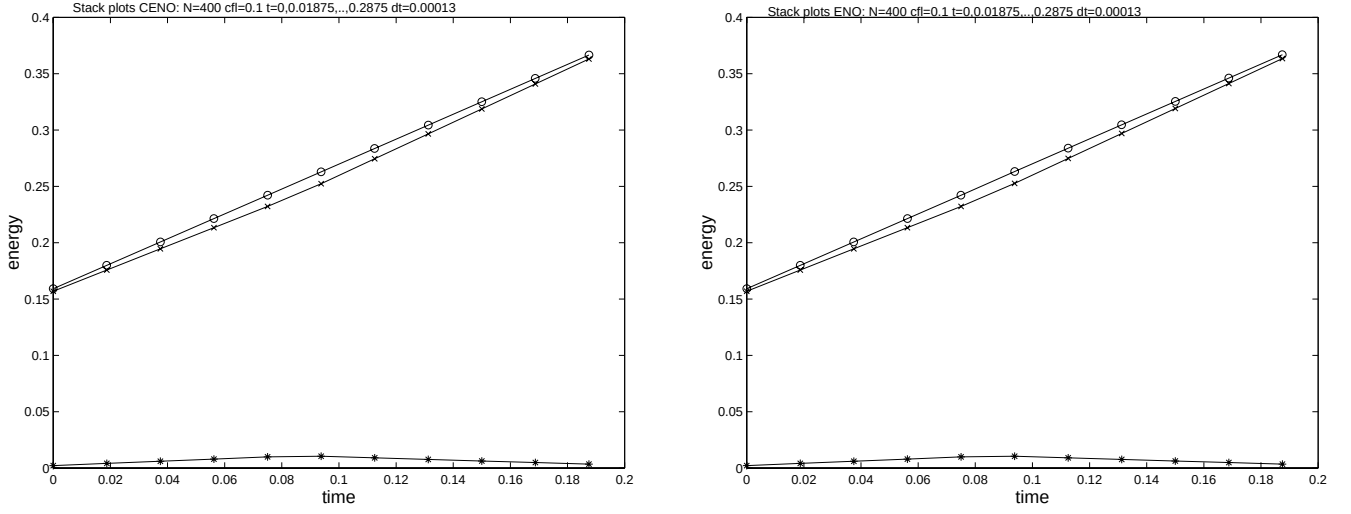


Figure 4.15: Kinetic (*), potential (x) and total energy (o) are shown for a) a LLF2 code and (b) an ENO code for a moving and reflected shock in air.

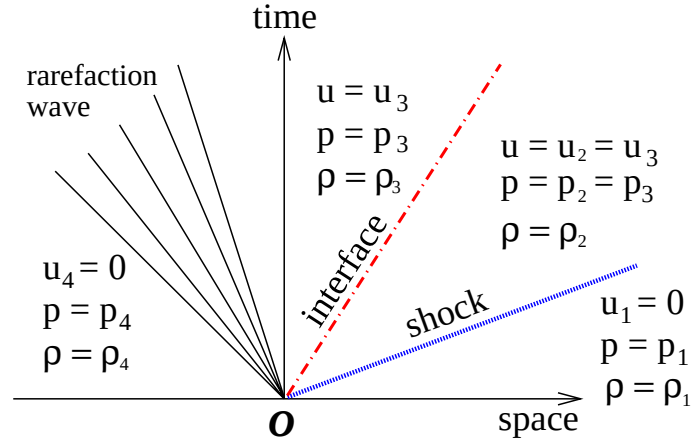


Figure 4.16: Four regions appear in this space time sketch for a shock tube in air.

of air into the ghost air/fluid region where the melt resides is a non-rigorous fix but other methods I have tried turned out to be unstable. The ghost fluid method is useful because it is easy to implement with little extra computational cost. Given these circumstances, the ghost fluid method is applied with brute force: increasing resolution clearly improves the performance, as may be seen in Fig. 4.20 where the resolution is increased by a factor of 2.5. The ENO simulation is worse than the LLF2 simulation due to oscillations around the interface. The ENO algorithm could be modified to switch to a convex ENO (or LLF2) algorithm near the interface, but this has not yet been implemented. Based on this performance, further calculations employ the faster LLF2 scheme.

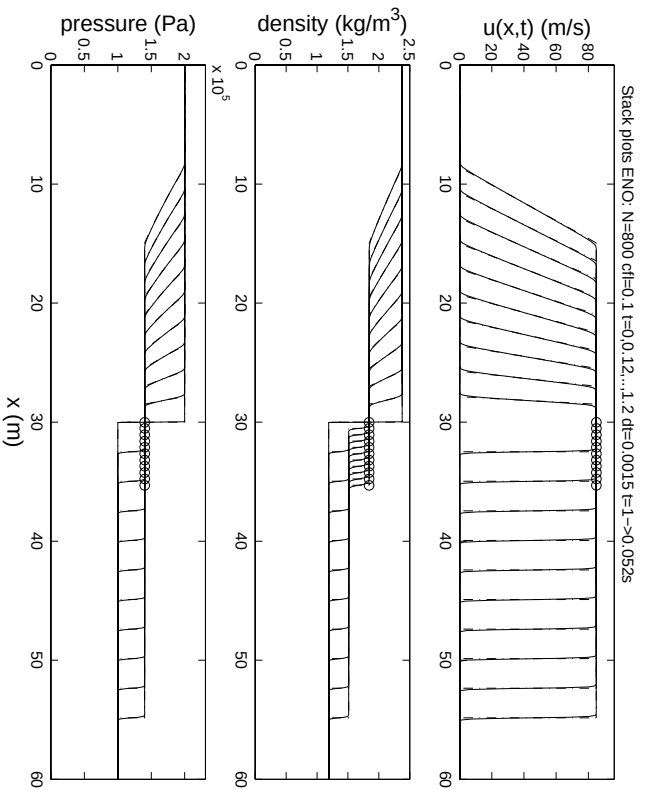
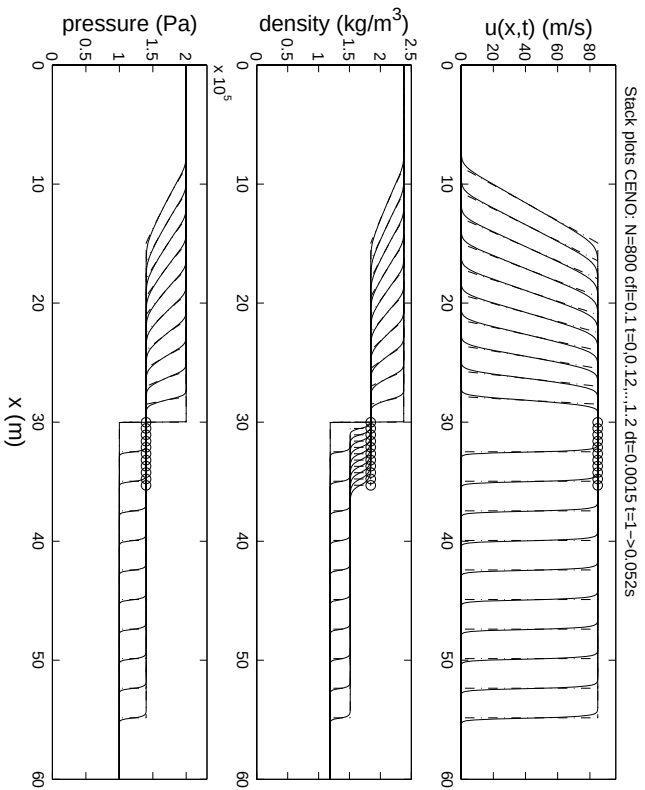


Figure 4.17: For a shock tube in air, velocity, density, and pressure profiles are shown for (a) a LLF2 code and (b) an ENO code. Open circles denote the exact positions of the interface.

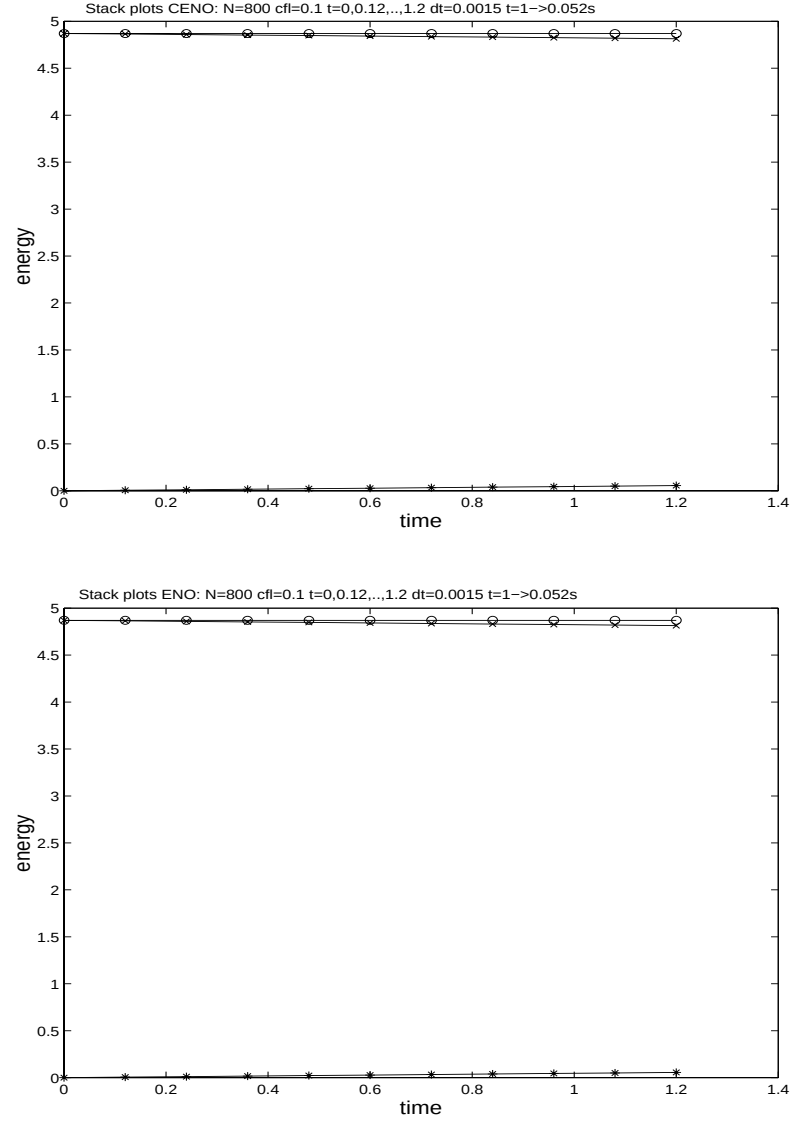


Figure 4.18: Kinetic (*), potential (x) and total energy (o) are shown for a) a LLF2 code and (b) an ENO code for a shock tube in air.

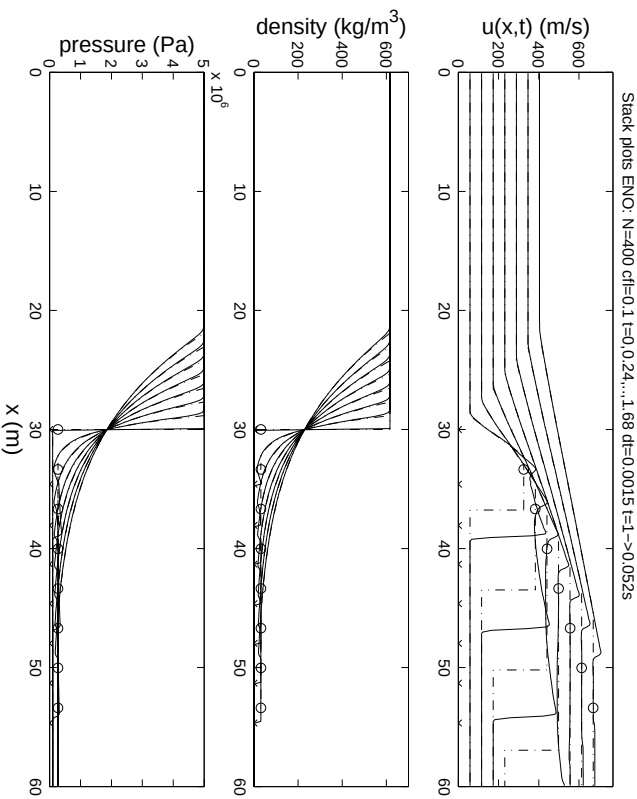
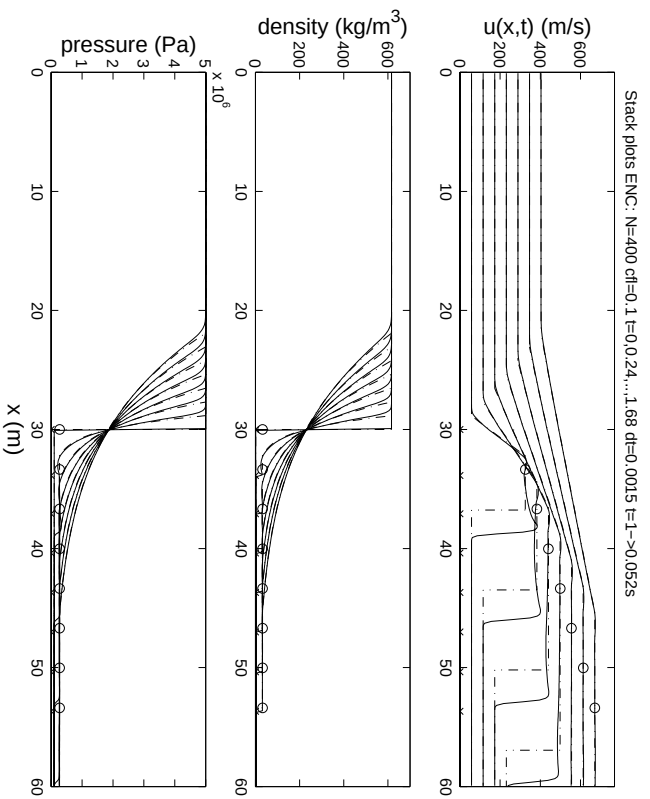


Figure 4.19: For a shock tube with a melt-volatile fluid left of the interface and air to the right, velocity, density, and pressure profiles are shown for (a) a LLF2 code and (b) an ENO code. Open circles denote the exact and crosses the simulated positions of the interface.

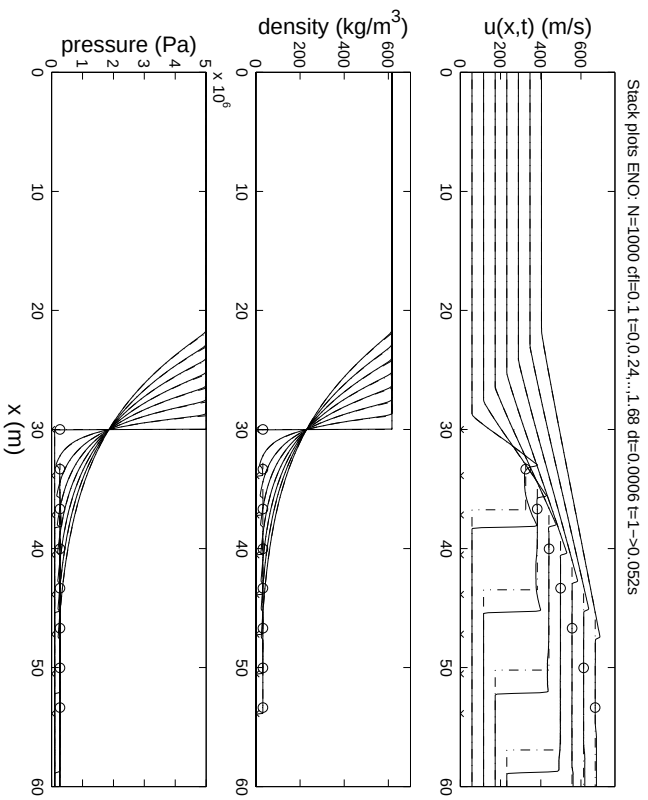
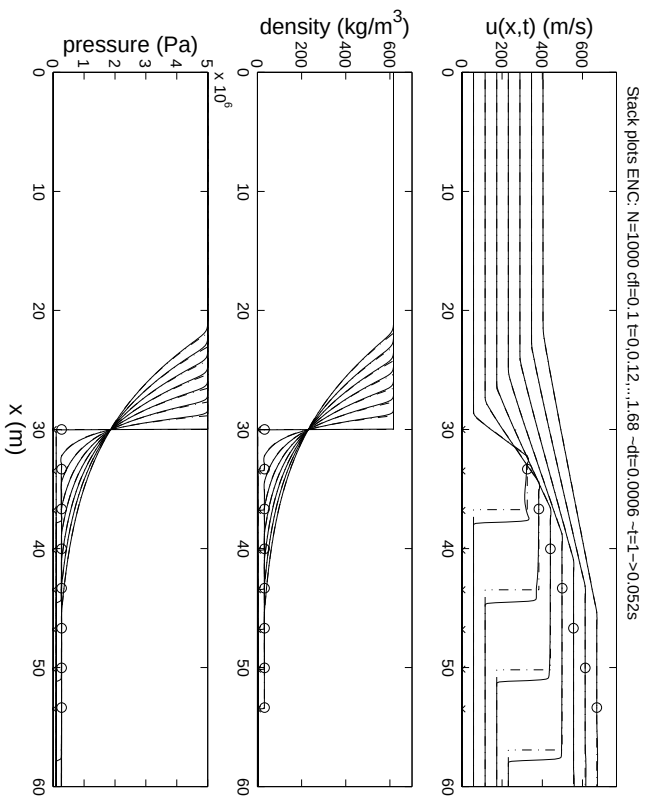


Figure 4.20: Same as in Fig. 4.19 with a resolution increase of 2.5.

4.3.2 Resonating shocks

For states with constant values of density, pressure, velocity and energy, values which may change across shocks and interface, the dynamics can be solved exactly by tracking each shock-shock, shock-wall and shock-interface interaction in space and time. The space-time diagram in Fig. 4.21 explains the evolution.

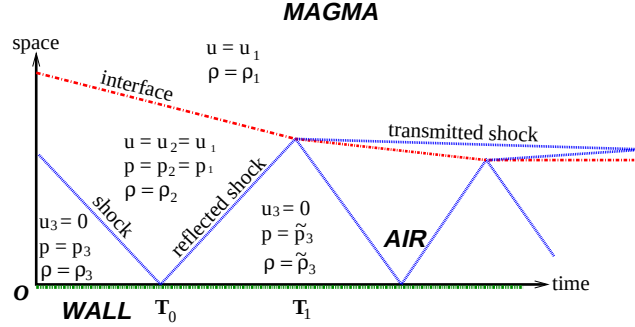


Figure 4.21: Space-time diagram of the resonating shock interactions between tunnel wall and melt-air interface. The vertical axis is space, the horizontal axis is time. The interface is denoted by a dashed-dotted line and the shocks by a solid line. Initially, there is an upper state 1 in the basaltic fluid, an intermediate air state 2, and a lower air state 3 at rest. (There is no gravity.)

Initially, basaltic fluid in constant state 1 flows towards the wall; across the interface in state 2, velocity and pressure in air are continuous and hence the same as in state 1, but air density has a lower value; the shock in air approaches the wall faster than the interface, and demarcates air state 2 from air state 3 while the latter is at rest. As time advances to T_0 , the shock in air reflects and amplifies against the wall, the reflected shock reflects against the interface at time T_1 , which has moved forward, and so on. The interface-shock reflection yields a reflected shock and a transmitted shock, the latter propagates in the basaltic fluid. The first couple of transmitted shocks are still swept towards the wall because they are unable to overcome the incoming fluid speed u_1 in state 1. They are, however, traveling away from the interface. Several transmitted shocks are thus generated in the basaltic fluid, the later ones overtake and annihilate the earlier ones and increase the speed of the shock front in the magma. Finally, a strongly-amplified shock in the magma propagates away from the wall with great speed. Numerical evidence in the stack plots of Fig. 4.22, Fig. 4.23 and the energetics of Fig. 4.24 supports this picture. In particular, pressure and velocity profiles indicate how the pressure rises in the air pocket following each shock reflection, while the movement of the interface emerges in the density profile; the interface position is marked on all horizontal axes. (The small density jump of the shock in air has almost disappeared in the line thickness, but a large jump across the transmitted shock in magma appears after the first shock-interface encounter in profile 6.)

As the magma-air interface slows down and finally arrests, the dynamics is no longer presented accurately in our flow-tube model. Upon slowing down, a gravity current must form, fill the floor at the end of the drift, and leave an air pocket in the upper corner, which will later start to spread across the ceiling of the drift. Gravity currents are clearly not captured by the flow-tube model. Based on a long-wave assumption, one could extend the flow-tube model to include a free-surface $h(\xi_1, t)$ as function of a horizontal spatial coordinate and time only, in analogy to shallow-water and gravity-current modeling in geophysical and geological fluid dynamics [e.g. Gill, 1982]. Alternatively, one could simulate the dynamics of a two-dimensional flow model, averaged over the lateral y -direction, in a vertical plane through the dike-drift system before resorting to computationally expensive three-dimensional simulations. While these more

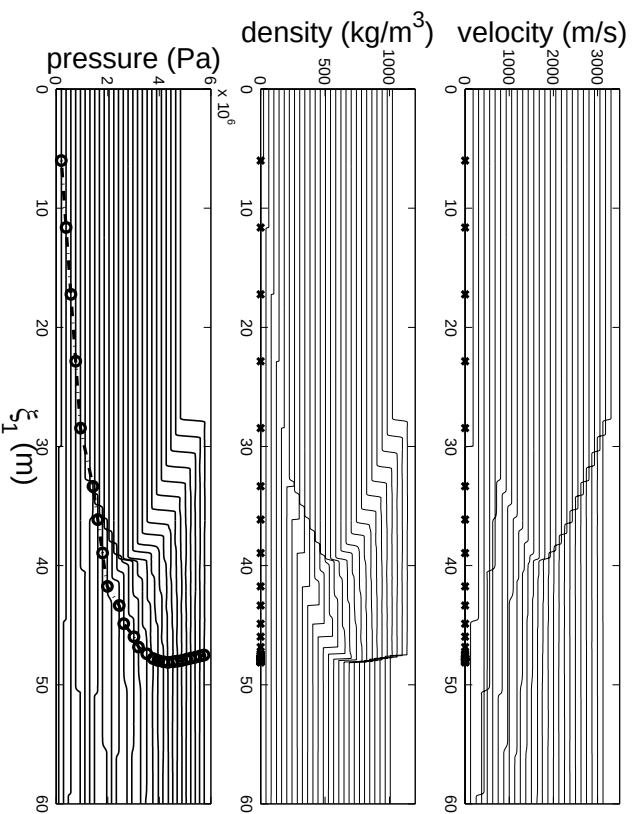


Figure 4.22: Velocity, density and pressure profiles are shown for a simulation of resonating shocks between a tunnel wall and melt-air interface. The tunnel wall is at the right. The crosses at the x -axis indicate the forward moving positions of the interface. The circles in the pressure plot indicate the position of the interface and the local pressure. Initial pressures are $p_1 = p_2 = 2 * 10^5 Pa$ and $p_3 = 10^5 Pa$. Observe the strong compression of air as time advances.

intricate details are left for future study, the shock amplification process is arguably generic because the high-speed reflected shock will remain essentially one-dimensional away from the wall

The above examination emphasizes that large shock amplification will occur at the drift wall in magma-air interactions. The reference simulation in the section 5.2 reveals that a similar shock amplification mechanism occurs in a dike-drift geometry. Further simulations in Bokhove and Woods [2000, 2001] show how the strongly amplified reflected shock is partially transmitted at the dike-drift transition, and descends into the magma dike, but also partially reflects back into the drift.

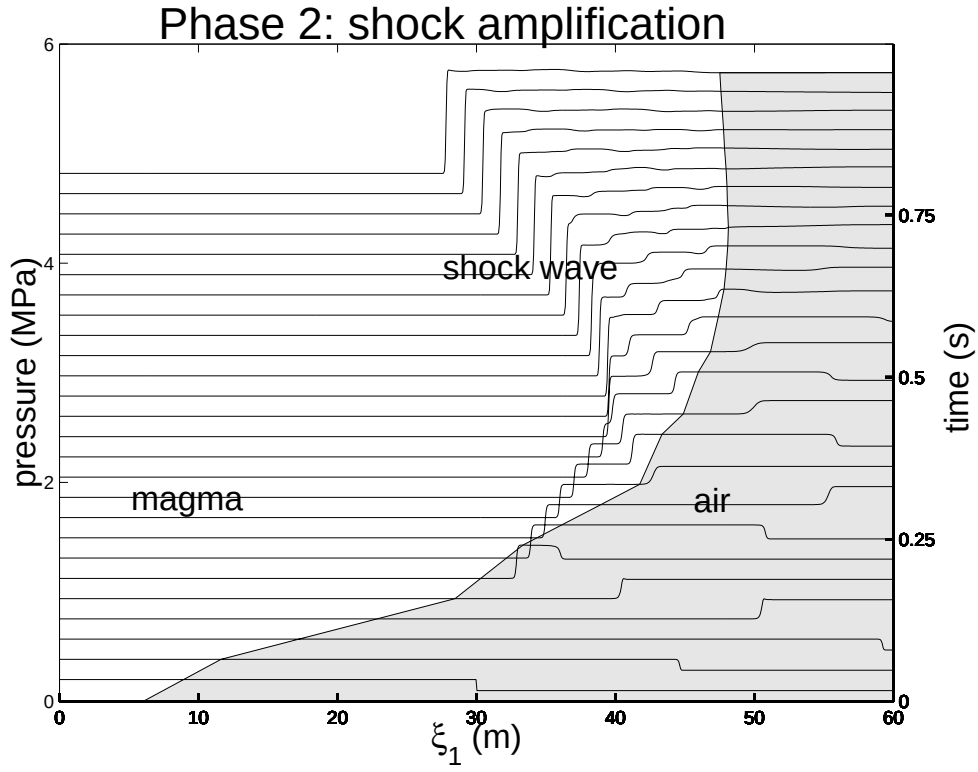


Figure 4.23: Detail of previous figure. Pressure profiles are shown for a simulation of resonating shocks between a tunnel wall and magma-air interface. The wall of the horizontal tunnel is at the right; the boundary on the left is open and allows inflow (and outflow). Initial values of the pressures are $p_1 = 0.2 \text{ MPa}$ (magma), $p_2 = 0.2 \text{ MPa}$ (air) and $p_3 = 0.1 \text{ MPa}$ (air). Observe the strong compression of air as time advances. Gravity plays no role.

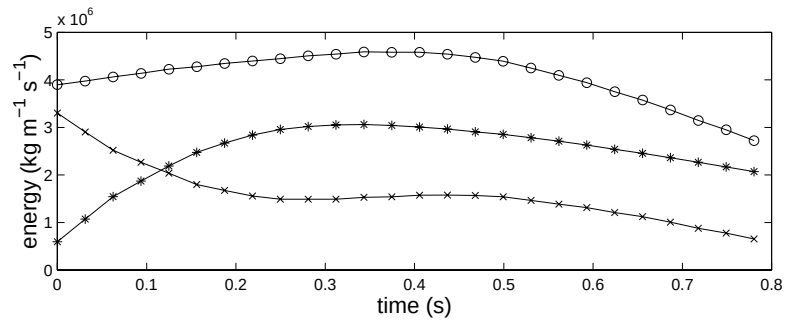


Figure 4.24: Kinetic (*), potential (x), and total (o) energy in a simulation of resonating shocks between the end wall of the tunnel and the magma-air interface (LLF2 code).

4.4 One block 2D magma dynamics tests

The following simulations are full two-dimensional simulations in which the dynamics should display no dependence on the vertical coordinate, and in which we can compare the two-dimensional numerical simulations with the one-dimensional numerical and exact solutions for the magma dynamics. It is seen that the i) stationary shock, ii) moving and reflected shock and iii) shock tube solutions with the LLF2 code `encmac2.c` in Fig. 4.25, Fig. 4.26, and Fig. 4.27 are the same as the numerical and exact solutions (LLF2 code) in Fig. 4.2, Fig. 4.4, and Fig. 4.9, respectively.

The boundary conditions in the three simulations are as follows:

- i) extrapolating East and West, solid North and South, e.g. set `EASTEXTRAP`, `EASTEXTRAP`, `NORTH-SOLID`, `SOUTHSOLID` in `encmac2.c` for simulation 200.
- ii) extrapolating West, solid North, South and East, e.g. set `EASTSOLID`, `EASTEXTRAP`, `NORTHSOLID`, `SOUTHSOLID` in `encmac2.c` for simulation 201.
- iii) extrapolating East and West, solid North, South and East, e.g. set `EASTSOLID`, `EASTEXTRAP`, `NORTH-SOLID`, `SOUTHSOLID` in `encmac2.c` for simulation 202.

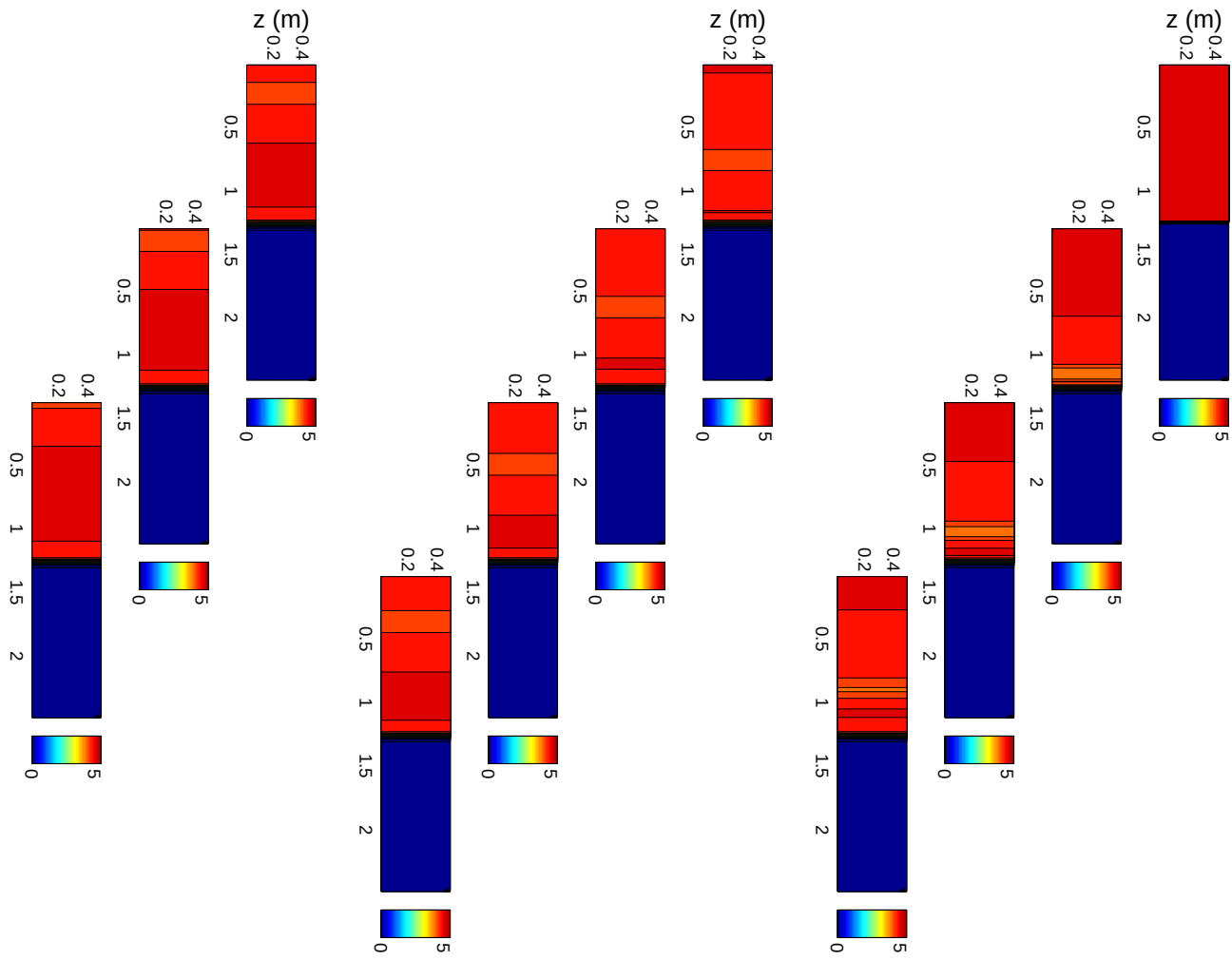


Figure 4.25: Pressure contour plots are shown for a stationary shock of a melt-volatile fluid with the LLF2 code. Time spacing is 0.025 s. Run 200. Compare with 1D simulation.

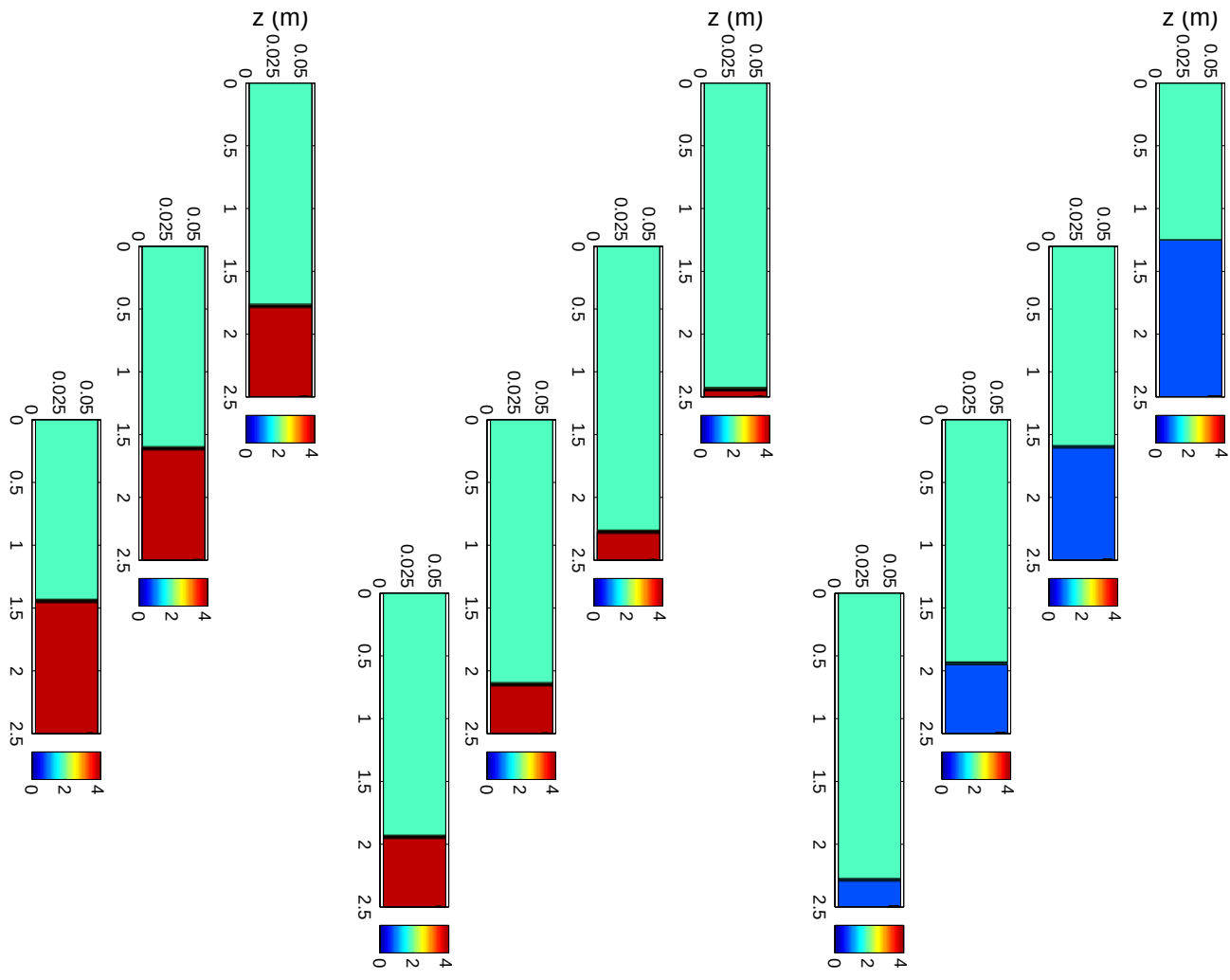


Figure 4.26: Pressure contour plots are shown for a moving and reflected shock of a melt-volatile fluid with the LLF2 code. Time spacing is 0.5 s. Run 201. Compare with 1D simulation.

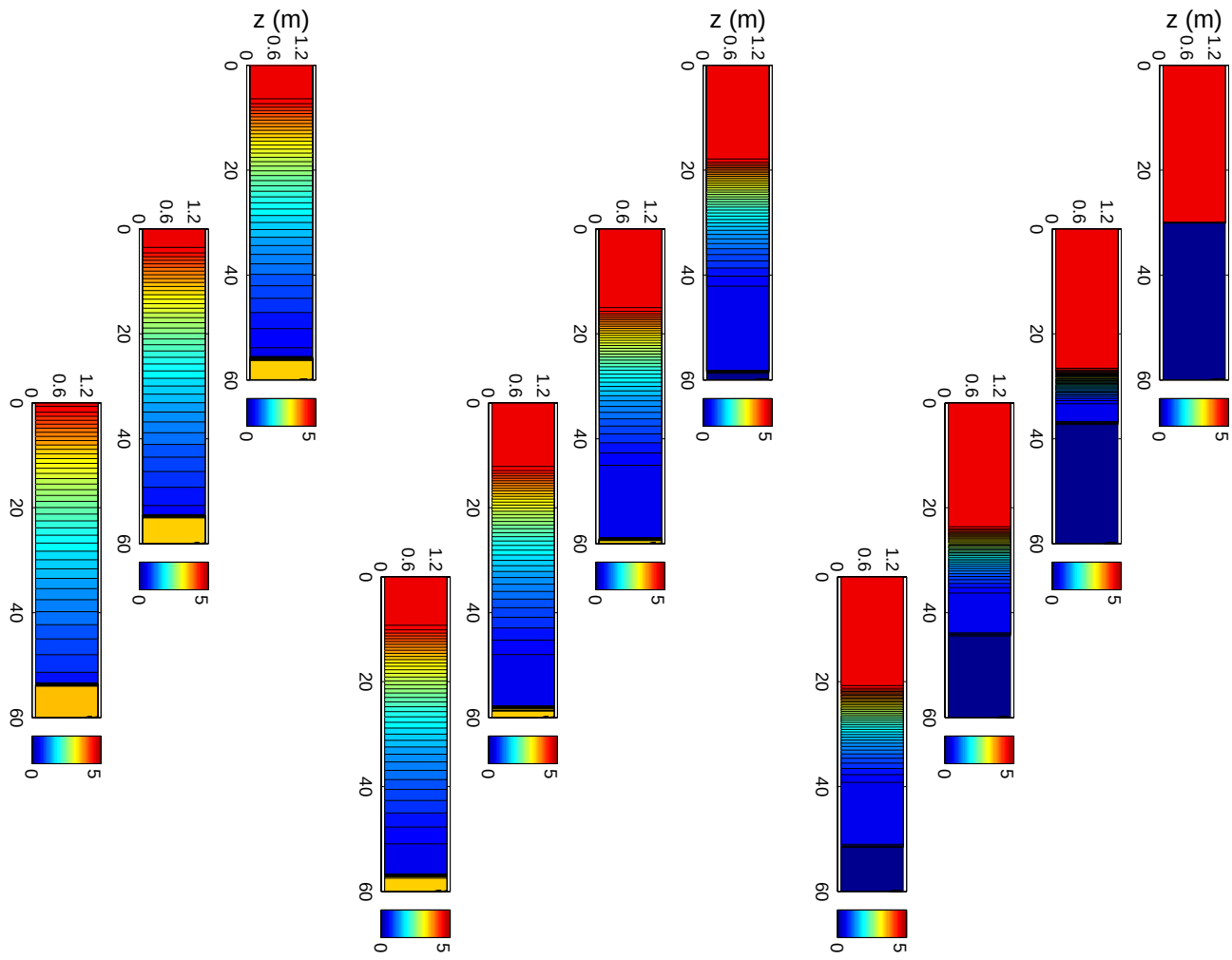


Figure 4.27: Shock tube tests for melt-volatile flow with a right wall. Pressure contour plots are shown with the LLF2 code. Time spacing 0.3. Run 202. Compare with 1D simulation.

4.5 Two block 2D magma dynamics tests

Inviscid magma dynamics is considered within two connected rectangular domains. The connection between the two domains is a horizontal strip at the bottom left corner of block 1 and the top of block 2. This configuration arises as a lateral (y -direction) average of the dike-drift geometry. In this section the weight $b(x, z) = 1$ in (18): we are considering purely two-dimensional simulations. The implication is that the volume fluxes can only be compared qualitatively with the three-dimensionally averaged one-dimensional simulations results. The test simulations have the following value of g and initial conditions (the number indicates the run):

- 204) $g = 0$ and we consider a shock tube solution with initially 5 MPa high pressure on the left and 0.1 MPa low pressure on the right. The initial condition is set up so that the initial flow at the top remains fairly one dimensional. Moreover the flow in the drift remains fairly one-dimensional and the distance between the gate position of the initial jump in pressure and the wall is equal to the one in the reflected one-dimensional shock tube problem. Hence, we can compare the results with Fig. 4.9.
- 205) $g = 0$ and we consider a moving shock with initially 0.2 MPa high pressure in the dike and 0.1 MPa low pressure in the drift. The moving shock will remain fairly one-dimensional on the left of the domain in block 1 and 2 and the height of block one is equal to the distance between the initial shock and the wall in the one-dimensional moving and reflected shock simulation. We can thus partially compare these one- and two-dimensional simulations.
- 212) $g = 9.81$ and we consider a shock tube solution with initially 5 MPa high pressure in the dike and 0.1 MPa low pressure in the drift.
- 213) $g = 9.81$ and we consider a shock tube solution with initially 5 MPa high pressure in the dike and 0.1 MPa low pressure in the drift. Simulation 213 is a long-time version of simulation 212.

The boundary conditions in the three simulations are as follows: the West, North, East boundaries of block 1 are impenetrable, part of the South boundary of block 1 is impenetrable, the East and West boundary of block 2 is impenetrable, the South boundary of block 2 is open, and the North boundary of block 2 has an open connection with the bottom-left boundary of block 1. In `encmac2.c`, we choose `EASTSOLID`, `WESTSOLID`, `NORTHSPEC1`, `SOUTHSPEC2`, `BLOCK12GLUE`.

The evolution of the pressure in Fig. 4.28 at the North boundary of block 1 compares well with the pressure evolution in Fig. 4.9. The evolution of the pressure in Fig. 4.29 at the West boundary compares well with the pressure evolution in Fig. 4.4.

Simulations 212 and 213 are computed on an irregular grid with extra resolution around the dike-drift breakthrough. This grid transformation is described in section 5.4. The simulations in Fig. 4.30 shows the dynamics after the magma breaks from dike into drift every 0.0104 s till time 0.1664 s . The one-dimensional reference simulation in Fig. 5.3 displayed three phases: the initial shock tube phase, the shock reflection phase, and finally the emerging decaying phase in which rarefaction and shock waves leave the domain and decay. These two-dimensional simulations show that there is an additional phase in which a vertical magma jet forms and reflects against the roof as in the one-dimensional shock tube with reflection simulation. Initially and near the symmetry line or wall on the left the speeds are high and the flow is more or less one dimensional. Due to the low-pressure region in the drift, the flow eventually becomes two dimensional, but further in the drift we can see in Fig. 4.32 that the flow becomes fairly one-dimensional again since the diameter of the drift is small relative to its length. Simulation 213 has a simple extrapolating boundary

condition on the South boundary of block 1, thus allowing for waves to leave the domain. This boundary condition works reasonably well but does contaminate the interior, especially in combination with the nonzero value of gravity. This effect is very slow and the downward propagating rarefaction waves diminish the effect completely. To check this we show pressure contour plots in Fig. 4.33 of the same simulation 213 but now with a solid South boundary in block 1: we see that there is no effect by changing the boundary. Hence the contamination around the South boundary is kept at bay from the key dike-drift areas of interest.

To assess the validity of the one-dimensional flow tube model, we show the pressure evolution of a one-dimensional model which is an average of the previous two-dimensional model. The connection at the breakthrough is smooth in contrast to the contract nozzle connection we considered before in Fig. 5.3. When we compare Fig. 4.32 with Fig. 4.34, we see that in the two-dimensional calculations it takes time for the magma to turn around the corner and become nearly one-dimensional again: the reflection process in the two-dimensional simulation starts at time 1.04 s while it begins at 0.65 s in the one-dimensional case.

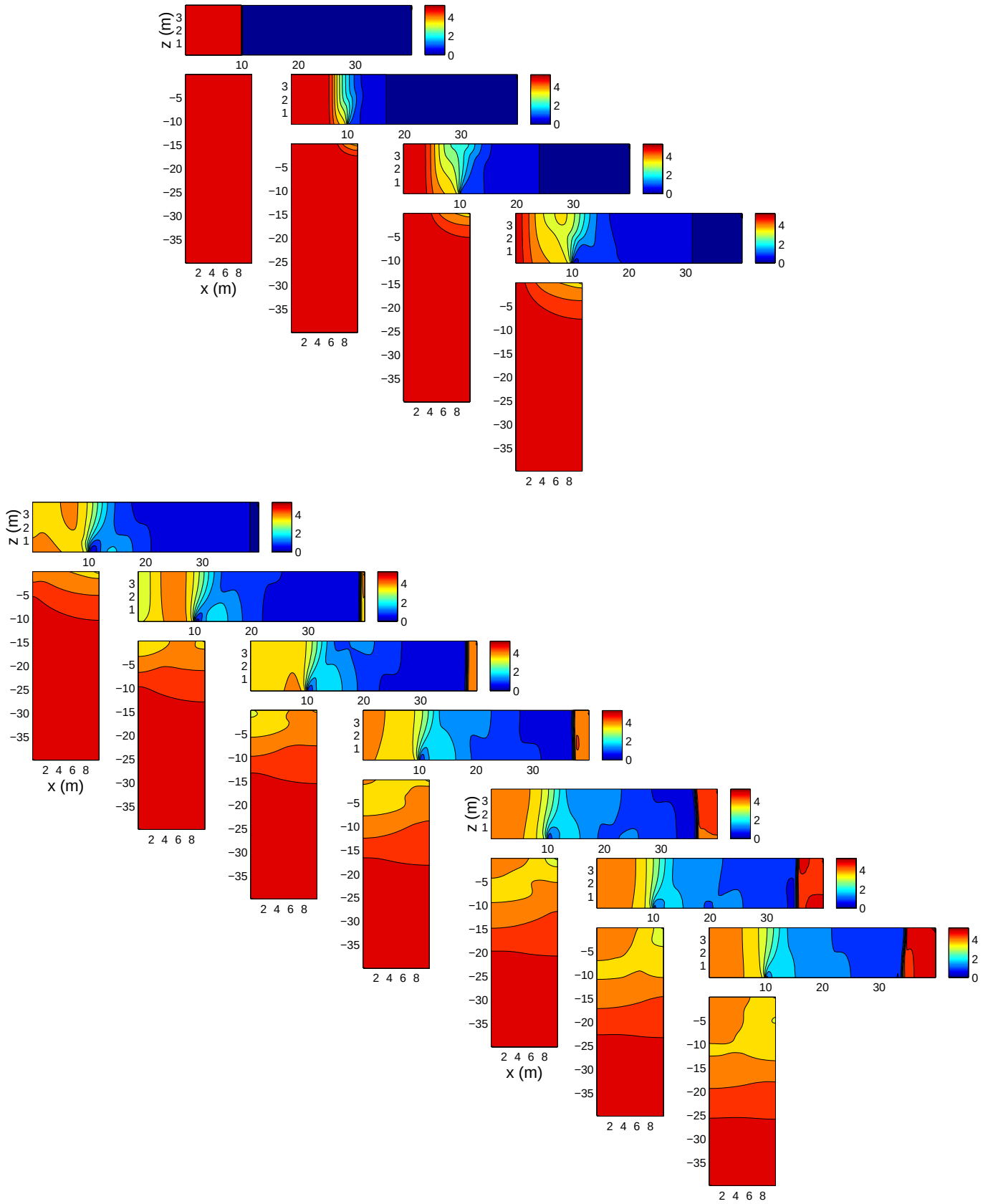


Figure 4.28: Shock tube with gate in block 1. Pressure contour plots are shown using a LLF2 code. Arrows indicate the flow direction. Time spacing is 0.5 s . Run 204. Time spacing 0.0312 s .

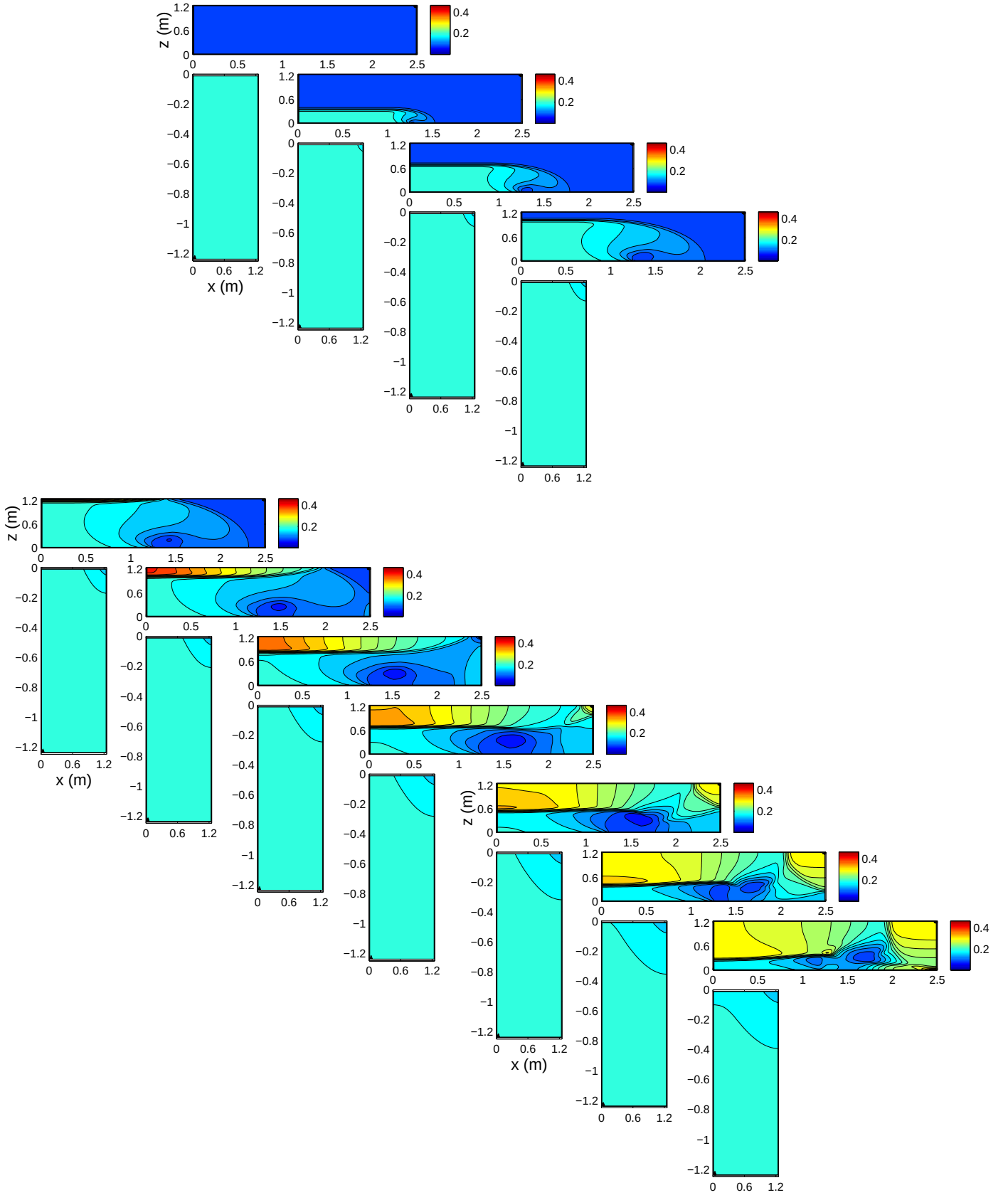


Figure 4.29: Shock tube with gate between block 1 and 2. Shock tube tests for melt-volatile flow with a right wall. Pressure contour plots are shown using a LLF2 code. Time spacing 0.026 s. Run 205.

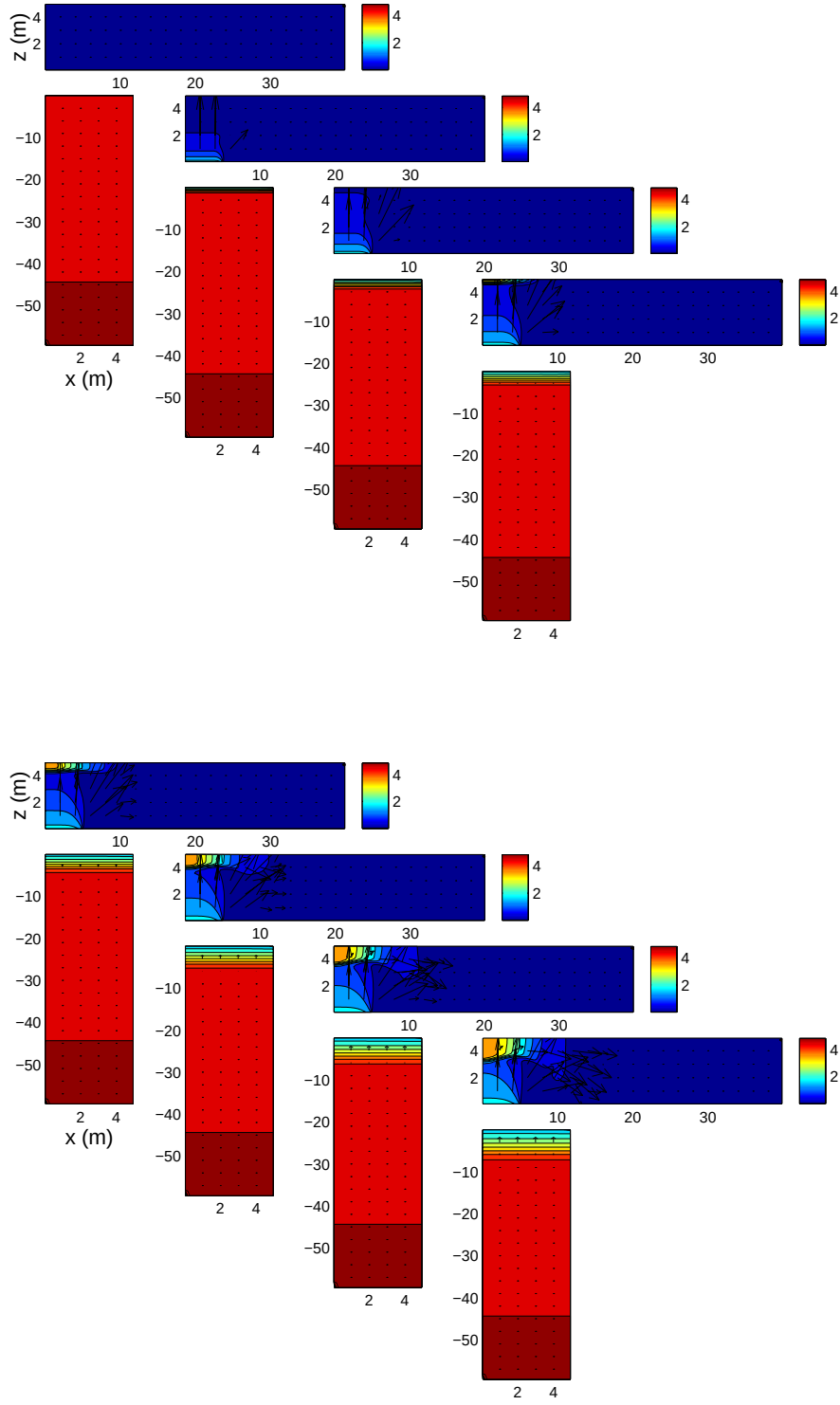


Figure 4.30: Shock tube with hydrostatic initial condition in dike and drift, *e.g.*, simulation with $g \neq 0$. Pressure contour plots are shown using a LLF2 code. Time spacing is 0.0104 s. Run 212.

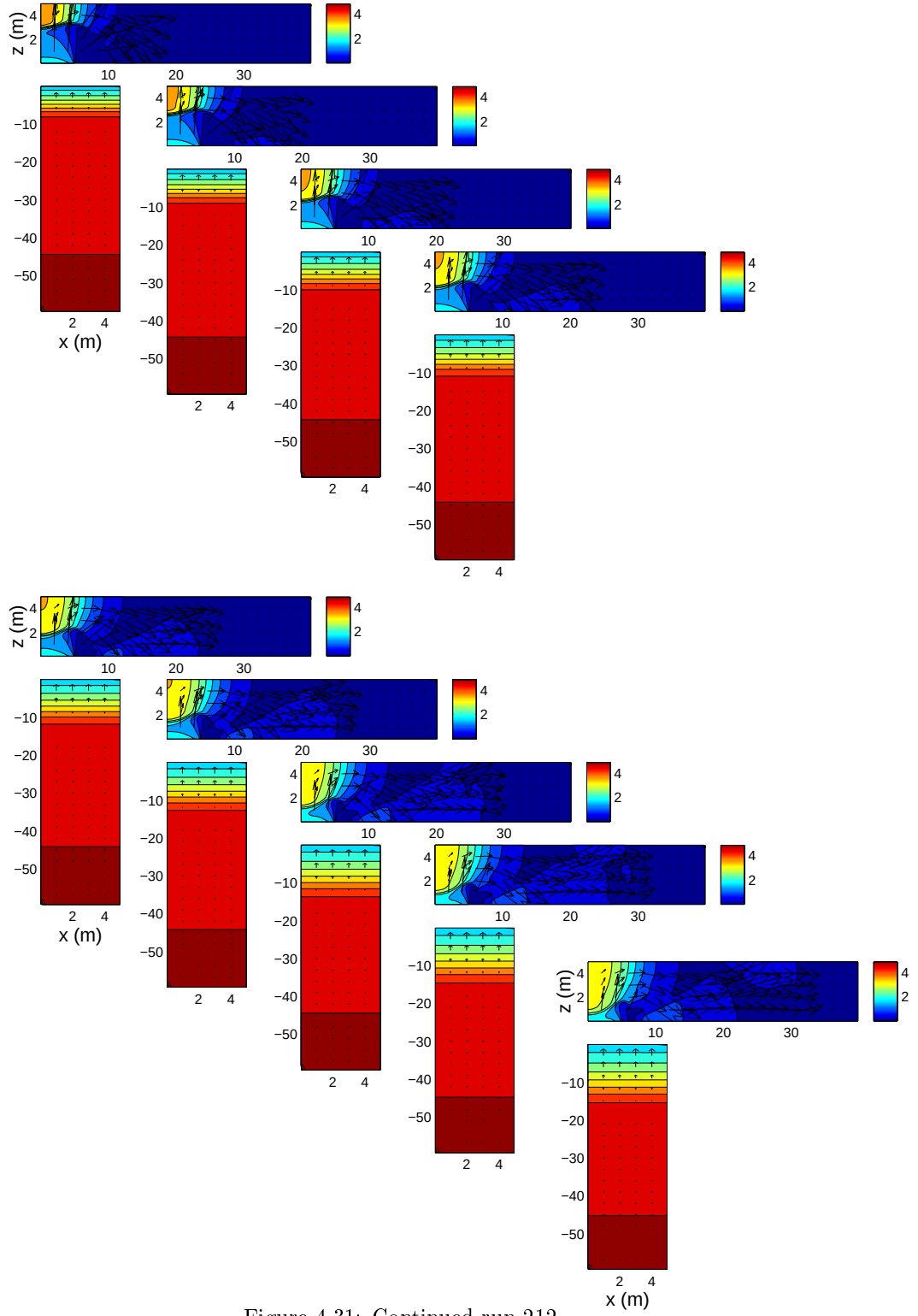


Figure 4.31: Continued run 212.

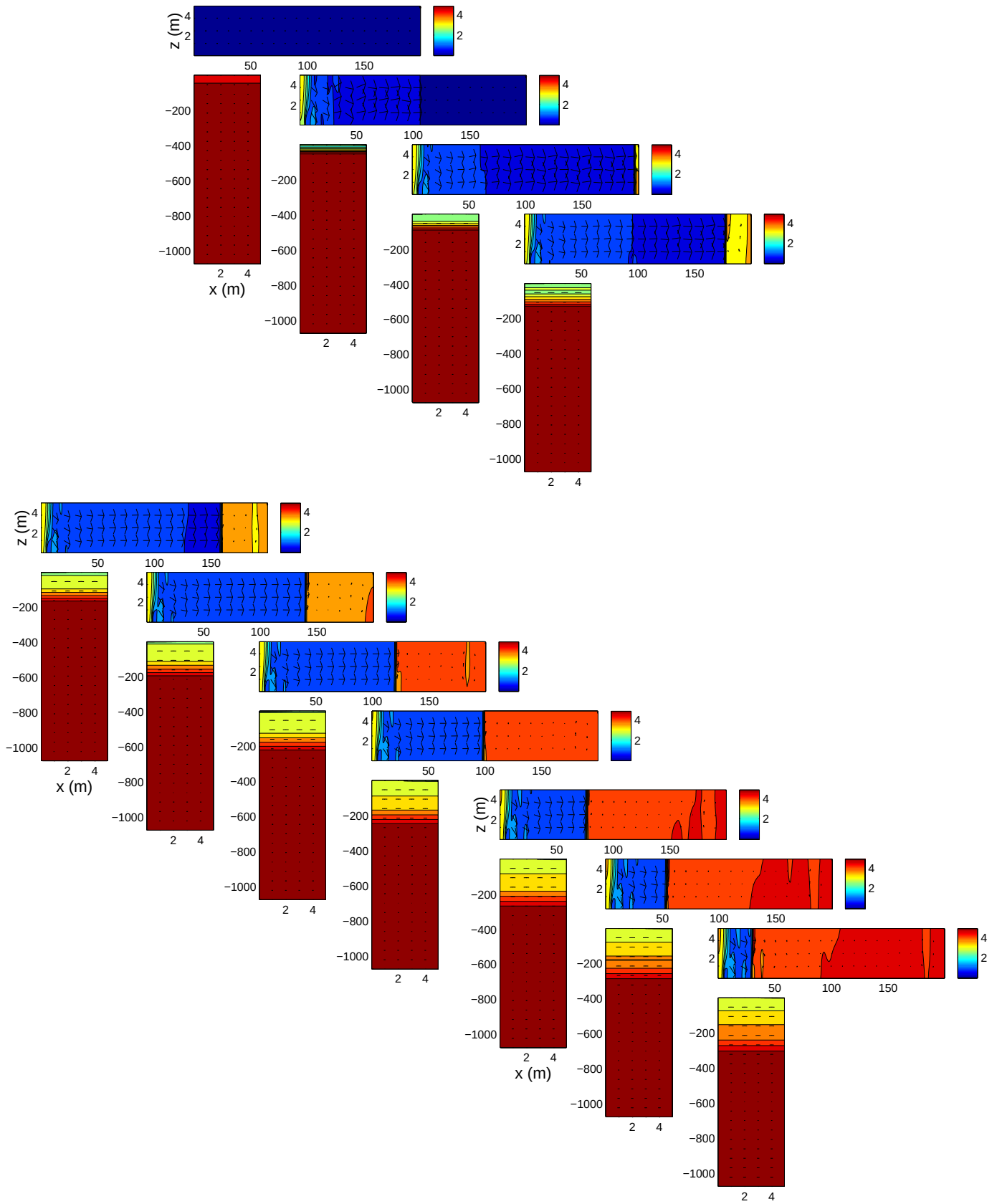


Figure 4.32: Shock tube with hydrostatic initial condition in dike and drift, *e.g.*, simulation with $g \neq 0$. Pressure contour plots are shown using a LLF2 code. Time spacing 0.5202 s . Run 213.

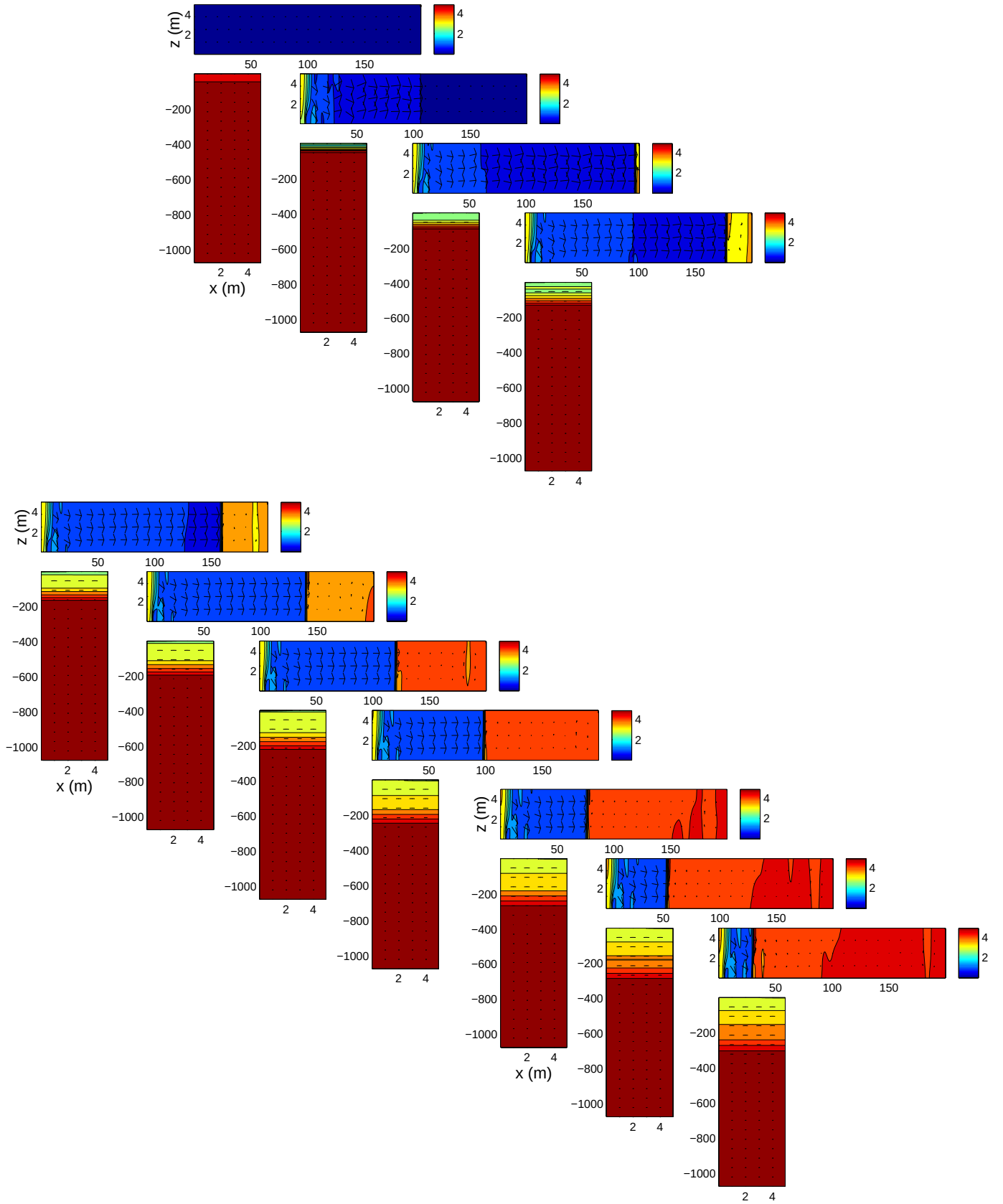


Figure 4.33: Shock tube with hydrostatic initial condition in dike and drift, *e.g.*, simulation with $g \neq 0$. Pressure contour plots are shown using a LLF2 code. Time spacing 0.5202 s . Run 214.

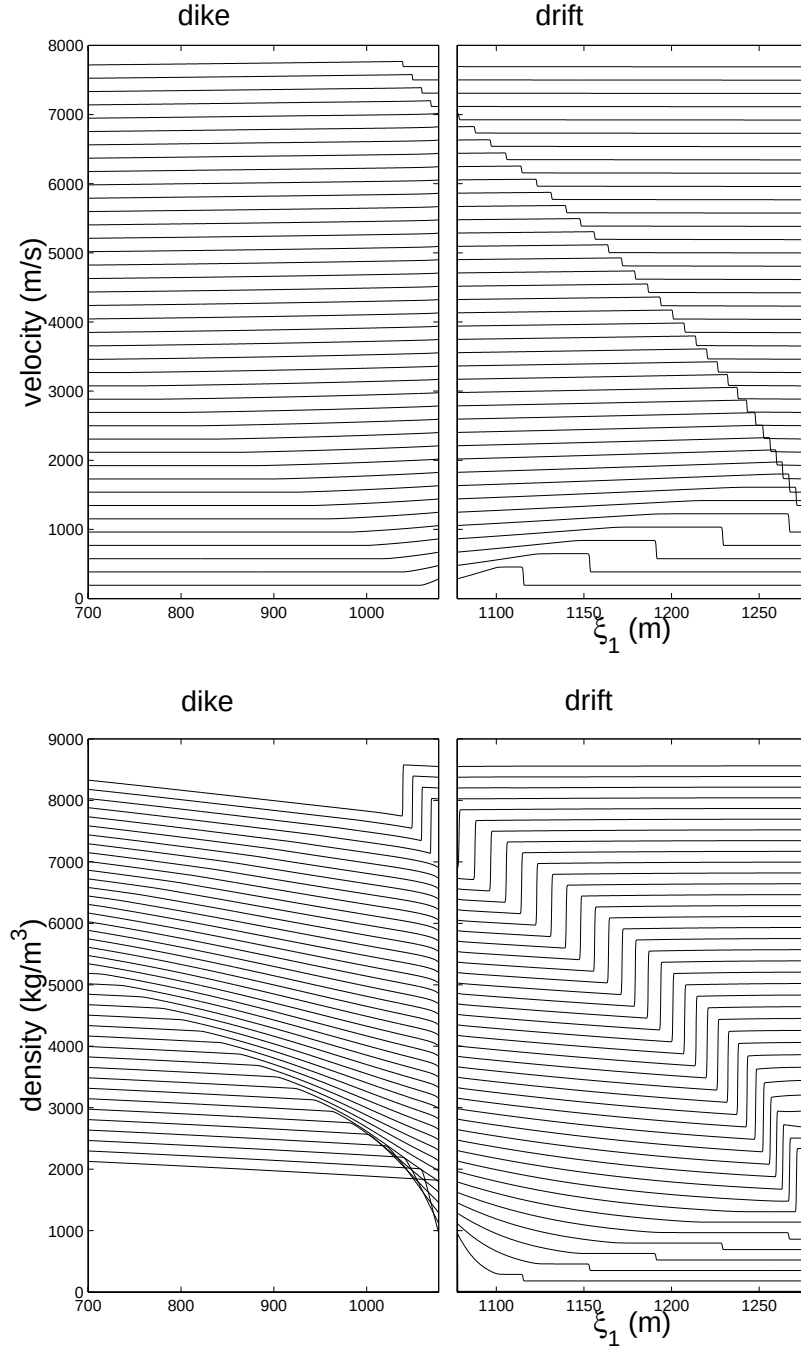


Figure 4.34: Velocity, density and pressure profiles are shown for magma flow in a dike-drift system. The one-dimensional model is an average over the two-dimensional geometry. Spacing 0.13 s . Simulation 161 with `entis1.c`. Compare the shock reflection phase in this simulation with Figure 4.23.

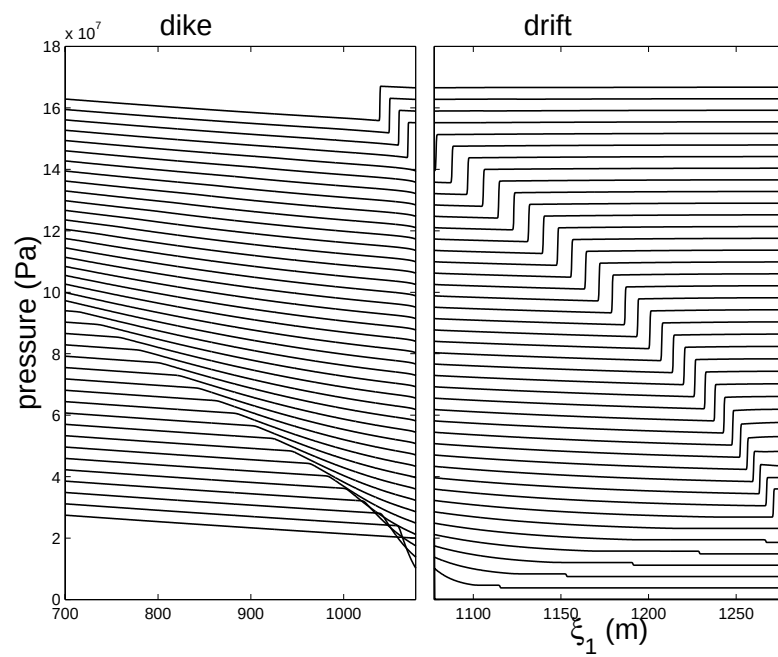


Figure 4.35: Continued simulation 161.

Chapter 5

Magma-repository interactions

In this chapter, we define the geometry of the dike and drift system used in the one- and two-dimensional simulations. We will also describe the one-dimensional reference simulation on magma-repository interactions used in Bokhove and Woods [2000, 2001], where one can also find the results of an extensive parameter study. In addition, we will outline several multi-block geometries which will be implemented in future simulations.

5.1 Geometry and grid transformation in 1D

A cross section of the magma dike and repository drift is given in Fig. 5.1. The connection between dike and drift is represented as an elliptical arc. With a coordinate origin $\mathcal{O}$ for x and z at the South-East intersection of dike and drift, the arc has coordinates: $x = -(1/2)w \cos[(\xi_1 - L_{dike}) \pi / (2 L_{arc})]$, $z = (1/2)d \cos[(\xi_1 - L_{dike}) \pi / (2 L_{arc})]$.

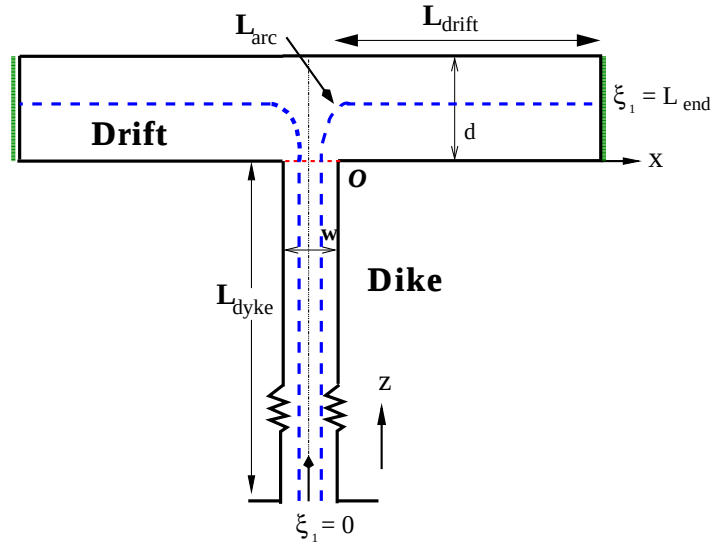


Figure 5.1: A vertical cross section of the magmatic dike and drift system defines the various length scales involved in all the flow-tube model simulations.

Characteristic scales in the configuration are: the length of the magma dike in the simulation L_{dike} , the distance from the origin $\mathcal{O}$ to the end of the drift L_{drift} , arc length $L_{arc} = \frac{1}{2} d E(m)$ where $E(m)$ is the

elliptical integral of the second kind with $m = (d^2 - w^2)/d^2$, and the total flow tube length L_{end} . For longer dike lengths the pressure rises above the critical pressure p_c for which all volatiles are dissolved in the then incompressible magma. Dike lengths are therefore restricted to stay below critical pressure p_c . Given the geometry, gravity in the flow-tube model is present in the dike, but diminishes in the arc to zero in the drift [cf. (10)].

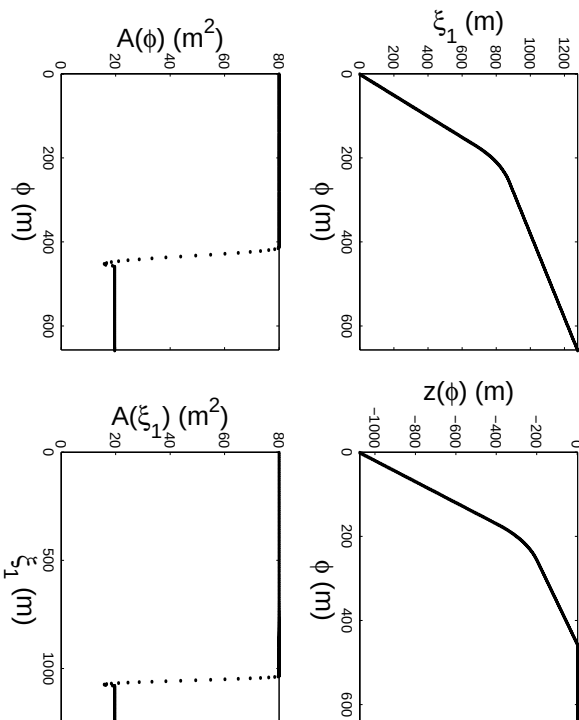


Figure 5.2: Dependence of (a) $\xi_1(\phi)$, (b) $z(\phi)$, (c) $A(\xi_1)$ and (d) $A(\phi)$ on the stretched coordinate ϕ for 300 ϕ -grid points. The numerical grid is coarser near the magmatic chamber. Figure made with Matlab program `stretch.m`, *i.e.* use `stretch(nchoice)` with `nchoice = 0`.

The cross-sectional area $A(\xi_1, t)$ of the flow tube is constant in the drift and in most of the dike, A_{dike} , for $\xi_1 < L_{end} - L_{drift} - (1/2)L_d - d$ but narrows in a nozzle or transition zone with minimum area $A = \pi dw$ before it enlarges to the constant drift area, A_{drift} . The characteristic spacing between drifts is L_d . This transition zone is a crude model of the characteristic three-dimensional flow path and volume between dike and drift. Most resolution is needed in the region of prime interest around the dike-drift intersection and in the drift, so instead of ξ_1 a smooth coordinate transformation to another coordinate ϕ is made. The regular ϕ -grid yields a coarse ξ_1 -grid near the bottom of the magma dike. The coordinate transformation is, however, one-to-one in the upper part of the magma dike, in the arc and in the drift. With some abuse of notation, the unit of ϕ is simply given in meters in some of the subsequent graphs. The dependence of $\xi_1(\phi)$, $z(\phi)$, $A(\phi)$ and $A(\xi_1)$ in the reference simulations and most other simulations is shown in Fig. 5.2.

The formulas for $\xi_1(\phi)$, $h_1(\phi)$, $z(\phi)$, $\partial z/\partial \phi$, and $A(\phi)$ are given by:

$$\xi_1(\phi) = \begin{cases} (1/\alpha)\phi, & 0 < \phi < \phi_L, \\ (1/\alpha)\left(e^{\pi(\phi-\phi_L)/\Delta\phi_L}(\phi-\phi_L) + \phi_L\right), & \phi_L < \phi < \phi_L + \Delta\phi_L, \\ \phi + L_{dike} - \phi_{dike}, & \phi_L + \Delta\phi_L < \phi < \phi_{end}, \end{cases} \quad (1)$$

$$h_1(\phi) = \begin{cases} (1/\alpha), & 0 < \phi < \phi_L, \\ (1/\alpha) e^{\eta(\phi-\phi_L)/\Delta\phi_L} \left(1 + \eta(\phi-\phi_L)/\Delta\phi_L\right), & \phi_L < \phi < \phi_L + \Delta\phi_L, \\ 1, & \phi_L + \Delta\phi_L < \phi < \phi_{end} - L_{drift} - L_{arc}, \\ \frac{\pi}{4L_{arc}} \sqrt{w^2 \sin^2\left(\frac{\pi(\phi-\phi_{dike})}{2L_{arc}}\right) + d^2 \cos^2\left(\frac{\pi(\phi-\phi_{dike})}{2L_{arc}}\right)}, & \phi_{end} - L_{drift} - L_{arc} < \phi < \phi_{end} - L_{drift}, \\ 1, & \phi_{end} - L_{drift} < \phi < \phi_{end}, \end{cases}$$

$$z(\phi) = \begin{cases} (1/\alpha)\phi - L_{dike}, & 0 < \phi < \phi_L, \\ (1/\alpha)(\phi - \phi_L) e^{\eta(\phi-\phi_L)/\Delta\phi_L} + (1/\alpha)\phi_L - L_{dike}, & \phi_L < \phi < \phi_L + \Delta\phi_L, \\ \phi - \phi_{dike}, & \phi_L + \Delta\phi_L < \phi < \phi_{dike}, \\ \frac{1}{2} d \sin\left(\frac{\pi(\phi-\phi_{dike})}{2L_{arc}}\right), & \phi_{dike} < \phi < \phi_{end} - L_{drift}, \\ \frac{1}{2} d, & \phi_{end} - L_{drift} < \phi < \phi_{end}, \end{cases}$$

$$\frac{dz(\phi)}{d\phi} = \begin{cases} (1/\alpha), & 0 < \phi < \phi_L, \\ (1/\alpha) e^{\eta(\phi-\phi_L)/\Delta\phi_L} \left(1 + \eta(\phi-\phi_L)/\Delta\phi_L\right), & \phi_L < \phi < \phi_L + \Delta\phi_L, \\ 1, & \phi_L + \Delta\phi_L < \phi < \phi_{dike}, \\ \frac{\pi d}{4L_{arc}} \cos\left(\frac{\pi(\phi-\phi_{dike})}{2L_{arc}}\right), & \phi_{dike} < \phi < \phi_{end} - L_{drift}, \\ 0, & \phi_{end} - L_{drift} < \phi < \phi_{end}, \end{cases}$$

$$A(\phi) = \begin{cases} A_{dike}, & 0 < \phi < \phi_{end} - L_{drift} - \frac{1}{2} L_d - d, \\ \left[1 - \cos\left(\frac{2\pi(\phi-\phi_{end}-L_{drift}-d-L/2)}{L_d}\right)\right], & \phi_{end} - L_{drift} - \frac{1}{2} L_d - d < \phi < \phi_{end} - L_{drift} - d, \\ \left[1 - \cos\left(\frac{\pi(\phi-\phi_{end}-L_{drift}-d)}{d}\right)\right], & \phi_{end} - L_{drift} - d < \phi < \phi_{end} - L_{drift}, \\ A_{drift}, & \phi_{end} - L_{drift} < \phi < \phi_{end}, \end{cases}$$

with $\phi_{dike} = \phi_{end} - L_{drift} - L_{arc}$; $\phi_{end} = L_{drift} + L_r - \Delta L_r + \phi_L + \Delta\phi_L$; $\phi_L = \alpha(L_{dike} + L_{arc} - L_r)$; and $\Delta\phi = \alpha \Delta L_r e^{-\eta}$. Typical values we have used are $\alpha = 0.25$, $L_r = 200\text{ m}$, $\Delta L_r = 200\text{ m}$ and η is found as the zero of $(1 + \gamma) \exp \gamma / \alpha = 0$, which assures that ξ_1 and its derivative are continuous in ϕ .

These functions $1/h_1(\phi)$, $z(\phi)$, $\partial z / \partial \phi$, $A(\phi)$ and $\partial A(\phi, t) / \partial \phi$ are found in the core programs `encmvs1.c`, `encmvt1.c` and `encmax1.c` as functions `H1scalef(x1)`, `Zhydro(x1)`, `Zdiff(x1)`, `Area(x1)` and `Diffarea(x1)`, respectively.

5.2 Reference 1D simulation

Characteristic parameter values for basaltic magma at Yucca Mountain are: $n_0 = 2\text{ wt}\%$, $L_d = 80\text{ m}$, $w = 2\text{ m}$, $d = 5\text{ m}$, $A_{drift} \approx 19.64\text{ m}^2$, $A_{dike} = 80\text{ m}^2$, $L_{drift} = 200\text{ m}$, $L_{dike} = 1075\text{ m}$; lithostatic pressure at $D = 400\text{ m}$ $p_L = \sigma g D = 10\text{ MPa}$ and overpressure $P_o = 10\text{ MPa}$ give a total pressure at the dike tip of $P_t = 20\text{ MPa}$; and the critical void fraction at the point of fragmentation is $\alpha = 0.7$.

The initial condition is quiescent flow on either side of the drift entrance, hydrostatic balance in the magma dike with a pressure $P_t = 20\text{ MPa}$ at the dike tip, and air at atmospheric pressure and room

temperature in the drift. The initial magma pressure and density can be determined by combining the equation of state and the hydrostatic balance condition in one relation, which root can be found numerically to yield the pressure. Except for hydrostatic balance, the initial condition is reminiscent of one for a classical shock tube.

The reference simulation advances 2.734 s . Its flow profiles are shown in Fig. 5.3 every 0.0683 s . Besides shock reflections between drift wall and magma-air interface similar to those studied in Fig. 4.22, a rarefaction wave is seen to travel into the magma dike. Both pressure and velocity profiles reveal the interplay of resonating shocks between drift wall and magma-air interface during the first tens of seconds. The density profile again strikingly marks the movement of the interface and the transmitted shock. Just before the end of the simulation a strongly amplified shock wave is seen to emerge in the magma. Its magnitude is about 28 times larger than the magnitude of the initial shock wave in air.

Further simulations and a parameter study, assessing the dependence of the rarefaction and shock wave propagation in the above and similar dike-drift systems, can be found in Bokhove and Woods [2000, 2001].

5.3 1D simulations for Science paper

The frictional parameterization for the simulations in Woods *et al.* [2001] differ from (13)-(15) and have the form

$$F_1 = f u = \left(\frac{\alpha_s \mu}{d^2} + \frac{2 C |u| \rho}{d} \right) u \quad (2)$$

with d the width of the dike or the diameter of the tunnel, $\alpha_s = 12$, $\mu = 10\text{ Pa s}$, turbulence factor $C = 0.01$. In scaled form, we get $f = [f] f'$, $d = L d'$ (see §2.2) and after dropping the primes we have

$$f = \frac{\nu_s}{d^2} + \gamma_s \frac{|m|}{A d} \quad (3)$$

with $\nu_s = \alpha_s \mu / (\rho_0 L U)$, $\gamma_s = 2 C = 0.02$, with A the relevant cross-sectional area at position ξ_1 , and with momentum area density $m = \rho A u$.

We take an isothermal temperature of $T = 1293\text{ K}$, $\rho_0 = 100\text{ kg/m}^3$, $L = 5\text{ m}$ in the simulations. The stretched grid and functions $A(\xi_1)$ and $z(\xi_1)$ are the same as in the reference simulations in Fig. 5.3 and in Bokhove and Woods [2000, 2001].

We show the pressure, velocity and density profiles for three different resolutions with 3000, 6000 and 12000 grid points, respectively, in Fig. 5.4, Fig. 5.5, and Fig. 5.6. The conclusions are that the modified frictional parameterization leads to a pressure peak in the nozzle area, which is not present in the parameterization (13)-(15) before. The present turbulent frictional parameterization is an order of magnitude stronger. We also notice that the dynamics is well-resolved: only in the sensitive velocity profiles we see any improvement with increasing resolution. Note that $m = \rho A u$ and ρA are the variables used in the computation and that $u = m / (\rho A)$ is a derived variable which is sensitive when ρA is small.

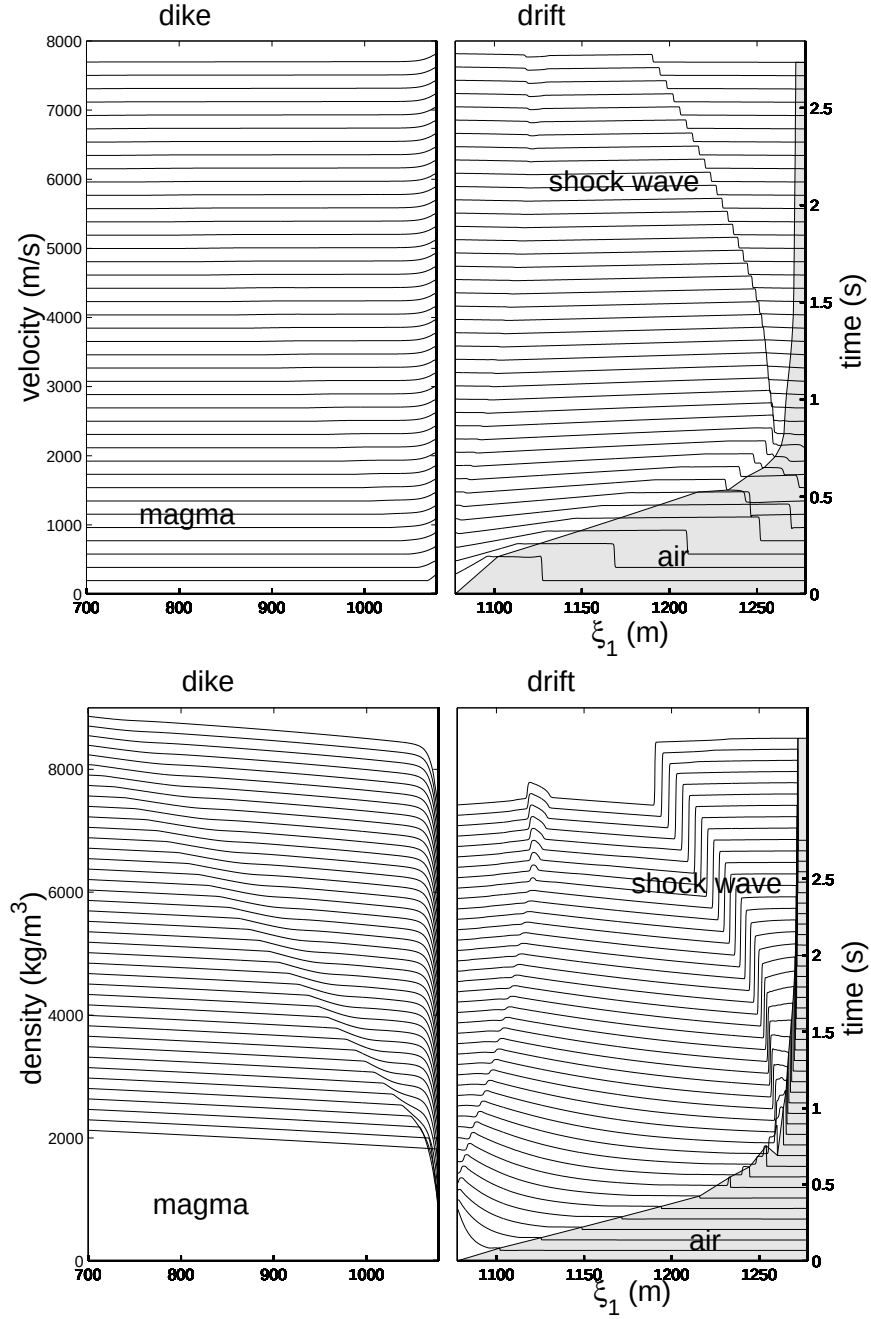


Figure 5.3: Velocity, density and pressure profiles are shown for magma-air interactions in a dike-drift system. The simulation encompasses 2.734 s and each of the 41 profiles is spaced 0.0683 s apart. Profiles in dike and drift have been separated to emphasize the different scaling in dike and drift. The rarefaction wave in the dike, the magma, the air, and the shock wave have been indicated. Short-time reference simulation 101 with encmvs1.c.

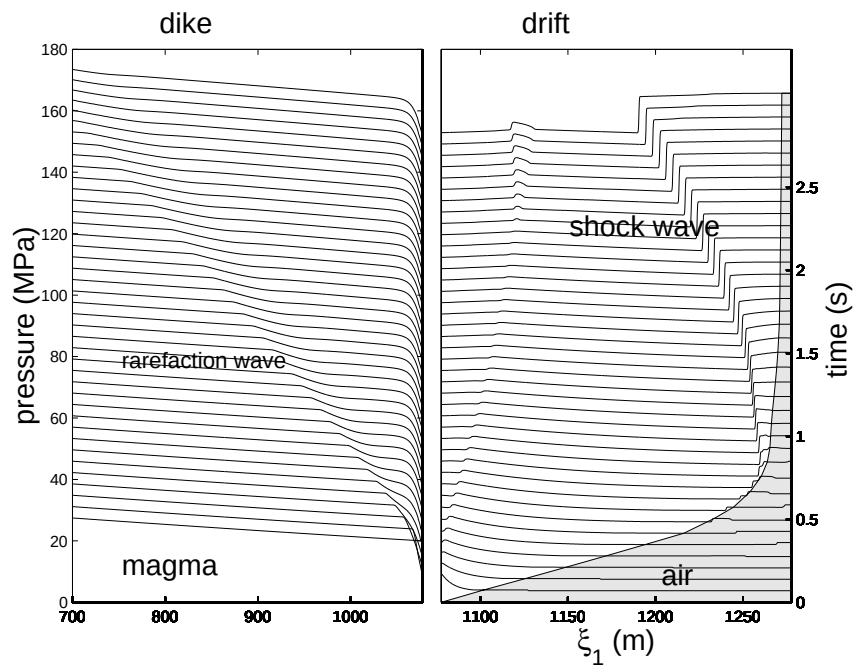


Figure 5.3: Continued, short-time reference simulation 101.

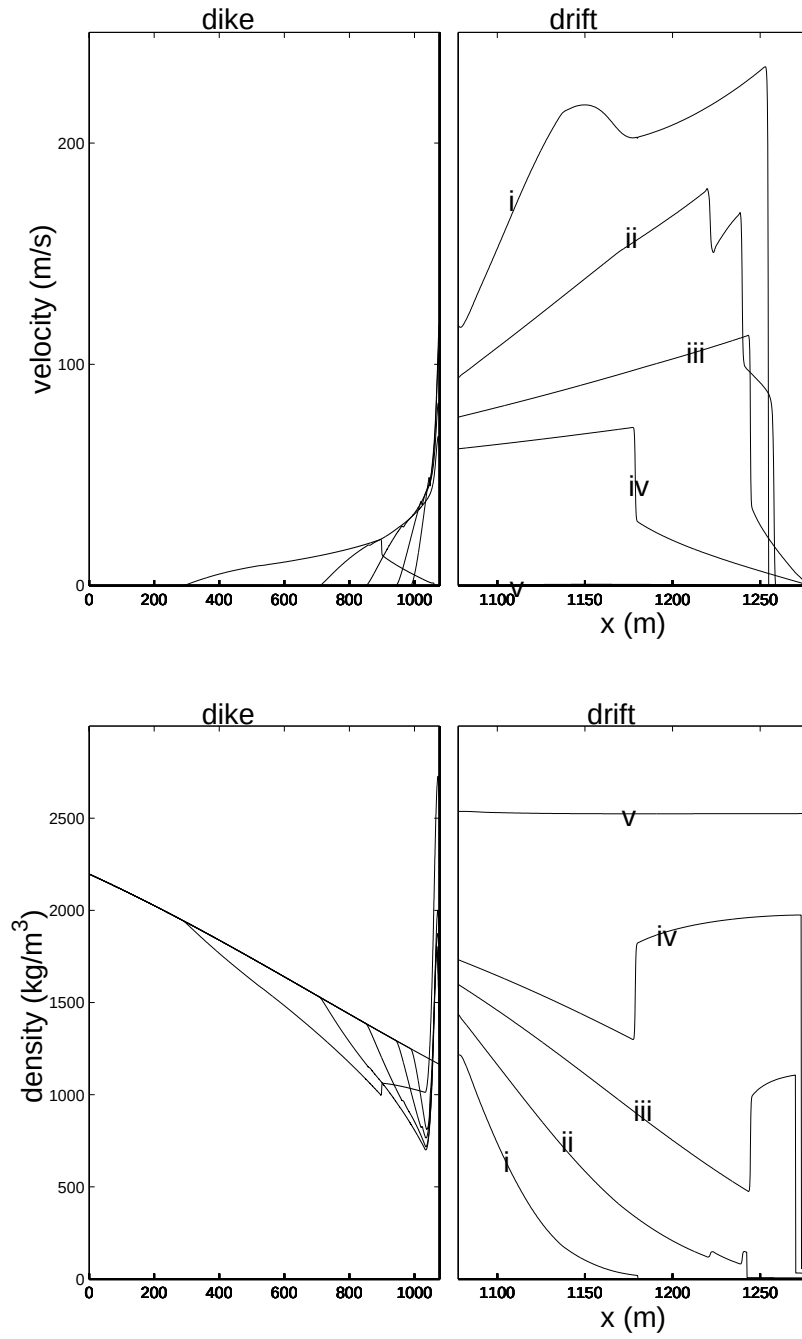


Figure 5.4: Velocity, density and pressure profiles are shown for magma-air interactions in a dike-drift system. Resolution 3000 grid points. Simulation 165 with `encmvs1.c`. Plot made with Matlab program `mva1science.m`.

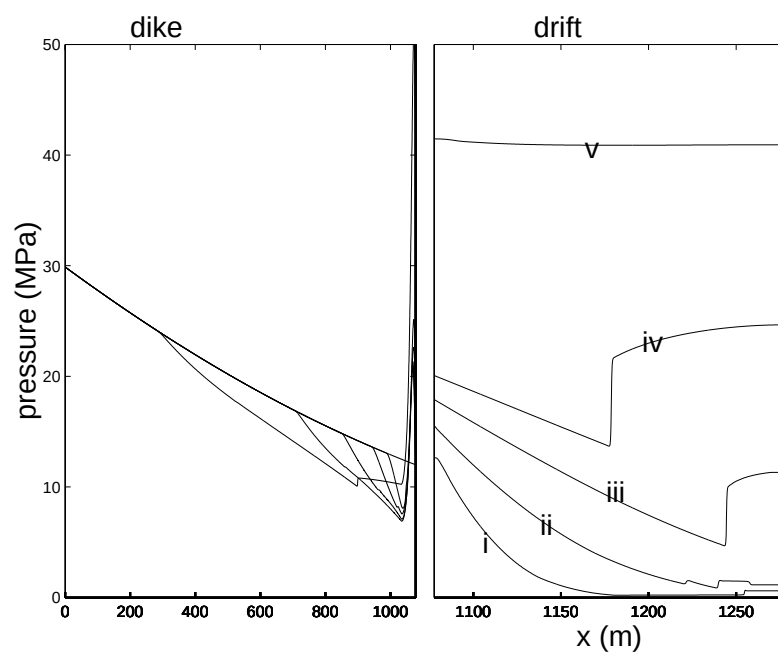


Figure 5.4: Continued simulation 165.

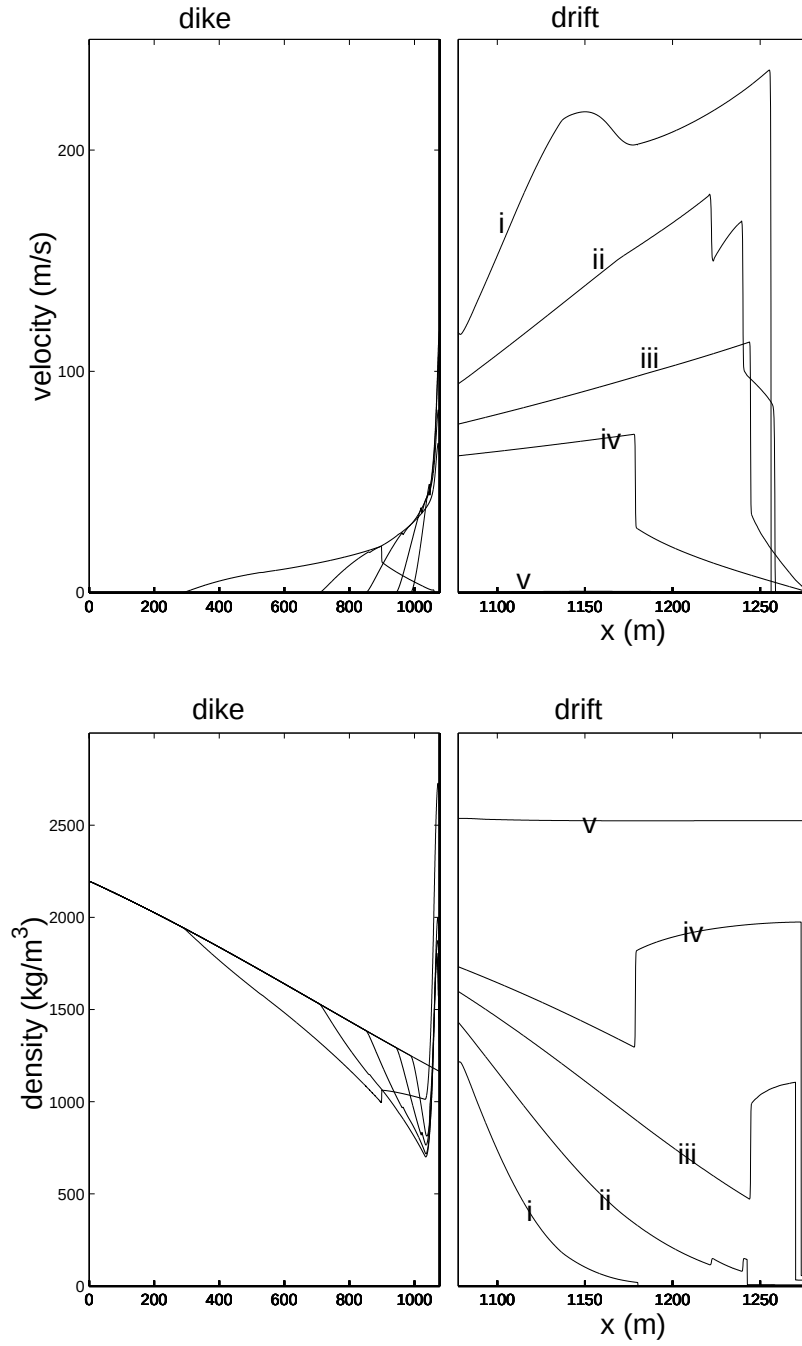


Figure 5.5: Velocity, density and pressure profiles are shown for magma-air interactions in a dike-drift system. Resolution 6000 grid points. Simulation 166 with encmvs1.c.

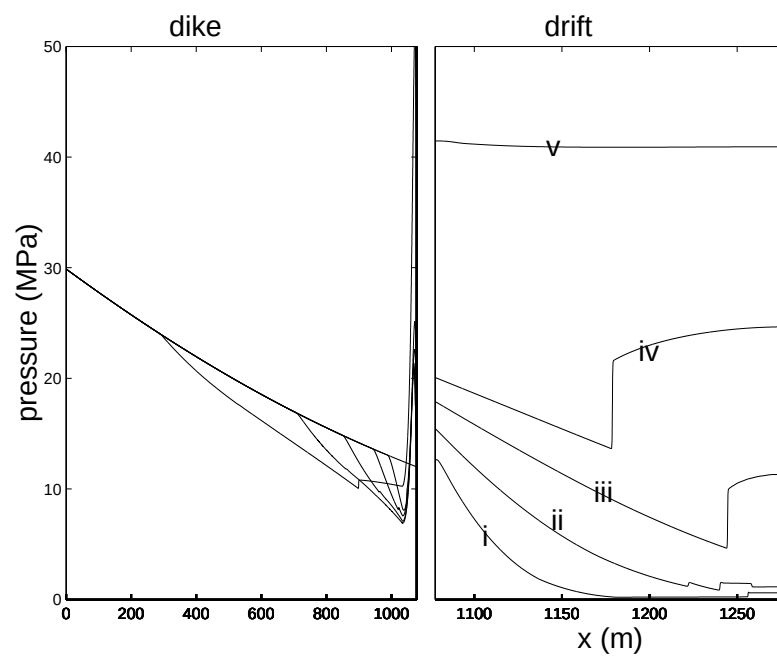


Figure 5.5: Continued simulation 166.

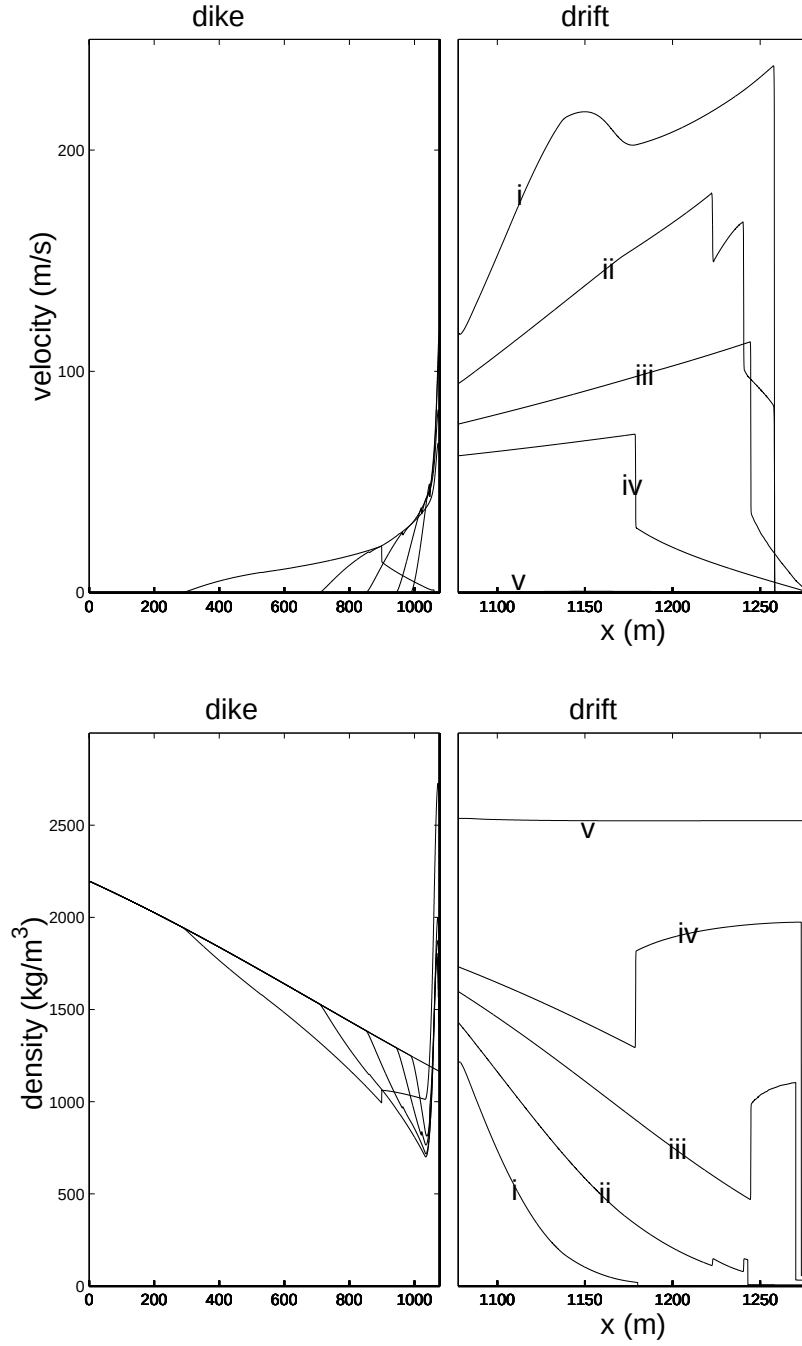


Figure 5.6: Velocity, density and pressure profiles are shown for magma-air interactions in a dike-drift system. Resolution 12000 grid points. Simulation 167 with encmvsl.c.

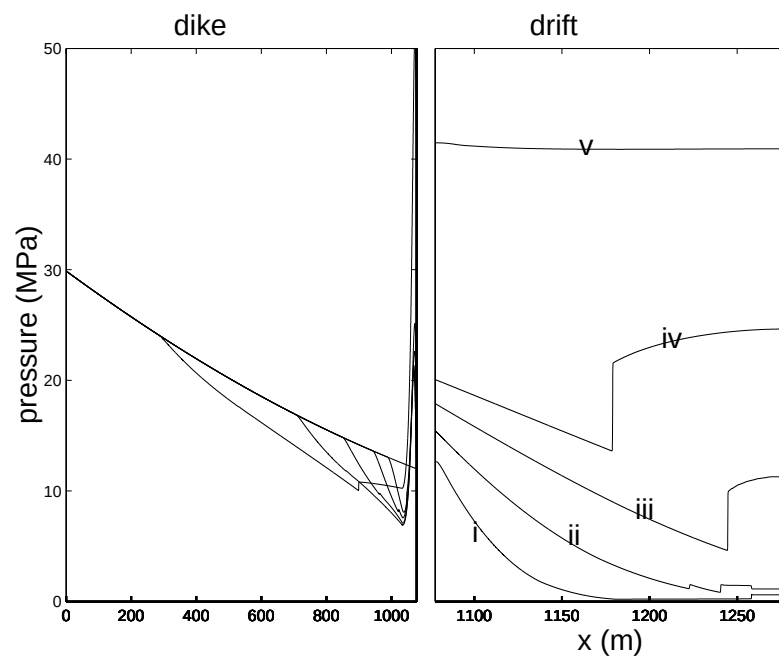


Figure 5.6: Continued simulation 167.

5.4 Geometry and grid transformation in 2D

It is anticipated that we require more resolution around the breakthrough region than in the dike and drift. The transformation $x = x(\phi)$, $z = z(\chi)$ transforms equations (18) into

$$\begin{aligned} \frac{\partial(\rho b u)}{\partial t} + \frac{\partial \phi}{\partial x} \frac{\partial(\rho b u^2 + p b)}{\partial \phi} + \frac{\partial \chi}{\partial z} \frac{\partial(\rho b u w)}{\partial \chi} &= \frac{\partial \phi}{\partial x} p \frac{\partial b}{\partial \phi} - F_1, \\ \frac{\partial(\rho b)}{\partial t} + \frac{\partial \phi}{\partial x} \frac{\partial(\rho u b)}{\partial \phi} + \frac{\partial \chi}{\partial z} \frac{\partial(\rho w b)}{\partial \chi} &= 0, \\ \frac{\partial(\rho b w)}{\partial t} + \frac{\partial \phi}{\partial x} \frac{\partial(\rho b u w)}{\partial \phi} + \frac{\partial \chi}{\partial z} \frac{\partial(\rho b w^2 + p b)}{\partial \chi} &= \frac{\partial \chi}{\partial z} p \frac{\partial b}{\partial \chi} - \frac{1}{F_l} \rho - F_3. \end{aligned} \quad (4)$$

We define a high-resolution grid in the area $(x, z) \in [0, w] \times [0, D]$ which smoothly turns into a coarser grid in the drift and an even coarser one in the dike, see the sketch in Fig. 5.7. For $x = x(\phi)$ we take

$$x(\phi) = \begin{cases} \phi, & 0 < \phi < \phi_w = w, \\ \left(e^{\eta(\phi - \phi_w)/\Delta\phi_w} (\phi - w) + w \right), & w < \phi < w + \Delta\phi_w, \\ (1/\alpha_2)(\phi - w - \phi_w) + e^\eta \Delta\phi_w + w, & w + \Delta\phi_w < \phi < \phi_{end} \end{cases} \quad (5)$$

with $x(\phi_w) = w$, the transition interval ending at $x = L_t = x(w + \Delta\phi_w) = (w + \Delta\phi_w) \exp[\eta]$, the length of the drift $L = (1/\alpha_2)(\phi_{end} - w - \Delta\phi_w) + \Delta\phi_w \exp[\eta] + w$, $\phi_{end} - w - \Delta\phi_w = \alpha_2(L - L_t)$ and $\alpha_2(1 + \eta) \exp[\eta] = 1$. Hence

$$\frac{\partial x}{\partial \phi} = \begin{cases} 1, & 0 < \phi < \phi_w = w, \\ (\eta/\Delta\phi_w) e^{\eta(\phi - \phi_w)/\Delta\phi_w} (\phi - w) + e^{\eta(\phi - \phi_w)/\Delta\phi_w}, & w < \phi < w + \Delta\phi_w, \\ 1/\alpha_2, & w + \Delta\phi_w < \phi < \phi_{end} \end{cases} \quad (6)$$

For $z = z(\chi)$ we take

$$z(\chi) = \begin{cases} \chi, & 0 < \chi < \chi_D = D, \\ e^{-\mu\chi/\Delta\phi_D} \chi, & -\Delta\chi_D < \chi < 0, \\ (1/\gamma_2)(\chi + \Delta\chi_D) - \Delta\chi_D e^\mu, & -\chi_{end} < \chi < -\Delta\chi_D \end{cases} \quad (7)$$

with $z(\chi_D) = D$, $\chi_{end} = \Delta\chi_D + \gamma_2(H - H_t)$, H the total length consisting of drift diameter plus the dike length considered, H_t the transition height with the fine and intermediate resolution near the drift, $\Delta\chi_D = H_t \exp[-\mu]$, and equation $\gamma_2(1 + \mu) \exp[\mu] = 1$ to determine μ . Hence

$$\frac{\partial z}{\partial \chi} = \begin{cases} 1, & 0 < \chi < \chi_D = D, \\ -(\mu/\Delta\chi_D) e^{-\mu\chi/\Delta\phi_D} \chi + e^{-\mu\chi/\Delta\phi_D}, & -\Delta\chi_D < \chi < 0, \\ 1/\gamma_2, & -\chi_{end} < \chi < -\Delta\chi_D. \end{cases} \quad (8)$$

Remember that w is the (half-)width of the dike and D the diameter of the drifts.

The above functions are implemented in functions `Xphi()`, `Zchi()`, `DphiDx()`, `DchiDz()` in file `encmac2.c`. The general initialization and set up are found in the beginning of function `Initial_Condition` in file `encmac2.c` for the choice of initial condition `icno = 6`.

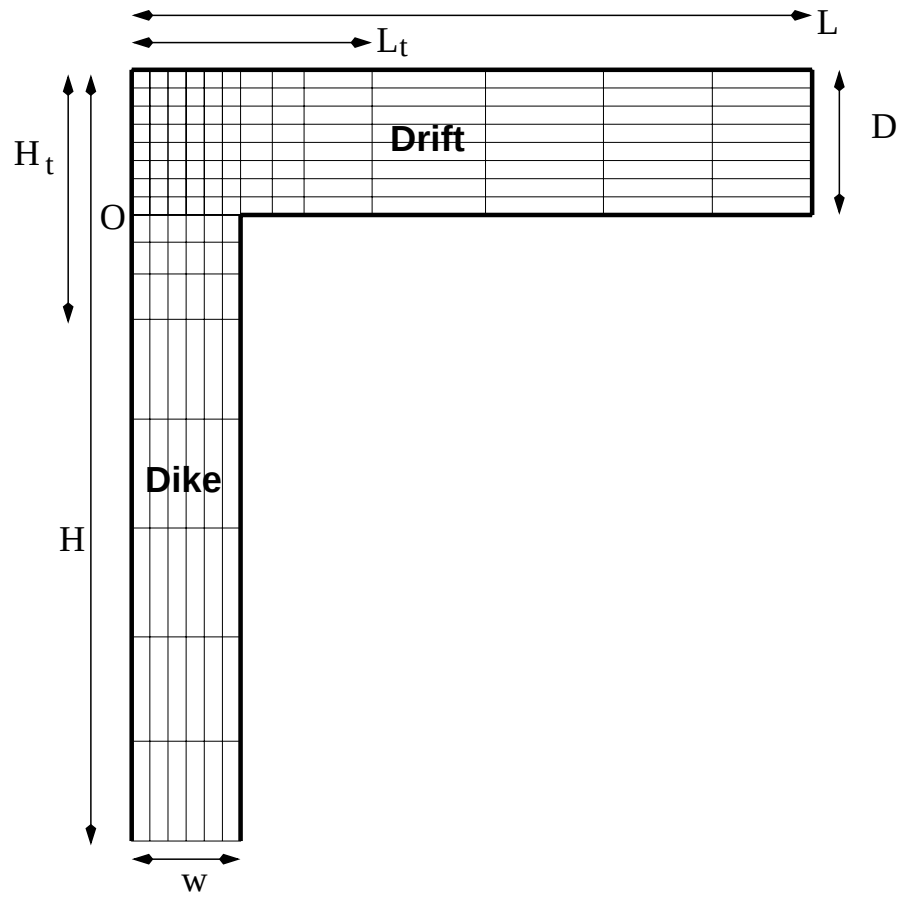


Figure 5.7: Definition sketch of the non-uniform two-dimensional grid in a computational domain consisting of two connected rectangular blocks, one for the drift and one for the dike.

5.5 Future dike-drift geometries

Since we don't model the feedback of the time- and space dependent magma pressure on the rock or vice versa explicitly, we have to prescribe the dike geometries as functions of space and time. Through this geometric description, the dike walls have an influence on the magma fluid dynamics in the dike and drift. Various possible dike-drift geometries under consideration are shown in Fig. 5.8, corresponding to the bulbous dike-tip model of Lister [1990] and the crack-tip model of Rubin [1993, 1995]. Our numerical model is restricted to geometries that consist of multiple, connected, weakly-deformable rectangular blocks, because there must exist a transformation to a rectangular domain in each computational block. Moreover, the resulting geometrical factors must not place too much restriction on the numerical performance. Only configurations Fig. 5.8a, b, d, and Fig. 5.8e are therefore acceptable. In principle, we can also consider asymmetric dike-drift intersections, which will of course require more computational effort.

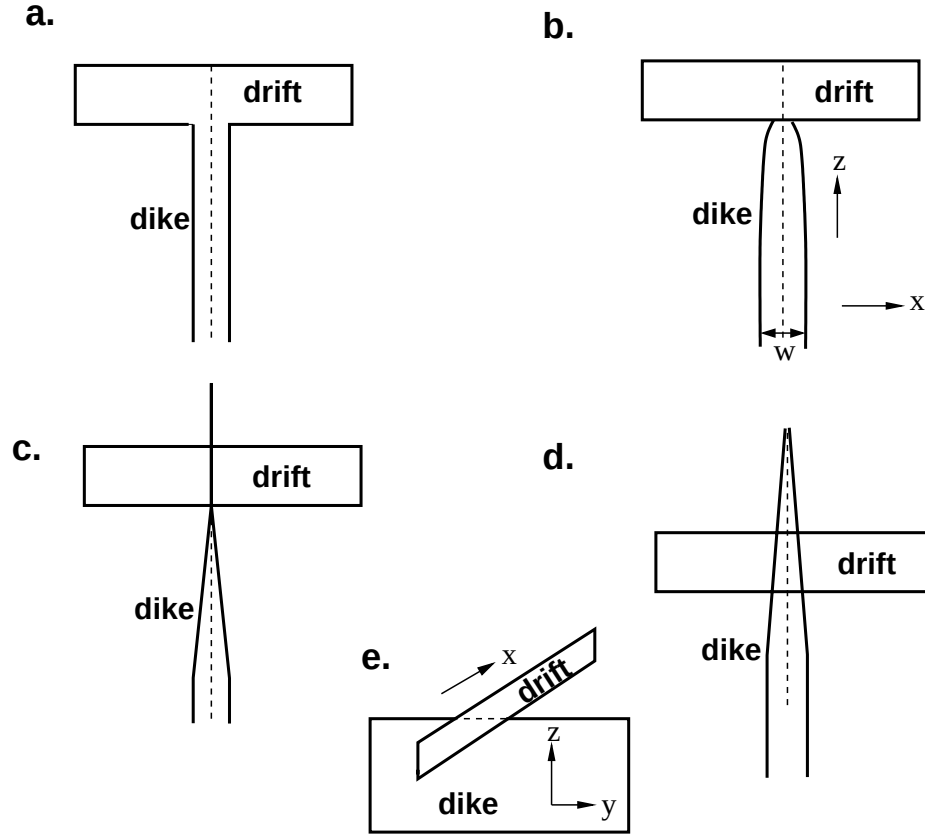


Figure 5.8: Various dike drift geometries under consideration are: laterally averaged models such as (a) the standard reference case, (b) a bulbous dike tip model, (c) a thin crack model, and (d) a thin crack model at a later time, and (e) a mixed model laterally averaged in the drift and averaged in the x -direction across the dike. The width $w = w(z, t)$ can be a function of time as long as the flow in the computational domain can be modeled reasonably well as two or three weakly deformed rectangular blocks. The dashed line denotes the line of symmetry.

Chapter 6

Conclusion

A detailed report has been given on numerical modeling of magma-repository interactions by explosive volcanic flows. In chapter 2, two mathematical models are introduced: a one-dimensional flow-tube and a two-dimensional flow area model, which model magma-air interactions in a dike drift geometry. The numerical discretization of these models for compressible magma and the gas dynamics of air are subsequently discretized in chapter 3. The discretizations used are shock-capturing numerical techniques since a major feature of magma-air interactions consists of interacting shocks. The way the numerical algorithms are translated into code has been described in section 3.8. Various one- and two-dimensional numerical tests in chapter 4 provide the validation of the developed programs. In chapter 5, we describe in detail the grid transformations which are used the one- and two-dimensional models, as well as the (reference) simulations on one-dimensional magma-repository interactions used in in Bokhove and Woods [2000, 2001] and Woods *et al.* [2001].

Potential users of the programs for magma and magma-air flow in flow tube models with adjustable geometries have been given detailed insights in the numerical algorithms and the dike-drift geometry. In parallel, it has also been indicated throughout the report how and where these aspects are implemented in the code. With the instructions in the Appendices, and the comments in the codes as well as the information in the package, one should be able to use and extend these programs.

There are several important numerical aspects that merit further research. These include the following. (i) Further comparisons of the numerical flow tube model can be made with stationary hydraulically-controlled flows in vertical, horizontal and curvilinear dike or conduit geometries; these stationary solutions are governed by ordinary differential equations [see Woods [1995] for explosive volcanic flows in vertical conduits]. (ii) Higher-order convex ENO schemes [Liu and Osher, 1998] are expected to perform better for the mixed hyperbolic and forced flows considered here; there is research potential in developing numerical schemes that are optimal under transient and quasi-steady conditions. (iii) The numerical treatment of interface dynamics should be improved on a fundamental level, rather than by increasing the resolution globally or locally. While the ghost-fluid method of Fedkiw *et al.* [1999] is simple and reasonably accurate, its mathematical basis is imperfect. (iv) The spatially one-dimensional flow tube model can be extended to include gravity-current formation at the end of the tunnel. The interface is then allowed to tilt, spread and thus form a compressible gravity-current, instead of remaining vertical after it slows down. (v) Two-dimensional modeling of the magma and air in separation requires numerical techniques that either follow the interface or incorporate the complicated mixing of hot ash, water vapor and air.

Acknowledgments The numerical work benefitted significantly from research of O.B. performed under an EC MAST-III Surf and Swash Zone Mechanics grant received by Professor D.H. Peregrine, who also kindly suggested involvement of the author in the project and shared his knowledge in several discussions. Dr. A. Roger-son assistance in the initial development of ENO numerical codes for shallow-water dynamics is gratefully acknowledged.

This manuscript is the result of work that the author performed for and in part at the Center for Nuclear Waste Regulatory Analyses (CNWRA) and for the U.S. Nuclear Regulatory Commission (NRC), through contracts of the CNWRA with the University of Bristol and the University of Twente. The presented results do not necessarily reflect the views or regulatory position of the NRC. It is a pleasure to thank L. Connor, C. Connor., B.E. Hill, R.H. Martin, R.S.J. Sparks, and A.W. Woods for all their comments and discussion. Last but not least, I wish to thank the staff at Southwest Research Institute for their hospitality during my various visits.

References

- Batchelor, G.K., 1988: *Fluid Dynamics*. Cambridge University Press, 615 pp.
- Bokhove, O., 1999: Numerical modeling of magma-repository interactions. School of Mathematics/Southwest Research Institute Internal report, 106 pp. Available upon request from the author and from: South West Research Institute, Center for Nuclear Waste Regulatory Analyses, 6220 Culebra Rd. San Antonio.
- Bokhove, O., & Woods, A.W., 2000: Explosive Magma-Air Interactions by Volatile-Rich Basaltic Melts in a Dike-Drift Geometry. NRC report, 55 pp.
- Bokhove, O., & Woods, A.W., 2001: The explosive decompression of basaltic magma into a sub-surface repository. Revision *J. Geophys. Res. Solid-Earth*, 57 pp.
- Connor, C.B., Lane-Magsino, S., Stamatakis, J.A., Martin, R.H., LaFemina, P.C., Hill, B.E., & Lieber, S., 1997: Magnetic surveys help reassess volcanic hazards at Yucca Mountain, Nevada. *Eos, Transactions, American Geophysical Union*, **7**, 73–78.
- Connor, C.B., J.A. Stamatakis, D.A. Ferrill, B.E. Hill, G.I. Ofoegbu, F.M. Conway, B. Sagar, & J.S. Trapp, 2000: Geologic factors controlling patterns of small-volume basaltic volcanism: Application to a volcanic hazards assessment at Yucca Mountain, Nevada. *J. Geophys. Res.* **105**, 417–432.
- Fedkiw, R.P., Aslam, T, Merriman, B., and Osher, S., 1999: A non-oscillatory Eulerian approach to interfaces in multimaterial flows (the ghost fluid method). *J. Comp. Phys.* **152**, 457–492.
- Gill, A.E., 1982: *Atmosphere-Ocean Dynamics*. Academic Press, 662 pp.
- Lamb, H., 1932: *Hydrodynamics*. Dover, Toronto, 508 pp.
- Leveque, R.J., 1997: Wave propagation algorithms for multidimensional hyperbolic systems. *J. Comp. Phys.* **131**, 327–353.
- Leveque, R.J., 1998: Balancing source terms and flux gradients in high-resolution Godunov schemes. The quasi-steady wave-propagation algorithm. *J. Comp. Phys.* **146**, 346–365.
- Lister, J.R., 1990: Buoyancy-driven fluid fracture - the effects of material toughness and of low-viscosity precursors. *J. Fluid Mech.* **210**, 263–280.
- Lister, J.R. & Kerr, R.C., 1991: Fluid-mechanical models of crack-propagation and their application to magma transport in dykes. *J. Geophys. Res. Solid-Earth* **96**, 10049–10077.
- Liu, X.-L. & Osher, S., 1998: Convex ENO high order multi-dimensional schemes without field by field decomposition on staggered grids. *J. Comp. Phys.* **142**, 304–330.
- Mériaux, C., Lister, J.R., Lyakhovsky, V., & Agnon, A., 1999: Dyke propagation with distributed damage of the host rock. *Earth Planet. Sci. Lett.* **165**, 177–185.
- Phillips, J.C., Lane, S.J., Lejeune, A.M., and Hilton, M., 1995: Gum rosin-acetone system as an analogue to the degassing behaviour of hydrated magmas. *Bull. Volc.* **57**, 263–268.
- Rubin, A.M., 1993: Dikes versus diapirs in viscoelastic rock. *Earth Planet. Sci. Lett.* **119**, 641–659.

- Rubin, A.M., 1995: Propagation of magma-filled cracks. *Ann. Rev. Earth Planet. Sci.* **23**, 287–336.
- Shepherd, T.G., 1993: A unified theory of available potential energy. *Atmos.-Ocean* **31**, 1–26.
- Shu, C-W., 1997: Essentially non-oscillatory and weighted non-oscillatory schemes for hyperbolic conservation laws. NASA Langley research Center. 77 pp.
- Shu, C-W. & Osher, S., 1988: Efficient implementation of essentially non-oscillatory shock-capturing schemes. *J. Comp. Phys.* **7**, 439–471.
- Shu, C-W. & Osher, S., 1989: Efficient implementation of essentially non-oscillatory shock-capturing schemes II. *J. Comp. Phys.* **83**, 32–78.
- Sparks, R.S.J., 1978: The dynamics of bubble formation and growth in magma. *J. Volcanol. Geotherm. Res.* **29**, 99–124.
- Toro, E.F., 1989: A weighted average flux method for hyperbolic conservation laws. *Proc. Roy. Soc. London, Ser. A* **423**, 401–418.
- Toro, E.F., 1997: *Riemann Solvers and Numerical Methods for Fluid Dynamics*. Springer-Verlag, 624 pp.
- Vergnolle, S., & Jaupart, C., 1986: Separated two-phase flow and basaltic eruptions. *J. Geophys. Res. Solid-Earth* **91**, 12842–12860.
- Whitham, G.B., 1974: *Linear and nonlinear waves*. John Wiley, Toronto, 636 pp.
- Wilson, L., Sparks, R.S.J., Huang, C.T., and Watkins, N.D., 1980: Explosive volcanic eruptions, IV, The control of magma properties and conduit geometry on eruption column behaviour. *Geophys. J. R. Astron. Soc.* **63**, 117–148.
- Wilson, L., & Head, J.W., 1981: Ascent and eruption of basaltic magma on Earth and moon. *J. Geophys. Res.* **86**, 2171–3001.
- Woods, A.W., 1995: The dynamics of explosive volcanic eruptions. *Reviews of Geophysics* **33**, 495–530.
- Woods, A.W. and Sparks, S., 1998: *Report on scoping calculations for magma-repository interaction at Yucca mountain*. Centre for Environmental and Geophysical Flows, University of Bristol, Bristol, 24 pp.
- Andrew W. Woods, Steve Sparks, Onno Bokhove, Anne-Marie Lejeune, Chuck Connor, and Brittain Hill, 2001: Modelling the explosive eruption of basaltic magma into the proposed high level radioactive waste repository at Yucca Mountain, Nevada. In preparation for *Science*, 8 pp.

Appendix A

Retrieval of package

The whole package of C-programs, header files, Matlab plotting programs, reports and html-files is available in a compressed form as `magrepint.tar.gz`. After retrieval it is best to place the file in the directory where you wish to unfold the package. Unfolding the package creates subdirectories. The retrieved tar-file can be uncompressed by the command `gzip -d magrepint.tar` on a LINUX or UNIX machine. The tar file is untarred by the command `tar xvf magrepint.tar` (see man-pages for `tar`). What will appear is an instruction file `READMEmagrepint`, a pdf version of this document `magnumrep01.pdf`, and directories with all the C-programs, Matlab-programs, and files to put the package onto a web page.

Tables 3.1, 3.2, 3.3, 3.4 and 3.5 summarize the purpose of all core functions. They define the name of the executable. After inspection of the `Makefiles` it will become clear which files belong together.

Appendix B

Examples

In this appendix, we will step-by-step go through a couple of examples. Once the user is familiar with these examples, the effects of a small parameter change within the same example can be investigated before larger changes are made within the actual program.

Compiling and linking on different compilers and machines may require modifications in: `iso1.h`, `encisolinp.c`, `encisolin.c`, `enoisolinp.c`, `enoisolin.c`, `iso2.h`, `enciso2inp.c`, and `enoiso2in.c`, see section 3.10. These programs are short and the changes for several compilers and machines can be found as comments in these programs.

B.1 Example 1: stationary shock in magma

The stationary shock in magma can be simulated with programs `enciso1` and `enoiso1` and should result in graphs equivalent to Fig. 4.2a,b. The initial conditions are found in the input files `enciso1.in` and `enoiso1.in`, respectively. The proper set of parameters can be found by searching for `simulation000` in either of these two input files. Copy, but don't move, these parameter sets to the top of the input file and save. Boundary conditions in both simulations are extrapolating ones. In programs `enciso1.c` and `enoiso1.c`, respectively, we have to `#define EXTRAP` near the top of these programs where we find the words `Boundary conditions`. Alternatively, we may `#define WESTEXTRAP` and `#define EASTEXTRAP` together.

Compile and link the programs on the command line with command: `< prompt >make enciso1< return >` or `< prompt >make enoiso1< return >`. At this point, the computer may either issue a warning or error message and the IO-treatment may have to be adjusted to fit your machine. The program will run with command `< prompt >enciso1< return >` or `< prompt >enoiso1< return >`. It is finished within a few seconds on a PC with Pentium-III processor.

Internal error message may now appear, for example because the boundary conditions were specified wrongly as `EASTSOLID`. But if all proceeds fine, four or five output files will be generated, either `e1ce000`, `e1ct000`, `e1cq000`, `e1cd000`, `e1ci000` or `e1se000`, `e1st000`, `e1sq000`, `e1si000` for LLF2 or ENO, respectively. Some of the two output files are not used in the magma simulations. A first check can be made by viewing the energetics in `e1ce000` or `e1se000`. The stationarity of the dynamics implies that energy should stay constant. The parameter set of the simulation is saved in `ec000/es000`, respectively.

The data format can be found in the program `encisolin.c/enoisolin.c` in the measurement routine `Measure()` in `enciso1.c/enoiso1.c`, respectively. At this point, you can use your own graphics package or the provided Matlab-routines. The Matlab-routines do provide insights into the data structure. Be sure that

the Matlab and data files can be accessed by specifying the right path names with the Matlab command: `addpath ....` A Matlab-animation can be made with `isot1bfilm.m` (e.g., set `nofile = 0` or `10` for LLF or ENO3, respectively). Stack plots, energetics and comparisons with exact solutions can be generated by `isot1.m`.

B.2 Example 2: shock tube in air

A shock tube in air can be simulated with programs `encair1` and `enoair1` and the graphs should result in graphs equivalent to Fig. 4.11a,b. The initial conditions are found in the input files `enciso1.in` and `enoiso1.in`, respectively. The proper set of parameters can be found under the phrase **Air tests** or by searching for `simulationair011` in either of the two input files. Copy, but don't move, these parameter sets to the top of the input file and save. Boundary conditions in these simulations are extrapolating ones;. In programs `encair1.c` and `enoair1.c`, respectively, we have to `#define EXTRAP` near the top of these programs where we find the words **Boundary conditions**. Alternatively, we may `#define WESTEXTRAP` and `#define EASTEXTRAP` together.

Compile and link the programs on the command line with command `< prompt >make encair1< return >` or `< prompt >make enoair1< return >`. At this point, the computer may either issue a warning or error message and the IO-treatment may have to be adjusted to fit your machine. The program will run with command `< prompt >encair1< return >` or `< prompt >enoair1< return >`. It is finished within a few seconds on a PC with Pentium-III processor.

Internal error messages may now appear, for example because the boundary conditions were specified wrongly as **EASTSOLID**. But if all proceeds fine, four or five output files will be generated, either `e1ce011`, `e1ct011`, `e1cq011`, `e1cd011`, `e1ci011` or `else011`, `elst011`, `elsq011`, `elsi011` for LLF2 or ENO, respectively. Not all of these files are used in air simulations. A first check of the program can be made by viewing the energetics in `e1ce011` or `else011`. Energy should stay constant as long as no information has left the domain. The parameter set of the simulation is saved in `ec011/es011`, respectively.

The data format can be found in the program `enciso1.in/c/enoiso1.in.c` in the measurement routine `Measure()` in `encair1.c/enoair1.c`, respectively. A Matlab-animation can be made with `isoair1bfilm.m` (e.g., with `nofile = 1, 11`, respectively). Stack plots, energetics and comparisons with exact solutions can be generated by `isoair1.m` (e.g., with `nb = 36, 32`, respectively).

B.3 Example 3: reference simulation run 101

The first reference simulation of magma-repository interactions is simulated with program `encmvs1`. Graphs should be equivalent to Fig. 5.3. The initial conditions are found in the input file `enciso1.in`. The proper set of parameters can be found by searching for the phrase `simulationair101` or simply `101`. Copy, but don't move, these sets to the top of the input file and save. Boundary conditions in these simulations are extrapolating ones; in programs `encmvs1.c` we have to `#define WESTCHAR`, `#define EASTSOLID` and `#define INTERFACE` near the top of program `encmvs1.c` where we find the words **Boundary conditions**.

Compile and link the program on the command line with command `< prompt >make encmvs1< return >`. At this point, the computer may either issue a warning or error message and the IO-treatment may have to be adjusted for your machine. The program will run with command `< prompt >encmvs1< return >`. It is finished within after a couple of hours on a PC with Pentium-III processor.

If all proceeds fine, four or five output files will be generated, either `e1ce101`, `e1ct101`, `e1cq101`, `e1cd101`, `e1ci101`. Check whether these files are generated immediately after the simulation has started. A first

check of the program can be made by checking the energetics in `e1ce101`. Note that the potential energy is dominating, we really should record the *pseudo-energy*. Pseudo energy is an exact combination of an energy and a mass invariant which measures the deviation from a (hydrostatic) basic state and is a second-order quantity in a small-amplitude perturbation expansion around the basic state (*e.g.* Shepherd 1993).

The data format can be found in the program `encisolin.c` in the measurement routine `Measure()` in `encmvs11.c`. A Matlab-animation may be made with `isoair1bfilm.m` (*e.g.*, with `nofile = 101`). Stack plots, energetics and comparisons with exact solutions can be generated by `mval.m`, which plots the velocity, density and pressure profiles against the computational coordinate ϕ , and `mvalsudet.m`, which plots these profiles against the curvilinear coordinates ξ_1 and adds shading where the air resides.

B.4 Example 4: science simulation run 165

The *Science* paper simulation of magma-repository interactions is simulated with program `encmvs1`. Graphs should be equivalent to Fig. 5.4. The initial conditions are found in the input file `enciso1.in`. The proper set of parameters can be found by searching for the phrase `simulationair165` or simply `165`. Copy, but don't move, these sets to the top of the input file and save. Boundary conditions in these simulations are extrapolating ones; in programs `encmvs1.c` we have to `#define WESTCHAR`, `#define EASTSOLID` and `#define INTERFACE` near the top of program `encmvs1.c` where we find the words `Boundary conditions`.

Compile and link the program on the command line with command `< prompt >make encmvs1< return >`. At this point, the computer may either issue a warning or error message and the IO-treatment may have to be adjusted for your machine. The program will run with command `< prompt >encmvs1< return >`. It is finished within after a couple of hours on a PC with Pentium-III processor.

If all proceeds fine, four or five output files will be generated, either `e1ce165`, `e1ct165`, `e1cq165`, `e1cd165`, `e1ci165`. Check whether these files are generated immediately after the simulation has started. A first check of the program can be made by checking the energetics in `e1ce165`.

The data format can be found in the program `encisolin.c` in the measurement routine `Measure()` in `encmvs1.c`. A Matlab-animation may be made with `isoair1bfilm.m`. Stack plots, energetics and comparisons with exact solutions can be generated by `mvalscience.m`, `mval.m` or `mvalsudet.m`. To use `mval.m` you have to edit the file. Edit the `mvalsudet.m` file appropriately since the tick marks need to be set individually. Search for all occurrences of `101` in the file `mvalsudet.m` to find out sample settings.

B.5 Example 5: one block 2D shocktube simulation run 202

We consider the magma dynamics simulation in one block of Fig. 4.27. The initial conditions are found in the input file `enciso2.in`. The proper set of parameters can be found by searching for the phrase `simulation202` or simply `202`. Copy, but don't move, these sets to the top of the input file and save. Boundary conditions in these simulations are `EASTSOLID`, `EASTEXTRAP`, `NORTHSOLID`, `SOUTHSOLID` which have to be set in `encmac2.c` at the top where we find the words `Boundary conditions`.

Compile and link the program on the command line with command `< prompt >make encmac2< return >`. At this point, the computer may either issue a warning or error message and the IO-treatment may have to be adjusted for your machine. The program will run with command `< prompt >encmac2< return >`. It is finished within after a couple of hours on a PC with Pentium-III processor.

If all proceeds fine, four or five output files will be generated, either `e2ce202`, `e2ct202`, `e2cu202`, `e2cq202`, `e2cd202`, `e2ci202`. Check whether these files are generated immediately after the simulation has started. A

first check of the program can be made by checking the energetics in e2ce202.

The data format can be found in the program enciso2in.c in the measurement routine Measure() in encmac2.c. Pressure contour plots with arrows for the velocity can be made with magtws.m. Edit file magtws.m to obtain right input directory. See under 202.

B.6 Example 6: two block 2D magma simulation run 212

We consider the magma dynamics simulation in one block of Fig. 4.27. The initial conditions are found in the input file enciso2.in. The proper set of parameters can be found by searching for the phrase simulation212 or simply 212. Copy, but don't move, these sets to the top of the input file and save. Boundary conditions in these simulations are EASTSOLID, WESTSOLID, NORTHSPEC1, SOUTHSPEC2, BLOCK12GLUE which have to be set in encmac2.c at the top where we find the words Boundary conditions.

Compile and link the program on the command line with command *< prompt >make encmac2< return >*. At this point, the computer may either issue a warning or error message and the IO-treatment may have to be adjusted for your machine. The program will run with command *< prompt >encmac2< return >*. It is finished within after a couple of hours on a PC with Pentium-III processor.

If all proceeds fine, four or five output files will be generated, either e2ce212, e2ct212, e2cu212, e2cq202, e2cd212, e2ci212. Check whether these files are generated immediately after the simulation has started. A first check of the program can be made by checking the energetics in e2ce212.

The data format can be found in the program enciso2in.c in the measurement routine Measure() in encmac2.c. Pressure contour plots with arrows for the velocity can be made with magtws.m. Edit file magtws.m to obtain right input directory. See under 212.